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ANTIBIOTIC POLICY 2026

Antibiotic Policy

2026

JIPMER

Puducherry

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जवाहरलाल स्नातकोत्तर आयुर्विज्ञान शिक्षा एवं अनुसंधान संस्थान
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FOREWORD

Antimicrobial resistance (AMR) has emerged as one of the most serious global public health challenges of the 21st century. Inappropriate use of antibiotics has significantly contributed to the growing burden of resistant infections, resulting in increased morbidity, mortality, healthcare costs, and prolonged hospital stays.

Healthcare institutions play a pivotal role in combating antimicrobial resistance by promoting rational antibiotic use. Institution-specific antimicrobial guidelines, informed by local microbiological data and resistance patterns, are essential tools to support clinicians in making evidence-based treatment decisions while preserving the effectiveness of existing antimicrobial agents.

The Antimicrobial Stewardship Plan of JIPMER have been developed to provide practical, concise, and locally relevant recommendations for the management of common infectious syndromes encountered in both community-acquired and hospital-acquired settings. These guidelines align with national priorities on antimicrobial stewardship and are intended to support safe, effective, and judicious use of antimicrobials empirically.

I commend the multidisciplinary team of clinicians, microbiologists, and clinical pharmacists who have contributed their expertise to the development of this guideline. I am confident that this document will serve as a valuable resource for healthcare professionals and will strengthen antimicrobial stewardship practices within our institution.

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Antibiotic Policy

2026

I. General principles of antibiotic therapy

Treatment of infections due to Gram-negative bacilli: Gram negative bacilli are intrinsically resistant to penicillin. However, the majority of gram-negative bacilli were originally susceptible to extended spectrum penicillins (ampicillin, amoxicillin, piperacillin) as well as cephalosporins. Later, these bacteria started producing road spectrum beta lactamases. These beta lactamases inactivate extended spectrum penicillins and first generation cephalosporins as well as some second generation cephalosporins. However, third and fourth generation cephalosporins are resistant to the action of these beta lactamases. So, the drugs of choice for these pathogens were 3rd or 4th generation cephalosporins. However, of late the emergence of very broad-spectrum beta lactamases has increasingly rendered 3rd and 4th generation cephalosporins largely ineffective against Gram negative bacilli

There are two types of very broad-spectrum beta lactamases.

Extended spectrum beta lactamases (ESBL) and

Amp C beta lactamases (Amp C BL)

ESBL confer resistance against 3rd and 4th generation cephalosporins. Ceftriaxone resistance is taken as the marker of ESBL production. ESBL production has become very common among gram negative bacilli such as E. coli and Klebsiella. In India, over 3/4th of even community acquired E. coli produce ESBL.

Amp C beta lactamases confer resistance against 3rd generation cephalosporins but the 4th generation cephalosporin cefepime is generally effective against Amp C BL producing bacteria.

Beta lactamase inhibitors are able to inhibit extended spectrum beta lactamases. So, a combination of beta lactam (BL) with beta lactamase inhibitor (BLI) is usually effective against ESBL producing pathogens. BL/BLI combinations are useful in treating mild to moderate infections caused by ESBL producing pathogens but in seriously ill patients they may be inferior to carbapenems.

Table 1. List of cephalosporins…

Gen	*Oral*	*Parenteral*
First	Cephalexin	Cefazolin
Second	Cefuroxime	Cefuroxime Cefoxitin
Third	Cefdinir Cefpodoxime Cefixime	Ceftriaxone Cefoperazone Ceftazidime
Fourth	-	Cefepime
…BL/BLI combinations		
Ampicillin/Sulbactam Amoxicillin/Clavulanic acid Piperacillin/Tazobactam Cefoperazone/Sulbactam		

Gram negative bacilli are important pathogens in UTI and intra-abdominal infections. They are usually caused by enteric Gram-negative bacilli such as E coli, Klebsiella etc. Over 3/4th of even community acquired enteric Gram-negative bacilli produce ESBL; 3rd or 4th generation cephalosporins are ineffective against such bacteria. Such infections should be treated with BL/BLI or carbapenems. BL/BLI combinations are generally effective, but they may be inferior to carbapenems in severe or life-threatening infections, particularly in high inoculum infections.

ESBL producing pathogens generally carry resistance genes conferring resistance to many other classes of antibiotics as well. They are commonly resistant to cotrimoxazole and quinolones. However, the vast majority of community acquired ESBL producing gram negative bacilli remain susceptible to amikacin.

Carbapenems are not inactivated by extended spectrum or Amp C beta lactamases. They are effective against even ESBL or Amp C producing bacteria. However, carbapenemase producing gram negative bacilli are increasing in prevalence, and we are beginning to lose these potent weapons as well. This is gradually leading us to a post-antibiotic era ruled by superbugs.

Note: Beta lactamase inhibitors have only limited efficacy against Amp C beta lactamases; hence BL/BLI combinations do not work well against Amp C producing bacteria

ESBL coverage is not required for community acquired infections above diaphragm, such as meningitis and pneumonia. There, BL/BLI or carbapenems are generally not required; 3rd generation cephalosporins remain the choice.

Table 2. Susceptibility of Gram-negative bacilli to Beta lactams

Beta lactam drug	*Susceptibility of enteric Gram-negative bacilli*	*Susceptibility of enteric Gram-negative bacilli producing broad spectrum beta lactamases*	*Susceptibility of Gram-negative bacilli producing Extended spectrum beta lactamases (ESBL)*	*Susceptibility of Gram-negative bacilli producing Amp C beta lactamases*
Penicillin G	Intrinsically resistant	-	-	-
Aminopenicillins: Ampicillin, Amoxycillin	Yes	No	No	No
Carboxy penicillin Piperacillin	Yes	No	No	No
1rst generation cephalosporins	Yes	No	No	No
2nd generation cephalosporins	Yes	May be	No	No
3rd generation cephalosporins (parenteral)	Yes	Yes	No	No
4th generation cephalosporins	Yes	Yes	No	Yes
BL/BLI combination	Yes	Yes	Yes, but may be inferior to carbapenems in severe or life-threatening infections	No
Carbapenems	Yes	Yes	Yes	Yes

Note: Haemophilus influenza and Moraxella catarrhalis are gram negative coccobacilli. They are important respiratory pathogens. They commonly produce broad-spectrum beta lactamase, but they do not produce ESBL. They remain susceptible to 3rd and 4th generation cephalosporins and BL/BLI.

Other antibiotics for infections with gram negative bacilli (irrespective of beta lactamase production): Even though originally cotrimoxazole was a very effective antibiotic against many gram-negative bacilli, well over 2/3rd of even community acquired E. coli and Klebsiella isolates are now resistant to co-trimoxazole.

Similarly, originally quinolones had excellent activity against most of the gram-negative bacilli, but now well over 2/3rd of even community acquired gram negative bacilli are quinolone resistant. Gentamicin resistance is common even among community acquired pathogens but most of the community acquired gram negative bacilli (>90%) remain susceptible to amikacin. However, amikacin resistance is becoming increasingly common among hospital acquired gram negative bacilli.

Streptococcal infections: Beta-haemolytic streptococci of all groups (A, B, C, F, G) remain uniformly susceptible to penicillin and amoxicillin. (AIIMS 2018).

Macrolide resistance is seen in about half the strains of group A beta haemolytic streptococci; clindamycin resistance is seen in about one third 33% (both as per AIIMS 2018, JIPMER 2019). Tetracycline resistance is very high

Resistance pattern of group A beta haemolytic streptococci (studies included invasive and non-invasive isolates).

Table 3. Resistance among group A beta haemolytic streptococci

	JIPMER 2019	*AIIMS 2018*
Penicillin	0%	0%
Erythromycin	53%	46%
Clindamycin	33%	33%
Tetracycline	58%	79%
Ciprofloxacin	NA	10%

Staphylococcal infections: Once upon a time, Staph. aureus strains were fully susceptible to penicillin. But over the years, they started producing penicillinase which inactivates penicillin and its broad spectrum and extended spectrum derivatives such as ampicillin, amoxicillin, and piperacillin. Now over 90% of the strains are resistant to penicillin. To overcome this problem, **penicillinase resistant penicillins** were developed. They are resistant to the action of penicillinase. This group is called **the methicillin group** and includes methicillin, oxacillin, cloxacillin, dicloxacillin, flucloxacillin and nafcillin. Methicillin is no longer used clinically. Only cloxacillin and flucloxacillin are available in India. It must be noted that the staphylococcal penicillinase is a narrow spectrum penicillinase and does not inactivate cephalosporins and BL/BLI combinations. So, drugs of choice for staphylococci today are penicillinase resistant penicillins, cephalosporins and BL/BLI combinations. Since these staphylococci are susceptible to methicillin, they are called MSSA.

Among cephalosporins, first generation cephalosporins are *highly* effective against MSSA. Many second generation cephalosporins and the third generation cephalosporins have only *modest* efficacy. Cefepime, the 4th generation cephalosporin and carbapenems also have *modest* activity against staphylococci. Some third generation cephalosporins such as cefixime and ceftazidime have *very poor* activity against staphylococci.

In the last several decades, staphylococci with altered penicillin binding protein (PBP) have emerged. Since the binding site is altered, none of the beta lactams are effective against these staphylococci. These strains are resistant to all beta lactams including the methicillin group, cephalosporins,

carbapenems as well as BL/BLI combinations. They are called MRSA. (Recently some fifth generation cephalosporins have been developed which are effective against MRSA).

Since penicillinase production and altered PBP affect only beta lactams, they do not influence the efficacy of other classes of antibiotics against Staphylococci. However, resistance to other classes of antibiotics such as macrolides, tetracyclines, cotrimoxazole, clindamycin and quinolones has become increasingly common among both MSSA and MRSA due to different resistant mechanisms. This is particularly true for MRSA.

Treatment of MSSA: Amoxicillin is ineffective against staphylococci. Cloxacillin and amoxycillin-clavulanate are effective against MSSA. Other BL/BLI combinations such as ampicillin/sulbactam, piperacillin/tazobactam and cefoperazone sulbactam are also active against staphylococci. First generation cephalosporins (cephalexin, cefazolin) are effective against MSSA. In contrast, second generation and third generation cephalosporins are less effective against staphylococci and are not the preferred agents. Cefixime, an oral third generation cephalosporin has very poor anti-staphylococcal effect.

Other antibiotics which were originally effective against staphylococci (both MSSA and MRSA) include tetracyclines, clindamycin, macrolides, cotrimoxazole, quinolones. They can be used for non-serious infections. However, resistance has become common for macrolides, quinolones, cotrimoxazole and to a lesser extent clindamycin.

If a SA strain (whether MSSA or MRSA) is macrolide resistant but clindamycin sensitive, such a strain may still be clindamycin resistant due to a phenomenon called inducible clindamycin resistance. Such strains may be identified by D test. Positive D test indicates inducible clindamycin resistance, and such strains will not respond to clindamycin therapy

Treatment of MRSA: Doxycycline, clindamycin and cotrimoxazole are commonly effective against MRSA and can be used for empirical therapy of mild to moderate MRSA infection. Doxycycline is the best choice among the three, in the Indian setting; ICMR surveillance data 2018 shows resistance of MSSA and MRSA to doxycycline is much lower as compared to clindamycin and cotrimoxazole.

Resistance among Staphylococci (ICMR 2021); includes OPD, IP, as well as ICU samples.

Table 4. Resistance among Staphylococci

	All SA	MSSA	MRSA
Tetracycline	11%	8%	15%
Clindamycin	26%	21%	34%
Cotrimoxazole	32%	27%	39%
Erythromycin	56%	46%	72%
Ciprofloxacin	83%	78%	90%

For empirical therapy of severe infections due to MRSA vancomycin is the choice and linezolid is an alternate choice (except for infective endocarditis, where the efficacy of linezolid is not well established). De-escalation should be considered if the isolate is sensitive to other drugs such as doxycycline, cotrimoxazole or clindamycin. Clindamycin should be used only if the isolate is sensitive to macrolides; in case of macrolide resistance, clindamycin should be used only if D test rules out inducible clindamycin resistance.

Macrolides are ineffective against MRSA even if sensitive as per AST in vitro and they should never be used to cover for MRSA. Resistance to fluoroquinolones in S. aureus is common and develops quickly. Monotherapy with fluoroquinolones for S. aureus infections is not recommended. Rifampicin should NEVER be used as monotherapy because resistance develops rapidly. There is no evidence that linezolid is superior to TMP/SMX, doxycycline, or clindamycin in the management of skin, soft tissue infections or osteomyelitis.

Pneumococcal infections: Originally, pneumococci were exquisitely sensitive to penicillin. However, resistance has slowly emerged over the years due to altered penicillin binding protein. Such strains are becoming increasingly common in India.

Table 5. Susceptibility breakpoint based antibiotic dosing for Pneumococcus			
Dose of penicillin G and amoxicillin depending on the MIC	*Meningitis, IV route*	*For non-meningeal infections, IV route (e.g., pneumonia)*	*Non meningeal infections, oral route (e.g., otitis media, sinusitis)*
Susceptible (MIC µg/mL)	**≤0.06** Penicillin G 24 MU per day in 6 divided doses	**≤2** Penicillin G 12 MU per day in 6 divided doses	**≤0.06** Amoxicillin 500 mg tid
Intermediate (MIC µg/mL)	-	**=4** Penicillin G 18 to 24 MU per day in 6 divided doses	**=0.12-1** Amoxicillin 1000 mg tid
Resistance (MIC µg/mL)	**≥0.12** Penicillin G is not effective	**≥8** Penicillin G is not effective	**≥2** Oral Amoxicillin not effective
MIC: Minimum inhibitory concentration (for penicillin), IV: Intravenous			

MIC is a measure of sensitivity of bacteria to antibiotics. The success of therapy depends on the MIC of the strain as well as the site of infection and the dose and route of the antibiotic.

Treatment of meningitis: Penicillin penetrates the uninflamed meninges poorly. CSF level of penicillin is only 0.02% of that of plasma, increasing to 20% with strong meningeal inflammation. Levels achieved in CSF are sufficient to kill the bacteria only if the MIC of the strain is very low (≤0.06 µg/mL). If the MIC is ≥0.12 µg/mL or above, penicillin becomes ineffective.

Treatment of infections outside the CNS with IV therapy: For infections outside CNS such as pneumonia, IV penicillin G is effective as long as the MIC of the strain is ≤4 µg/mL.

Oral therapy for otitis media, sinusitis and CAP: Since oral therapy can achieve only much lower plasma levels, oral therapy is effective only if the MIC is low; strains with MIC of ≤0.06 µg/mL are considered susceptible and respond to usual doses of amoxicillin (500 mg tid in adults and 45mg/kg/day in children); strains with MIC of 0.12 to 1 µg/mL are considered intermediate and respond to double the usual doses of amoxicillin (1000 mg tid or 90mg/kg/day in children). Strains with MIC of 2 µg/mL or above will not respond to oral therapy.

Indian data: Varghese et al found penicillin non-susceptibility of 42.8% and ceftriaxone non-susceptibility of 28.5% among meningitis strains at CMC, Vellore in the year 2016. A multicentre Indian study by Manoharan et al showed that 17% penicillin non-susceptibility among meningitis isolates collected during the period 2011 to 2015. In a study from JIPMER, 8 of the 14 CSF isolates (57%) were penicillin non-susceptible (Murthy et al, 2014 to 2017, 134 total isolates, 11% from CSF and 8% from blood). However, all the isolates had MIC less than 1 µg/mL.

ICMR AMR surveillance data for the year 2021 shows that, among meningeal isolates penicillin susceptibility is only 23% and ceftriaxone susceptibility is only 67%. So, empiric therapy for community acquired pyogenic meningitis should be ceftriaxone with vancomycin. The same data showed that penicillin and ceftriaxone susceptibility among non-meningeal invasive isolates was 98%

and 98%, respectively. For pneumococcal infections outside CNS, high dose penicillin or ceftriaxone is highly effective, and addition of vancomycin is generally not warranted.

Oral therapy for otitis media, sinusitis, and CAP: The usual dose of amoxicillin for the above infections is 500 mg tid in adults and 45 mg/kg/day in children. Such a dose is effective only if the MIC of the strain is 0.06 µg/mL or less. However, pneumococci with intermediate penicillin resistance (MIC of 0.12-1 µg/mL) have become increasingly common in India. For the outpatient treatment of CAP or sinusitis or otitis media due to such strains, usual dose of amoxicillin is ineffective and double the usual doses are recommended. Oral third generation cephalosporins such as cefpodoxime or cefdinir are also effective but high dose amoxicillin remains the most effective oral beta lactam against these strains. Resistance in pneumococci is due to modification of the binding site and not due to production of beta lactamase; hence, BL/BLI combinations offer no advantage against pneumococci as compared to BL alone.

II. Principles of treatment of infection and sepsis

Treatment of infection vs sepsis: It is important to differentiate infection from sepsis. Sepsis refers to infection leading to life threatening organ dysfunction; it is a medical emergency; antibiotics need to be started as soon as possible ideally within one hour. However, infection without organ dysfunction is usually not a medical emergency.

Sepsis = infection + life threatening organ dysfunction caused by infection

Infection vs sepsis*:* In the early stages of infection, the organ function is normal, but over time, the interplay of infection and host response may lead to organ dysfunction. Note that even when infection is accompanied by bacteraemia there is no organ dysfunction in the early stages, but untreated infection and bacteraemia can potentially lead to organ dysfunction. When infection leads to organ dysfunction, it is called sepsis. Organ dysfunction is usually the result of dysregulated host response. Whether infection proceeds to sepsis/organ dysfunction depends on several factors such as virulence of the pathogen, microbial load, immune status of the host, physiological reserve of the host, co-morbidities etc. Source control and effective antimicrobial therapy can prevent or limit the organ dysfunction. Development of sepsis/organ dysfunction significantly increases morbidity and mortality; so, it is critically important that sepsis is recognized early and treated effectively.

qSOFA score has been used to identify sepsis. qSOFA score consists of 3 components.

- Respiratory rate ≥22/minute
- Altered mentation.
- Systolic blood pressure ≤100 mmHg]

A score of ≥2 suggests sepsis. Septic shock is defined as hypotension requiring vasopressor therapy to maintain a MAP of ≥65 mmHg plus elevated lactate level (>2 mmol/L or >18 mg/dL). Mortality is much higher in those with shock as compared to those without (≥40 versus ≥10 percent)

There are two approaches to antibiotic therapy:

De-escalation approach: Start with broad spectrum and or combination therapy and deescalate after culture reports or clinical improvement. This approach should be limited for serious or life-threatening situations where inappropriate empirical therapy has serious consequences.

Escalation approach: Start with limited spectrum antibiotic appropriate for the infection; Switch to broader spectrum drug or add additional drugs only if initial therapy fails. This approach is appropriate for mild to moderate infections where failure of initial therapy does not have serious consequences. This helps in limiting the use of broad spectrum and reserve drugs.

So, it is important to strike a balance between the need to provide appropriate empirical therapy and the need to minimize the use of broad spectrum and reserve drugs.

Sepsis represents medical emergency and needs to be treated aggressively; broad spectrum antimicrobial therapy should be initiated to cover for likely pathogens while waiting for culture reports; de-escalation to the antibiotic with the narrowest spectrum possible should be done once culture reports become available or once the patient improves significantly. Other measures such as source reduction (drainage of abscess), relief of obstruction (relief of ureteric obstruction and hydronephrosis with stone removal or stent) are important.

Infection without sepsis is not generally a medical emergency and, in such patients, aggressive broad spectrum antibiotic coverage is generally not warranted; in such cases one may initiate appropriate narrow spectrum antibiotic therapy; escalation to broader spectrum therapy may be needed in

selected patients as per clinical response or culture results. In many patients, it may be possible to withhold empirical therapy, until culture results become available. Some patients may respond to source reduction alone.

- Currently, qSOFA score is used to identify sepsis; it has ***limited sensitivity*** but good specificity.
- Sepsis 3 definition ***does not identify all patients at high risk*** for mortality or disease progression.
- A subset of patients with infection do not meet criteria for sepsis 3 definition but are at high risk; they also need aggressive therapy.

Table 6. Common cause of sepsis and systemic febrile illness	
Common pathogens causing sepsis/septic shock (non-neutropenic adults)	
Gram negative bacteria	E. coli, Klebsiella (community acquired) Acinetobacter, Pseudomonas (usually HCAI)
Gram positive bacteria	Staphylococcus aureus, pneumococcus, Streptococcus pyogenes, other streptococci
Fungi	Candida, others (particularly if there is immunocompromise, prolonged broad spectrum antibiotic therapy, etc.)
Toxin mediated	Streptococcal TSS Staphylococcal TSS
Systemic febrile illnesses due to specific pathogens	
Malaria	*Plasmodium spp.*
Scrub typhus	*Orientia tsutsugamushi*
Leptospirosis	*Leptospira*
Enteric fever	*Salmonella typhi*

III. Getting your antibiotic right

Empirical therapy vs definitive therapy: In patients with infection, we start antibiotics after sending cultures. The initial choice of antibiotics depends on several factors such as site of infection, severity of illness, place of acquisition, host factors, etc. This is called **empirical therapy**. Once culture reports become available, the antibiotics are adjusted as per the organism and sensitivity pattern. Such culture guided therapy is called **definitive therapy**.

The 5 Rs of antibiotic therapy: The clinician should make every attempt to ensure all the following

- Right diagnosis (The ID syndrome)
- Right drug
- Right time
- Right dose
- Right duration

Right diagnosis (identifying the ID syndrome)

Right diagnosis is extremely important. First, we need to identify the focus of infection or the ID syndrome. This is primarily based on history and physical examination. It is supplemented by initial laboratory tests that are immediately available. Based on the ID syndrome, the necessary samples must be sent; making the microbiological diagnosis is crucial. As additional test results become available the ID syndrome should be further defined.

Blood culture: Take blood cultures before initiating antibiotics; Take at least 2 samples for blood culture, preferably through two different venipunctures. If the patient has a central line, take one sample through the central line and one through peripheral venipuncture. Take 3 or 4 samples in special situations (suspected infective endocarditis, patient already received antibiotic, highly resistant organisms suspected). If only aerobic cultures are sent, draw 10 ml for each sample and put in the aerobic bottle. (If both aerobic and anaerobic cultures are to be sent, draw 20 ml for each sample; put 10 ml in the aerobic bottle and 10 ml in the anaerobic bottle; this is called 1 set of blood culture).

Other cultures: Send other relevant cultures based on the suspected source (eg. urine, sputum, pus etc.)

Urine culture: Urine sample for culture should be processed immediately (within 1 to 2 hours). Otherwise, it may be refrigerated at 4 degree C up to 24 hours; but once removed from refrigeration, it should be processed immediately.

Right time: Antibiotic should be administered as early as possible. In patients with sepsis, the antibiotic should be given within one hour, especially in those with septic shock. Delay in initiation of antibiotic therapy significantly increases mortality. Inappropriate initial empiric therapy also significantly increases the mortality rate (34% vs 18%). Every effort must be taken to ensure early administration of appropriate initial empirical antibiotic therapy.

Right drug: Administering appropriate empirical therapy is crucial. Mortality significantly increases with inappropriate therapy. Several factors should be considered in the selection of empiric therapy. It should cover the likely pathogens. Which pathogens need to be covered depends on several factors. First and foremost, we should identify the focus of infection or the ID syndrome (UTI, pneumonia, meningitis, etc). Next, we should consider the site of acquisition of infection and the local prevalence of antibiotic resistance in the community, hospital, or the unit under consideration. Among community acquired infections, recent hospitalization of at least 5 days within the last 90

days and recent use of parenteral antibiotics within the last 90 days increase the risk of resistant pathogens. Resistant pathogens are more likely in hospital acquired infections, particularly with ICU acquired infections as compared to ward acquired infections. Immune status of the patient strongly influences the likely pathogens and the need for aggressive therapy. Factors such as the presence of central venous catheters etc increase the probability of Staphylococcal infection. It is important to consider the entry of the selected drug into the relevant compartment. For example, aminoglycosides do not enter the CNS and hence should not be used to cover Gram negative meningitis; neonates with immature blood brain barrier are an exception.

Factors influencing the choice of empiric therapy:

1. source of infection (ID syndrome)
2. severity of illness
3. community or hospital acquired (ward acquired, or ICU acquired in case of later)
4. local prevalence of resistant pathogens in the community/hospital/unit
5. risk factors for MDR pathogens in the individual (recent antibiotic use, recent hospitalization, prior colonization/infection with MDR pathogens)
6. presence of devices (urinary catheter, central venous catheter etc)
7. adequateness of source control, in applicable patients
8. age, comorbidity
9. immunocompetent vs compromised.
10. renal or liver disease (safety as well as need for dose adjustment)
11. safety of the drug in pregnancy or lactation, if applicable
12. penetration of the drug into the site of infection

Table 7. Spectrum of coverage for the common broad-spectrum antibiotics

Bacterial pathogen	*Gram negative*		*Gram positive*			
	Gram negative bacilli	*ESBL gram negative bacilli*	*Pneumococcus, Streptococcus*	*Anti- MSSA effect*	*Enterococcus*	*MRSA*
3rd or 4th generation cephalosporin	Yes	No	Yes	Yes	No	No
BL/BLI	Yes	Yes*	Yes	Yes	Yes	No
Carbapenem	Yes	Yes	Yes	Yes	Yes	No

Yes, but may be inferior to carbapenem in severe or life threatening infections*

In sepsis of unknown focus broad spectrum coverage should be given.

Gram negative coverage:

1. 3rd and 4th generation cephalosporin provide Gram-negative coverage, but with the emergence of ESBL Gram negative pathogens, their use in sepsis has become limited.
2. BL/BLI combinations provide better Gram-negative coverage including moderate efficacy against ESBL pathogens.

3. Carbapenems provide excellent Gram-negative coverage against ESBL pathogens

Gram positive coverage:

1. All the above provide good coverage against pneumococci and streptococci
2. All provide reasonable MSSA coverage; but penicillinase resistant penicillin and first generation cephalosporins are the choice for MSSA infections. None of these antibiotics provide MRSA coverage
3. Additional MRSA coverage should be provided in selected cases
4. None of the cephalosporins are effective against enterococci
5. BL/BLI and meropenem provide good enterococcal coverage
6. Selected patients will require antifungal coverage

Tropical infections:

Patients with sepsis secondary to tropical infections such as scrub typhus, leptospirosis or malaria need specific therapy. Additional broad-spectrum therapy may be needed if the diagnosis is not certain, and the patient is severely ill.

Right dose: Appropriate dosing of the selected antibiotic is extremely important. Dose should be adjusted as needed in case of renal disease and liver disease. For most antibiotics that are renally eliminated, the initial dose should not be reduced even in the presence of renal failure. Only the maintenance dose needs to be reduced as per GFR. Failure to administer the full dose initially may seriously compromise therapy. Only for some nephrotoxic drugs such as aminoglycosides the first dose itself may have to be reduced. For patients undergoing renal replacement therapy, suitable additional doses should be given for drugs that are removed during RRT. For example, many beta lactam drugs are removed during dialysis necessitating post dialysis supplementation.

Dose should be adjusted as per age and body weight in neonates and children. Neonatal and paediatric doses of drugs are different from adult doses expressed as per kg. Often children need higher dose per kg. Some drugs need a loading dose that is higher than the maintenance dose. Failure to give the loading dose may compromise the therapy. Dose adjustment should also be done for significant obesity. In obese individuals the weight used depends on the individual drug. Actual weight or ideal weight or adjusted weight should be used as per the drug class.

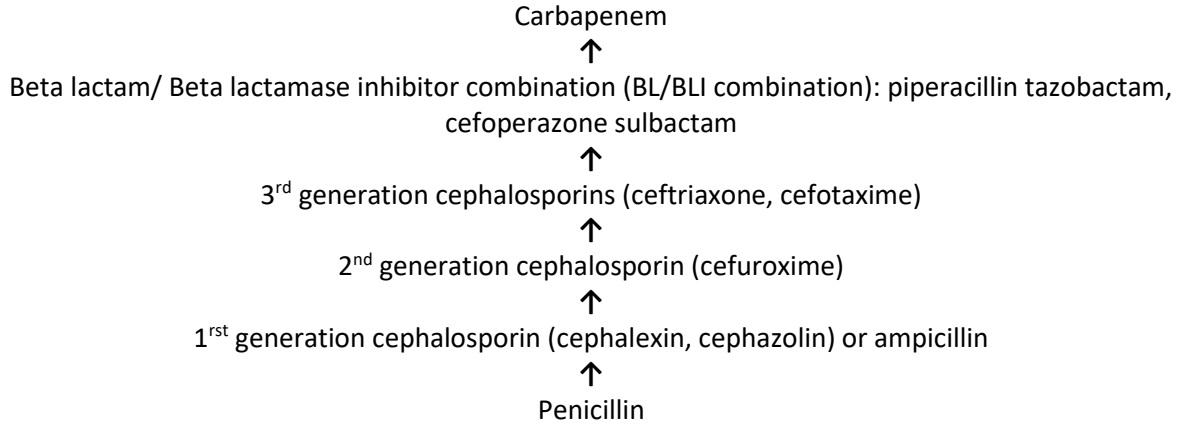
Right duration:

Empirical therapy vs definitive therapy: In patients with infection, we start antibiotics after sending cultures. The initial choice of antibiotics depends on several factors such as site of infection, severity of illness, place of acquisition, host factors, etc. This is called empirical therapy. Once culture reports become available, the antibiotics are adjusted as per the organism and sensitivity pattern. Such culture guided therapy is called definitive therapy

The concept of de-escalation and escalation (Method to adjust therapy once culture reports become available): Empirical antibiotic therapy may consist of one or more antibiotics. A beta-lactam antibiotic generally forms the backbone of empirical therapy. This is called the pivotal antibiotic. An aminoglycoside is often added as a second antibiotic to provide additional coverage against gram negative bacilli. Vancomycin may be added if there is a need to provide MRSA coverage.

After culture report becomes available the treatment is modified as per the organism and sensitivity. This often involves de-escalation (switching to a narrower spectrum antibiotic from the same or different class, withdrawal of additional antibiotics) or escalation (switching to a broader spectrum antibiotic from the same or different class).

Figure 1: Ranking of beta lactam antibiotics in the order of spectrum (broadest spectrum above and narrowest spectrum below)



Switching from a higher rank to a lower rank is de-escalation. Switching from a lower rank antibiotic to a higher rank is escalation.

Examples of escalation and de-escalation:

Case vignette: A patient with febrile neutropenia is started on piperacillin tazobactam and amikacin. Vancomycin is added as he has a central line in place and empirical MRSA coverage is deemed necessary. Once blood culture results become available, decisions regarding definitive therapy need to be taken. Depending upon what the blood culture grows, several scenarios are possible. Let us look at some of these possible scenarios and what should be the definitive therapy in each of those scenarios.

Table 8. Possible scenarios pertaining to the case vignette

	Culture report	*Definitive therapy*	*Comment*
Scenario 1	Blood culture grows E. coli sensitive to piperacillin tazobactam but not sensitive to ceftriaxone or any of the narrower spectrum antibiotics such as quinolones or cotrimoxazole.	Continue piperacillin tazobactam; stop amikacin; stop vancomycin	Stopping of unnecessary antibiotics is called de-escalation
Scenario 2	Blood culture grows E. coli resistant to piperacillin tazobactam, cefepime, quinolones etc but sensitive to amikacin, meropenem	Switch from piperacillin tazobactam to meropenem; stop amikacin and vancomycin	Broadening of the spectrum from piperacillin- tazobactam to meropenem is escalation; withdrawal of amikacin and vancomycin is de-escalation
Scenario 3	Blood culture grows E. coli sensitive to piperacillin tazobactam, ceftriaxone but resistant to quinolone and cotrimoxazole	Switch from piperacillin tazobactam to ceftriaxone; stop amikacin and vancomycin	Narrowing of the spectrum from piperacillin-tazobactam to ceftriaxone is de-escalation; withdrawal of amikacin and vancomycin is de-escalation
Scenario 4	Blood culture grows E. coli sensitive to piperacillin tazobactam, ceftriaxone, quinolone and cotrimoxazole	Switch from piperacillin tazobactam to cotrimoxazole or quinolone	Narrowing of the spectrum from piperacillin-tazobactam to cotrimoxazole or quinolone is de-escalation; withdrawal of amikacin and vancomycin is de-escalation
Scenario 5	Blood culture grows Staph aureus resistant to cloxacillin and sensitive to vancomycin	Continue vancomycin; stop piperacillin tazobactam and amikacin	Stopping piperacillin and amikacin is de-escalation.
Scenario 6	Blood culture grows Staph aureus sensitive to cloxacillin and vancomycin	Switch form vancomycin to cloxacillin; stop piperacillin and amikacin.	Narrowing down the staphylococcal coverage form vancomycin to cloxacillin is de-escalation. Stopping piperacillin and amikacin is also de-escalation.

Additional comments

Scenario 2: Even though the strain is sensitive to amikacin, amikacin is not recommended as monotherapy for the treatment of bacteraemic infections. So, escalation to meropenem is warranted. However, for non-bacteraemic infections amikacin monotherapy can be used. For example, in case of non-bacteraemic pyelonephritis amikacin monotherapy may be used.

Scenarios 3 and 4: The narrowest spectrum antibiotic should be used. In example 2 ceftriaxone is the narrowest spectrum antibiotic to which the organism is sensitive. In example 4, cotrimoxazole is the narrowest spectrum antibiotic.

Scenario 6: The Staph aureus strain is susceptible to cloxacillin and vancomycin (MSSA). In the treatment of serious MSSA infections, the mortality is lower with cloxacillin as compared to vancomycin. So MSSA should not be treated with vancomycin.

IV. PK/PD optimization

Antimicrobial use may be improved by pharmacokinetic and pharmacodynamic optimization.

MIC: MIC is the minimum concentration of the antibiotic that is capable of inhibiting the growth of the bacteria. Once the concentration of the antibiotic exceeds the MIC bacterial killing happens. The rate at which the bacterial counts come down over time can be plotted.

Plasma concentration of an antibiotic and rate of bacterial killing: For some antibiotics higher the plasma level faster is the rate of killing (e.g., aminoglycosides). For other antibiotics, once the concentration of the antibiotic in the tissue exceeds MIC, bacterial killing occurs at a relatively same rate; increasing the concentration of the antibiotic at the site over and above the MIC does not further increase the rate of bacterial killing (e.g., beta lactams). This is illustrated below;

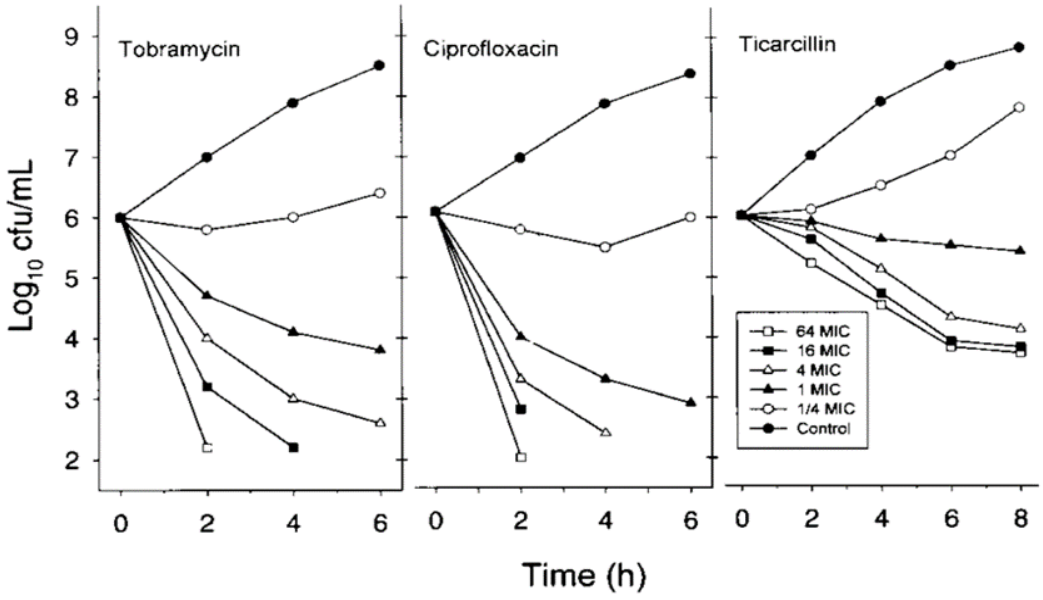


Figure 2: The figure shows the rate of bacterial killing when the plasma concentration of the antibiotic increases (shown in multiples of MIC). X axis depicts the time and Y axis depicts the colony count. The slope of the lines indicates rate of bacterial killing. Upsloping lines indicate the colony count goes up over time indicating insufficient antibiotic concentrations. The slope of the down sloping lines indicates the rate of bacterial killing.

For Tobramycin (aminoglycosides), as the plasma concentration increases the rate of bacterial killing increases (downslopes become progressively steeper). For Ciprofloxacin(quinolones), as the plasma concentration increases, the rate of bacterial killing increases (downslopes become progressively steeper). For Ticarcillin (beta lactams), increase in the plasma level from 1 MIC to levels up to 64 MIC only slightly increases the rate of bacterial killing (downslopes do not become much steeper))

Determinants of the efficacy of an antibiotic: After an antibiotic is given as a bolus or a short infusion the plasma level of antibiotic reaches a peak concentration (Cmax) and falls in an exponential manner over time. After some time, the level falls below the MIC and remains so till the next dose is given.

There are three parameters that influence the rapidity of bacterial killing

1. The Cmax/MIC ratio
2. The fT>MIC
3. AUC/MIC

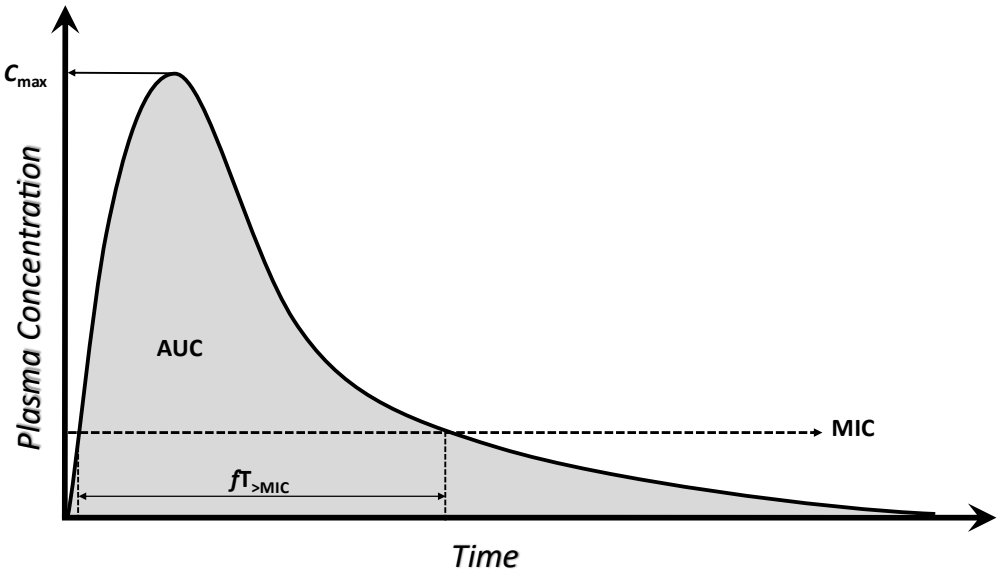


Figure 3: The plasma concentration of the orally administered antibiotic shows a rapid raise followed by a gradual fall***. C_{max} refers to the peak plasma concentration. $fT_{>MIC}$ refers to the time during which the plasma concentration is above MIC.***

The Cmax/MIC ratio: For some antibiotics higher the peak concentration achieved, faster is the rate of bacterial killing. That is, higher the Cmax/MIC ratio, higher is the rate of bacterial killing. Such antibiotics are called said to exhibit concentration dependent killing. (e.g., aminoglycosides)

The fT>MIC: fT>MIC refers to the fraction or proportion of the dosing interval during which the plasma concentration of the antibiotic remains above the MIC.

For some antibiotics, once the concentration of the antibiotic in the plasma exceeds MIC, bacterial killing occurs at a relatively same rate. Increasing the plasma concentration of the antibiotic over and above the MIC does not further increase the rate of bacterial killing. For such antibiotics, the efficacy is related to what proportion of the dosing interval the plasma concentration remains above MIC. Such antibiotics are said to exhibit time dependent killing. For such antibiotics it does not matter how high is the plasma concentration that is achieved. What matters is how long the plasma concentration exceeds the MIC (e.g., beta lactams).

- For penicillins, the plasma concentration should exceed MIC at least 50% of the time, that is, fT>MIC of 50%.
- For cephalosporins the plasma concentration should exceed MIC at least 70% of the time, that is, fT>MIC of 70%
- For carbapenems the plasma concentration should exceed MIC at least 40% of the time, that is, fT>MIC of 40%

Table 9. Bactericidal and bacteriostatic effect determined by $fT_{>MIC}$

	$fT_{>MIC}$ for bacteriostatic effect	*$fT_{>MIC}$ for bactericidal effect*
Penicillins	30%	50%
Cephalosporins	40%	70%
Carbapenems	20%	40%

Penicillins and cephalosporins have poor post antibiotic effect and require higher fT>MIC. In contrast, carbapenems have significant post antibiotic effect and so require a lower fT>MIC. Lower fT>MIC is sufficient for bacteriostatic effects and higher fT>MIC is necessary for bactericidal effect. Much higher f>MIC is needed in immunocompromised individuals.

AUC/MIC: For some antibiotics the rate of killing best correlates with the area of the drug concentration curve that is above MIC. They are said to exhibit AUC/MIC dependent killing (e.g., quinolones)

Extended infusion and continuous infusion: Conventionally the antibiotic is given as a short infusion over 30 minutes. When the same drug is given as an extended infusion over 4 hours, it achieves a delayed and lower peak, and the rate of elimination occurs at a much lower rate. So, the drug concentration remains above MIC for a much longer period. In other words, a higher fT>MIC is achieved, thus increasing the efficacy of the antibiotic.
Note that extended infusion achieves longer fT>MIC.

Look at two situations:

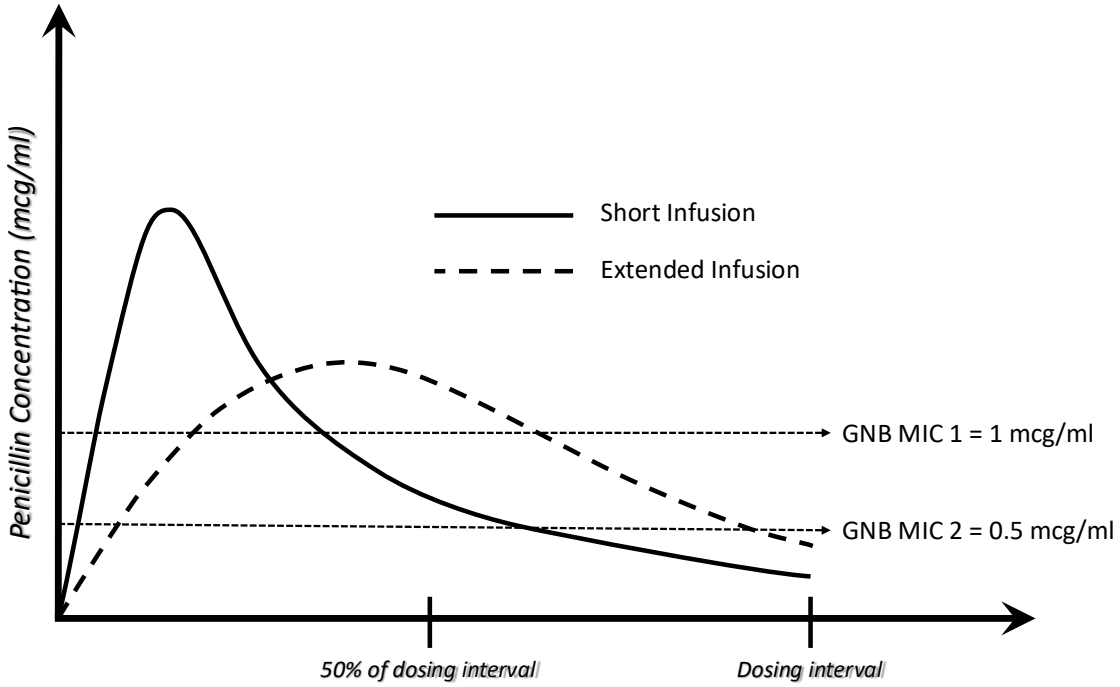


Figure 4: MIC is 0.5: Here even conventional infusion achieves fT>MIC of over 50%

MIC is 0.5mcg/ml: Here conventional infusion itself achieves fT>MIC of more than 50%.

MIC is 1mcg/ml: Here conventional infusion achieves fT>MIC of less than 50%; achieving a target fT>MIC of at least 50% is possible only with extended infusion

So extended infusion of antibiotics is particularly important with relatively resistant organisms.

If an extended infusion is given the plasma concentration of the antibiotic rises slowly and it takes some time before the plasma concentration exceeds MIC. This lag in achieving a concentration above MIC is a disadvantage; this can be overcome by giving the first dose as a 30-minute infusion

and subsequent doses may be given as extended infusion. This is particularly important in patients who have severe or critical illness.

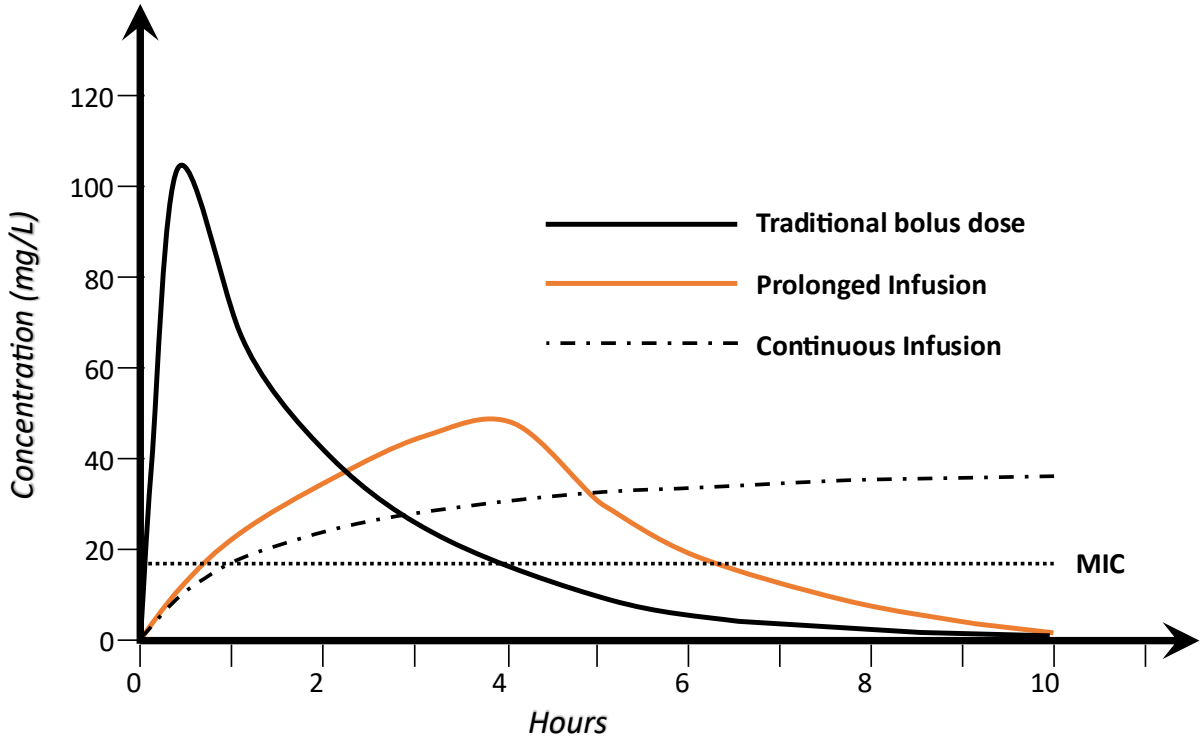


Figure 5: The above diagram shows that extended infusion and continuous infusion lead to longer time above MIC (Adapted from Antimicrobial stewardship Edited by Matthew Laundy).

Mortality benefit with prolonged infusions: Most simulation studies demonstrate a 5–20% improvement in CFR (case fatality rate) for cephalosporins and carbapenems when the same dose is simulated as a 3–4 h infusion compared with a standard 0.5 h infusion.

Problems with continuous infusions: The antibiotic should be dissolved in a compatible IV fluid. The duration of stability of the antibiotic depends on the fluid chosen for dilution, the volume of dilution and the concentration achieved as well as the room temperature at which it is stored or infused. The time from the preparation of the diluted solution to the completion of the infusion should not exceed this limit. Some beta-lactam agents, such as carbapenems, are not stable at room temperature for long durations. So, the duration of infusion of imipenem or meropenem should not exceed 3 hours. In a patient on continuous infusion of an antibiotic, compatibility of other drugs administered through the same IV line should be verified.

Table 10. Extended and continuous infusion regimen for Beta lactam antibiotics			
Antibiotic and stability of the infusion solution ***(Stability data from Sanford)***	***Traditional dosing***	***Extended dosing***	***Continuous infusion***
Piperacillin tazobactam @ 37°C: 24 hours @ 25°C: 24 hours @ 4°C: No data	3.75 g to 4.5 g q 6h as 30 min infusion	Same dose given as 4-hour infusion	
Cefoperazone sulbactam	Cefoperazone dose is 2 to 4 g per day (increase to 6 to 8 g per day for relatively resistant pathogens; ***up to 16 g per day*** have been used) Sulbactam 2 to 4 g per day (***up to 12 g per day*** have been used for Acinetobacter infections) Given as a 30 min infusion	Cefoperazone sulbactam may be given as an infusion over 4h (Cefoperazone sulbactam in the ratio of 1:1 or 2:1; 2:1 ratio is preferable with higher doses of Cefoperazone)	Total daily dose may be given as a 24 h infusion
Meropenem @ 37°C: < 4 hours @ 25°C: 4 hours @ 4°C: 24 hours	1 to 2 g IV Q8H as 30-minute infusion	Same dose given as 3-hour infusion	
Imipenem	500mg to 1 g IV Q6H as 30-minute infusion	Same dose given as 3-hour infusion	
Cefepime @ 37°C: 8 hours @ 25°C: 24 hours @ 4°C: > 24 hours	2 g IV Q8H given as 30-minute infusion	Same dose given as 3-to-4-hour infusion	

V. Sepsis – General Principles

1. Obtain appropriate cultures	2. Ensure early and appropriate antibiotic therapy	3. Timely review of antibiotic therapy
Blood culture: Take blood cultures before initiating antibiotics. Take at least 2 samples for blood culture, preferably through two different venipunctures. If the patient has a central line, take one sample through the central line and one through peripheral venipuncture. Take 3 or 4 samples in special situations (suspected infective endocarditis, patient already received antibiotic, highly resistant organisms suspected) If only aerobic cultures are sent, draw 10 ml for each sample and put in the aerobic bottle. (If both aerobic and anaerobic cultures are to be sent, draw 20 ml for each sample. Put 10 ml in the aerobic bottle and 10 ml in the anaerobic bottle; this is called 1 set of blood culture) Other cultures: Other relevant cultures based on the suspected source (eg. urine, sputum, pus etc.)	Patients should receive the first dose of antimicrobials as soon as possible, preferably within 1 hour of presentation. Keep the antibiotic ready, so that the antibiotic may be administered once blood cultures have been drawn (compete test dose while cultures are being drawn; the role of test dose for antibiotics is questionable except in case of known penicillin allergy) The choice of antibiotic as empiric therapy depends on several factors (*see* **Right Drug** and **Right Dose** @page 11) Ensure appropriate dose modification based on renal or liver function. Ensure adequate dosing in the obese. ***Important:*** In patients with renal impairment, the first dose of antibiotic should be the full dose; only the maintenance dose needs to be reduced as per creatinine clearance; The first dose itself should be reduced only in case of some drugs such as aminoglycosides (e.g. the first dose of amikacin reduced from 15 mg/kg to 7.5 mg/ kg if creatinine clearance is less than 30ml). In addition, for patients on haemodialysis, post-dialysis supplementation dose should be given after each dialysis in case of many drugs.	Review the situation after 48 to 72 hours and deescalate or escalate as required as per culture reports and clinical response. De-escalate to the narrowest spectrum antibiotic as early as possible. Ceftriaxone resistance should be taken as evidence of ESBL production. For severe, life-threatening infections caused by ESBL producing pathogens, BL/BLI combinations such as piperacillin/tazobactam or cefoperazone/sulbactam may be inferior to carbapenems. In such situations, carbapenem may be used instead of BL/BLI combination Add clindamycin when toxic shock syndrome is suspected. Add vancomycin when MRSA is likely.

Sepsis – Empirical Therapy

#	Clin. Diagnosis	Etiology	Empirical Antibiotic Therapy							Remarks
SYS-01	***Sepsis (Community acquired)*** *A qSOFA score≥2 suggests sepsis. (see remarks)*	***Gram negative bacteria*** *E. coli, Klebsiella (community acquired) Acinetobacter, Pseudomonas (usually HCAI)* ***Gram positive bacteria*** *Staphylococcus aureus, pneumococcus, Streptococcus pyogenes, other streptococci* ***Fungi*** *Candida, others* *(Particularly in immunocompromised, prolonged broad-spectrum antibiotic therapy)* ***Toxin mediated*** *Streptococcal TSS* *Staphylococcal TSS* ***Systemic febrile illnesses due to specific pathogens*** *Malaria* *Scrub typhus* *Leptospirosis* *Enteric fever*	**I**	Piperacillin-tazobactam 4.5g IV Q6H **OR** Cefoperazone-Sulbactam 2g IV TDS-QID or 3g IV Q12H	±	Amikacin 15 to 20mg/kg IV OD *(particularly if BL/BLI is the primary antibiotic)*	±	Vancomycin (for MRSA coverage) 15 to 20mg/Kg IV Q8H with TDM; *see vancomycin dosing @page 77*	• Sepsis is defined as life threatening organ dysfunction resulting from infection. • Organ dysfunction is represented by increase in SOFA score by 2 or more points from baseline. • Mortality is much higher in those with shock as compared to those without (≥40 versus ≥10 percent) • qSOFA score is used for screening. It consists of 3 components. (RR≥22, altered mentation, SBP≤100mmHg).
			II	Meropenem 1-2g IV Q8H						
SYS-02	***Septic Shock (Community acquired)*** *Sepsis with hypotension requiring vasopressor to maintain MAP≥65mmHg plus elevated lactate level (>2 mmol/L or >18 mg/dL).*		**I**	Cefoperazone-Sulbactam 2g IV TDS-QID or 3g Q12H **OR** Piperacillin-tazobactam 4.5g IV Q6H **OR** Meropenem 1-2g IV Q8H	±	Amikacin 15 to 20mg/kg IV OD *(particularly if BL/BLI is the primary antibiotic)*	±	Vancomycin (for MRSA coverage) 15 to 20mg/Kg IV Q8H with TDM; *see vancomycin dosing @page 77*	• Mortality is much higher in those with shock as compared to those without (≥40 versus ≥10 percent)
SYS-03	***Septic Shock (Hospital acquired)***	*Klebsiella, Pseudomonas, Acinetobacter, etc.*	**I**	Meropenem 1-2g IV Q8H **OR** Polymyxin B 20,000-25,000 IU/kg IV loading followed by 12,500-15,000 IU/kg IV Q12H **OR** Colistin 9 million units IV loading followed by 4.5 million units IV Q12H	±	Amikacin 15 to 20mg/kg IV OD *(particularly if meropenem is the primary antibiotic)* **OR** Another antibiotic depending upon susceptibility	±	Vancomycin (for MRSA coverage) 15 to 20mg/Kg IV Q8H with TDM; *see vancomycin dosing @page 77*	• Consider polymyxin or colistin if there are risk factors for carbapenem resistant pathogen such as ICU acquired infection, current or recent carbapenem therapy • Polymyxins, amikacin, vancomycin all carry the risk of nephrotoxicity; this should be considered in designing the combination therapy • **See separate document for treatment of carbapenem resistant GNB**

VI. BACTERAEMIA - Principles of antibiotic therapy for Gram-negative bacilli bacteraemia

Enterobacteriaceae bacteremia

- For empiric therapy only a single agent therapy is indicated. However, when the probability of resistance to the selected agent is > 10 to 20%, it is better to add a second drug, particularly in those with sepsis or septic shock.
- Only a single agent therapy is indicated for definitive therapy. However, aminoglycosides should not be used as monotherapy for bacteraemic illness.
- A 7-day duration of therapy is adequate for uncomplicated infection (patient responds promptly in 48 to 72 hours, no uncontrolled source of infection, no major immunocompromising condition, no endovascular, bone, joint, or CNS infection)
- Oral switch is safe once the patient has responded and has remained afebrile for 48 hours, provided an effective oral drug with excellent bioavailability is available to which the pathogen is susceptible (e.g., quinolone, cotrimoxazole)
- Severe or life-threatening infection with ESBL producing bacteria BL/BLI may be inferior to carbapenems, particularly in patients with septic shock. In such situations, a non-beta lactam drug may be used if the bacteria is susceptible (e.g., quinolone, cotrimoxazole). However, aminoglycosides should not be used as monotherapy for bacteraemic illness.

Pseudomonas bacteremia

- For empirical therapy generally combination therapy is indicated (to ensure that the regimen contains at least one active drug).
- For definitive therapy only a single drug is indicated except in a few situations. However, aminoglycosides should never be used as monotherapy.

- Combination therapy for Pseudomonas infections is indicated in only a few situations.
  - Endocarditis due to Pseudomonas should be treated with two effective drugs from two different classes.
  - Highly resistant strain which is only susceptible to one drug (e.g. colistin): to prevent emergence of resistance a second agent may be added (no definite evidence for this practice)
  - Highly resistant strain for which no effective drug is available: two drugs each of which are not effective may be combined for obtaining synergism.
  - Some treat bacteraemia in neutropenic patients with a second agent for initial 3 to 5 days. (no definite evidence for this practice)

Empiric therapy: Initial therapy before culture results are available.

Definitive therapy: Therapy guided by culture and sensitivity reports.

VII. CRBSI – General Principles to Patients with CRBSI

Diagnosis of CRBSI (Catheter related blood stream infection):

- For the diagnosis of CRBSI, take a two blood cultures, one through the central venous catheter (CVC) and one from a peripheral vein. The two samples should be taken **within 15 minutes of each other**. Mark the bottles clearly as central and peripheral.
- Ensure proper cleaning of the skin and catheter hub before taking the sample.
- If a blood sample cannot be drawn from a peripheral vein, it is recommended that 2 blood samples should be drawn through different catheter lumens.
- CRBSI is confirmed if microbial growth occurs in the CVC sample at least 2h prior to growth in the peripheral vein sample using automated blood culture detection systems (DTP, differential time to positivity of more than 2 hours with earlier growth occurring in the CVC sample**).** DTP of > two hours has been evaluated primarily the setting of febrile neutropenia in haematology patients. It has been shown to perform poorly for *Staphylococcus aureus* and common Gram-negative pathogens.

Prepare the skin/clean the catheter hub, before drawing sample for blood culture

- Use of either alcohol or tincture of iodine or alcoholic chlorhexidine (> 0.5%); avoid povidone-iodine; allow adequate contact and drying times to avoid blood culture contamination

Exit site infection

- If exit site is red, painful, or discharging pus then take swabs for culture
- Presence of inflammation or purulence at the exit site has very low sensitivity (<5 percent) for CRBSI but has high specificity (94 to 99 percent)

Catheter removal:

Indications for removing the catheter (any of the following):

1. CRBSI related complications

- Sepsis (organ dysfunction resulting from infection), hemodynamic instability, presence of concomitant endocarditis or evidence of metastatic infection, presence of suppurative thrombophlebitis, presence of a propagating clot, persistent bacteraemia after 72 hours of appropriate antimicrobial therapy

2. Pathogen related factors

- Infection with the following pathogens: S. aureus, P. aeruginosa, drug-resistant gram-negative bacilli, Candida spp.
- In the absence of complications, catheter salvage is reasonable in the setting of CRBSI due to CoNS and drug-susceptible Enterobacteriaceae (e.g., Escherichia coli, Klebsiella species, Enterobacter species).
- In patients with CRBSI due to Enterococcus spp, catheter removal is preferred; however, catheter salvage may be attempted in patients for whom catheter removal is not readily feasible.
- Catheter removal based on the above pathogen-related considerations must be weighed against the clinical indication for the catheter and the risks associated with removal and replacement of a catheter at another site.

Complicated vs uncomplicated CRBSI:

Uncomplicated CRBSI should meet all the following: catheter removed within five days of diagnosis: negative follow-up blood cultures within 24 to 72 hours after initial positive culture, defervescence within 72 hours of initial positive culture, no symptoms or signs of metastatic infection, no other indwelling intravascular prosthetic devices (e.g, vascular grafts). In addition, for Staph aureus infection infective endocarditis should be ruled out.

CRBSI – Empirical Therapy								
#	**Clinical Diagnosis**	**Etiology**	**Empirical Antibiotic Therapy**					**Remarks**
SYS-04	***CRBSI*** *Fever, chills or hypotension. Symptom onset at least 48 hours after catheter placement. At symptom onset, catheter still present or has been removed within last 48h.*	*Staphylococcus aureus, CoNS, Enterococcus, E. Coli, Klebsiella, Pseudomonas, Enterobacter, Acinetobacter, Stenotrophomonas maltophilia, Candida*	**I**	Piperacillin-tazobactam 4.5g IV Q6H		±	Vancomycin (for MRSA coverage) 15 to 20mg/Kg IV Q8H with TDM; *see vancomycin dosing @page 77*	• Treatment of uncomplicated CRBSI is up to 14 days and complicated CRBSI is up to 4 to 6 weeks. See CRBSI duration table below. • **See separate document for treatment of carbapenem resistant GNB**
				OR	Cefoperazone-Sulbactam 2g IV TDS-QID or 3g Q12H			
				OR	Meropenem 1-2g IV Q8H (if the patient is critically ill)			

CRBSI – Duration of Therapy (*for <u>uncomplicated</u> bacteraemia: see definition above*)		
	Catheter Removed	***Catheter not removed***
CoNS	Endovascular implant or orthopaedic hardware not present - 5 to 7 days. Endovascular implant or orthopaedic hardware present (but not infected) - 14 days	Endovascular implant or orthopaedic hardware not present: 10 to 14 days with ALT Endovascular implant or orthopaedic hardware present (but not infected): 14 to 21 days with ALT, if antibiotic started within 48 h of onset of symptoms of bacteraemia and bacteraemia clears within 48 hours of therapy; otherwise extend therapy to 4 to 6 weeks
Staph aureus	14 days; ≥ 7 days in children	28 days with ALT
Enterococcus	7 to 14 days	14 days with ALT
GNB	7 to 14 days (Monotherapy is sufficient for directed therapy if bacteremia clears within 48 hours)	14 days with ALT (Monotherapy is sufficient for directed therapy if bacteraemia clears within 48 hours)
Candida	At least 14 days after blood cultures become negative (repeat cultures daily or on alternate days till negative); in neutropenic patients continue until ANC > 500	At least 14 days after blood cultures become negative (repeat cultures daily or on alternate days till negative); in neutropenic patients continue until ANC > 500
Note: Catheter removal is recommended in case of Staph. aureus, Pseudomonas, drug resistant GNB, Candida and preferred in case of Enterococcus. ALT refers to antibiotic lock therapy		

VIII. Acute Undifferentiated Febrile Illness – General Principles

Acute Febrile illness may be divided into

- fever with localizing features and
- fever without localizing features (Acute Undifferentiated febrile Illness - AUFI)

The five most common causes for acute undifferentiated febrile illness in most regions of India are: Scrub typhus, Dengue, Malaria, Enteric fever, and Leptospirosis.

Of course, Covid 19 is an important differential diagnosis. Covid commonly presents with fever; upper respiratory symptoms may not occur, and cough may develop only after several days.

History: Careful analysis of the symptoms is essential for correct diagnosis. Prominent loose stools from the onset of fever, often with vomiting, abdominal pain or tenesmus suggests AGE. Loose stools occurring after a few days of fever suggests systemic febrile illness rather than AGE. Nasal symptoms, cough, or throat discomfort at the onset of fever indicate respiratory tract infection (RTI). Cough developing after a few days of fever is often seen in systemic febrile infections. That should not lead to a mistaken diagnosis of RTI or pneumonia.

Potential clues that are easily missed: Patients with skin or soft tissue infections such as furuncles or abscesses may not always volunteer the information. Perianal, testicular or scrotal infections are often missed as patients may be too shy to talk about them.

Patients also may not always volunteer regarding skin rash or eschar. Specific questioning and thorough examination is always essential.

Special points:

- Drug fever: Always keep drug fever in mind especially if the patient has been started on the drug recently (e.g., phenytoin, dapsone, allopurinol etc.). Early identification and withdrawal of the offending drug may be lifesaving.
- Infective endocarditis: Always consider infective endocarditis in those with underlying valvular heart disease or prosthetic valve or cardiac implanted electrical device (CIED)
- Deep seated abscess: Also note that in deep seated abscess localizing signs and symptoms may not always be obvious. (Liver abscess, splenic abscess, etc.)

Physical examination: Specifically look for lymphadenopathy, eschar, rash, splenomegaly. hepatomegaly on physical examination

Patients with eschar are often unaware of it; even if they are aware, they do not report it to the clinician. Specific questioning and thorough examination is always essential. The eschar is often hidden in the breast, axilla, groin or genital region. *Leave no cloth unturned!*

Laboratory examination:

Leukocytosis may be seen in scrub typhus and leptospirosis but makes malaria and dengue unlikely. In enteric fever, the WBC count is either normal or low but leukocytosis may occur in those with bowel perforation.

Thrombocytopenia can occur with all five; severe thrombocytopenia is seen with dengue and less commonly with malaria.

Specific diagnosis:

Out of the 5 common causes for AUFI, dengue NS1 testing by RDT or ELISA and RDT for malarial antigen are of great value.

For the other three, immediate specific diagnosis is not possible. Obtaining a blood culture before giving any antibiotic is extremely important for diagnosing enteric fever. Presence of eschar is virtually diagnostic of scrub typhus; a careful search for eschar is necessary. Lymphadenopathy also strongly suggests scrub typhus or less commonly leptospirosis. It is not a feature of enteric fever, malaria or dengue.

Note: RDT for scrub typhus has become recently available and has good sensitivity and specificity. There are no good rapid diagnostic tests for enteric fever or leptospirosis.

Acute Undifferentiated Febrile Illness – General Principles to AUFI *(continued)*

Diagnosis of dengue:
Sensitivity of dengue NS1antigen is nearly 100% on day 1 but falls gradually thereafter; by day 5 the sensitivity is about 60%. NS1 is generally negative after 7 to 9 days. The specificity of NS1 antigen test is 95 to 100%.

Dengue IgM antibody becomes detectable only after 3 to 5 days. IgM antibodies are detectable in 50% of patients by days 3-5 after onset of illness, increasing to 80% by day 5 and nearly 100 % by day 10.

It should be noted that dengue IgM may remain positive up to 3 months after dengue infection. So, in a patient with AUFI, a positive dengue IgM may represent recent infection with dengue and the current febrile illness may be due to another cause. So always correlate with the clinical and laboratory features and keep an open mind.

If there are no features of sepsis, wait for 2 to 4 days.
On day 3 to 5, consider CBC, urine routine.
Consider further testing.
Dengue NS1 antigen testing, malarial antigen, peripheral smear for malaria.
Consider empirical therapy after 4 to 7 days.

Malaria requires specific therapy; decide on the basis of smear and antigen testing
Dengue does not require specific therapy
Doxycycline covers scrub and leptospirosis
Ceftriaxone covers enteric fever and leptospirosis
Azithromycin covers all three
For severe or complicated enteric fever, ceftriaxone is the drug of choice.

Obtain blood culture *and other cultures as indicated, before empirical therapy (except in very resource poor situations)*

Options for empiric therapy:

Empirical doxycycline
In scrub typhus, usually there is a dramatic response in 24 to 48 hours. Leptospirosis will also respond. But enteric fever will not respond.

Empirical azithromycin
In scrub typhus, usually there is a dramatic response in 24 to 48 hours. Leptospirosis will also respond. But in enteric fever the response is often slow, taking several days and sometimes there is failure. Also, higher dose is need in enteric fever.

Empirical ceftriaxone
This is the choice for severe enteric fever or leptospirosis but does not cover scrub typhus.

Empirical therapy in AUFI patients:
For stable patients, monotherapy with doxy or azithromycin is preferred. For sick patients, ceftriaxone should be given with doxycycline or azithromycin.

Empirical therapy in AUFI patients with features of sepsis: If there are features of sepsis any time, obtain appropriate cultures (blood culture, other cultures such as urine etc., as appropriate) and start empirical therapy as early as possible. This is extremely important if there is septic shock. (Obtain basic tests such as complete blood count, urine routine etc. also).

Dengue requires only supportive therapy.
Malaria requires specific therapy. Start ceftriaxone with doxycycline or azithromycin if one of enteric fever, scrub typhus or leptospirosis is suspected.
If community acquired pneumonia is also a differential, then ceftriaxone with azithromycin is preferable.
If urinary or intra-abdominal or pelvic source of infection is also in the differential diagnosis, then one the following regimens may be used to cover adequately for gram negative bacilli.

Piperacillin tazobactam (amikacin may also be added for additional gram-negative coverage)
Cefoperazone sulbactam (amikacin may also be added for additional gram-negative coverage)

Meropenem (especially for patients with septic shock)
For severe or complicated enteric fever, ceftriaxone is the drug of choice.
If scrub typhus is a consideration, then doxycycline or azithromycin should be added to the above regimen.

Acute Undifferentiated Febrile Illness – Empirical Therapy					
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**		**Remarks**
SYS-05	***Acute Undifferentiated Febrile Illness (AUFI)- Without features of sepsis / organ dysfunction***	*Bacteria:* *Scrub typhus, Enteric fever and Leptospirosis.* *Non-bacterial:* *Dengue requires supportive therapy only.* *Malaria requires specific therapy*	**I**	Doxycycline 100mg PO BD for 5-7d (covers Scrub and Leptospirosis) **OR** Azithromycin 500mg PO OD for 3-5d (for Scrub and Leptospirosis) (Single dose therapy has been effective in some studies). Azithromycin 1g PO OD x 5d (for Enteric).	
SYS-06	***Acute Undifferentiated Febrile Illness (AUFI)- With features of sepsis / organ dysfunction***	*Scrub typhus, Enteric fever and Leptospirosis.* *Dengue requires supportive therapy only.* *Malaria requires specific therapy*	**I**	Ceftriaxone 2g IV OD **PLUS** Azithromycin 500mg IV/PO OD **OR** Doxy 100mg IV/PO BD	If diagnosis of scrub, enteric or leptospirosis is not definite and other conditions such as UTI etc. are in consideration, broad spectrum antibiotic such as piperacillin/tazobactam or cefoperazone/sulbactam will have to used instead of ceftriaxone

Acute Undifferentiated Febrile Illness – Empirical Therapy *(continued)*						
#	**Clinical Diagnosis**	**Etiology**		**Empirical Antibiotic Therapy**		**Remarks**
SYS-07	***AFI - Suspected Enteric Fever*** *Step ladder fever is characteristic: initial low-grade fever gradually increases to become high days later*	*Salmonella typhi, Salmonella paratyphi*		Azithromycin 1g PO on day 1 f/b 500mg PO OD for next 6 days OR 1g PO OD x 5d (for mild to moderate disease)		• Ceftriaxone is the preferred antibiotic for severe enteric fever. • Defervescence may take several days even with appropriate therapy. • Ciprofloxacin resistance is very common in India. • Amoxicillin 1 g PO TID or 2 g IV Q6H for 14 days OR Cotrimoxazole DS 1 tab BD for 7 to 14 days may be used after AST report as most strains have become sensitive now.
				OR	Ceftriaxone 2g IV BD for 10 x 14d (the choice for severe enteric fever)	
				OR	Cefixime 400mg PO OD x 14d (defervescence may be slower and failure rates may be higher as compared to azithromycin)	
				OR	Amoxicillin 1g PO TID or 2g IV Q6H x 14d	
				OR	Cotrimoxazole DS 1 tab BD x 7-14d	
SYS-08	***AFI - Suspected Scrub Typhus***			Azithromycin 500mg IV/PO OD x 3-5d		• Doxycycline may be used for ≤21 days in children of all ages
				OR	Doxycycline 100mg IV/PO BD x 5-7d	
SYS-09	***AFI - Suspected Leptospirosis*** ***Mild***		**I**	Doxy 100mg PO BD x 7d		• Leptospires are highly susceptible to a broad range of antibiotics, including penicillins, cephalosporins, aminoglycosides, macrolides, tetracyclines. In vitro evidence shows excellent susceptibility to carbapenems as well. • Doxycycline may be used for ≤21 days in children of all ages • Duration of therapy in severe leptospirosis is seven days
				OR	Azithromycin 500mg PO OD x 3d	
				OR	Amoxicillin 500mg PO TID x 7d	
SYS-09	***AFI - Suspected Leptospirosis Moderate to Severe***		**I**	Penicillin G 20MU IV Q6H x 7d		
				OR	Ceftriaxone 2g IV OD x 7d	
				OR	Doxycycline 200mg IV/PO STAT f/b 100mg BD x 7d	
				OR	Azithromycin 500mg IV/PO OD on day 1 f/b 250mg PO/IV OD x next 6d	

IX. Respiratory Infections – Community Acquired Pneumonia

Covid 19 infection

Covid-19 infection should always be considered in patients with fever, malaise, headache with or without respiratory symptoms or diarrhoea. Loss of smell or taste is characteristic but uncommon

Patients suspected to have Covid infection should always be evaluated in an isolation unit. Appropriate PPE should be worn by HCP. They should preferably be admitted to an isolation ward/ICU till diagnosis is confirmed or ruled out

Common bacterial pathogens

Pneumococcus
Chlamydia pneumoniae
Mycoplasma pneumoniae

Less common bacterial pathogens

Haemophilus influenza
Legionella pneumoniae
Staphylococcus aureus
Klebsiella pneumonia

Viral pathogens

Influenza
Covid 19

Blood culture

Blood culture is not indicated in all patients with CAP

Indications for blood culture in CAP

Severe CAP
Pneumonia leading to neutropenia
Immunocompromised patients (including asplenia)
Chronic liver disease

CAP managed as outpatient

Amoxicillin is the preferred drug in India. Low level penicillin resistance among pneumococci is increasing in India and oral therapy with low dose amoxicillin (500mg tid) may be inadequate; however high dose amoxicillin (1g tid) is still effective.

A macrolide or doxycycline may be used as the first choice in some regions of the world.
A macrolide or doxycycline will cover atypical pneumonia pathogens and pneumococcus. However, overall macrolide resistance is over 50% and doxycycline resistance is over 25% in India and are not the preferred options in India. Locally relevant data should be used to guide the decision.

Following influenza virus infection, secondary bacterial infection with staphylococcus or haemophilus influenza may occur

Consider amoxicillin/clavulanic acid instead of amoxicillin if staphylococcus or Haemophilus influenza is suspected

Respiratory fluoroquinolones (levofloxacin and moxifloxacin) cover pneumococcus and atypical pneumonia pathogens and may be used for empirical therapy. But it is better to avoid them and reserve them for their anti TB effect. They may have a role in penicillin allergic patients. There is increasing pneumococcal resistance to ciprofloxacin, and it should **not** be used for pneumococcal coverage.

CAP management in Hospitalized patients

Beta lactam PLUS macrolide combination should be used

Ceftriaxone is effective against pneumococcus. It is effective against Haemophilus influenza. It is effective against *Klebsiella* but not against ESBL producing Klebsiella. It also covers MSSA and anaerobic bacteria reasonably.

Piperacillin tazobactam is a good choice for severe pneumonia if resistant pathogens are considered. It is effective against ESBL producing Klebsiella but a carbapenem may be the choice particularly in seriously ill patients. Piperacillin tazobactam covers Pseudomonas. It covers MSSA well but **does not cover MRSA.** It has excellent anaerobic coverage as well.

Cefoperazone sulbactam is similar but inferior to piperacillin for Pseudomonas coverage. However, at JIPMER the susceptibility rates are slightly higher for cefoperazone sulbactam as compared to piperacillin tazobactam

Respiratory Infections – CAP *(continued)*		
Duration of therapy 5 to 7 days; but may be shorter if they fulfil criteria for stopping the antibiotic **Indications for stopping antibiotic.** Afebrile for 48 to 72 hours AND Has ***no more than one*** of the following HR > 100 BP < 90 RR > 24 O2 sat < 90 Altered mental status. Cough and chest X ray abnormalities may take 4 to 6 weeks to resolve There is NO need to extend the antibiotics if the patient is doing well otherwise Patients who have not responded to therapy after 48 to 72 hours should be re-evaluated. Longer duration may be warranted in selected patients with any of the following: necrotizing pneumonia, empyema, or lung abscess, or if extrapulmonary infection is identified (eg, meningitis or endocarditis)	**Indications for MRSA coverage** Empiric therapy for MRSA should be given to CAP patients with septic shock or respiratory failure requiring mechanical ventilation. In such patients, treatment for MRSA coverage with the addition of vancomycin (15 mg/kg IV every 12 hours, adjusted to a trough level of 15 to 20 mcg/mL and for renal function; in seriously ill patients, a loading dose of 25 to 35 mg/kg may be given). Linezolid (600 mg IV every 12 hours) is an alternative. Empirical coverage for MRSA may be indicated in other patients as well (see the following 2 tables) If MRSA is not isolated, coverage for this organism should be discontinued **Risk Factors for MDR gram negative bacteria and MRSA in CAP** Hospitalization in the previous 90 days Use of antibiotics in the previous 90 days Immunosuppression Non-ambulatory status Tube feeding Gastric acid suppression Severe COPD or bronchiectasis The last is a risk factor for CAP due to Pseudomonas.	**The incidence of MDR pathogens generally is not significantly increased *unless three or more risk factors* are present.** However, MRSA is an exception: the presence of one MRSA-specific risk factor (prior MRSA infection or colonization, long-term haemodialysis, or heart failure) and another pneumonia-specific risk factor may warrant MRSA coverage. **Clinical Features Suggesting Community Acquired MRSA Pneumonia** Cavitary infiltrate or necrosis Rapidly increasing pleural effusion Gross hemoptysis (not just blood-streaked sputum) Concurrent influenza Neutropenia Erythematous rash Skin pustules Young, previously healthy patient Severe pneumonia during summer months

Respiratory Infections – CAP *(continued)*										
	ORGANISM	***Penicillin***	***Amoxicillin or ampicillin***	***Amoxicillin clavulanate***	***Cefoperazone-sulbactam***	***Piperacillin tazobactam***	***Ceftriaxone***	***Macrolide***	***Doxycycline***	***Levofloxacin Moxifloxacin***
1	*Pneumococcus*	Yes (high dose needed)	Yes (high dose needed)	Yes (high dose of amoxicillin component needed)	✓	✓	✓	Resistance is common	Resistance is common	Yes (Ciprofloxacin resistance is common)
2	*Atypical pathogens (Mycoplasma, Chlamydophila, Legionella)*	✗	✗	✗	✗	✗	✗	✓	✓	✓
3	*Haemophilus influenza*	✗	R is common due to beta lactamase production	✓	✓	✓	✓	Resistance is common	✓	✓
4	*Staph aureus MSSA*	✗	✗	✓	✓	✓	Modest effect	Resistance is common	✓	✓
5	*Staph aureus MRSA*	✗	✗	✗	✗	✗	✗	✗	✓	Resistance is very common
6	*Klebsiella, E. coli (Significant proportion produce ESBL)*	✗	Klebsiella intrinsically resistant. E. coli resistance has become very common	Yes, but in severe or life-threatening infections, BL/BLI may be inferior to carbapenems			Yes (But, ESBL producing organisms will not respond)	✗	Resistance is common	Resistance is very common
7	*Pseudomonas*	✗	✗	✗	Yes (but inferior to piperacillin)	✗	✗	✗	✗	Resistance is very common

Respiratory Infections – CAP – Empiric Therapy							
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**				**Remarks**
RES-01	***Community Acquired Pneumonia Treated on OPD basis*** *(CURB-65 = 0 – 1)*	• *Streptococcus pneumoniae* • *Mycoplasma pneumoniae* • *Chlamydophila pneumoniae* • *Respiratory viruses* • *Haemophilus influenzae* • *Staph. aureus (especially secondary bacterial infection atter influenza virus infection)*	**I**	Amoxicillin 500mg or 1g PO TDS x 5d *(higher dose is preferred)*			• Pneumococcal resistance to erythromycin was 35% and to tetracycline was 35% (JIPMER 2018). Erythromycin resistant strains are always resistant to azithromycin as well. About 80% of tetracycline resistant strains are also likely to be resistant to doxycycline. • Resistance to oral amoxicillin may be significant and can be overcome by high dose amoxicillin.
				OR	Co-amoxiclav 625 mg or 875 mg TID		
				OR	Cefpodoxime 200mg PO BD x 5d		
				OR	Cefdinir 300mg PO BD x 5d		
			II	Azithromycin 500mg PO OD x 3-5d (alternative)			
				OR	Doxycycline 100mg PO BD x 5d(alternative)		
			III	Levofloxacin 750mg PO OD x 5d(alternative)			
RES-02	***Community Acquired Pneumonia Treated as In-Patient (non-ICU)*** *(CURB-65 = 2)*	• *Streptococcus pneumoniae* • *Mycoplasma pneumoniae* • *Chlamydophila pneumoniae* • *Respiratory viruses* • *Legionella* • *Haemophilus influenzae*	**I**	Ceftriaxone 2g IV OD x 5-7d	**AND**	Azithromycin 500mg IV/PO OD x 3-5d **±** Amikacin 15mg/kg IV OD (if risk factors for MDR Gram -ve pathogen present) **±** Oseltamivir	• For MRSA, add vancomycin (15 to 20mg/kg Q8-12H) or linezolid (600mg IV Q12H). See General Principles. • If influenza pneumonia is suspected, Oseltamivir may be added empirically. • Consider piperacillin-tazobactam or cefoperazone sulbactam instead of ceftriaxone, if there are risk factors for resistant gram-negative pathogens.
RES-03	***Community Acquired Pneumonia requiring ICU care*** *(CURB-65≥ 3)*	• *Streptococcus pneumoniae* *Staph. aureus* • *Legionella* • *Klebsiella pneumonia, other GNB* • *Haemophilus influenzae* • *Respiratory viruses*	**I**	Piperacillin-tazobactam 4.5g IV Q8H/6H	**AND**	Azithromycin 500mg IV/PO OD x 3-5d **±** Oseltamivir	• For MRSA add vancomycin (15 to 20mg/kg Q8-12H)) or linezolid (600mg IV Q12H) • If influenza pneumonia is suspected, Oseltamivir may be added empirically. • In severe or life-threatening infections by ESBL producing gram negative pathogens • BL/BLI may be inferior to carbapenems
				OR Cefoperazone-Sulbactam 2g IV, TID-QID or 3g Q12H			

Respiratory Infections - VAP and HAP – General Principles

- ✓ The absence of bacteria in the gram-stain of tracheal aspirate makes pneumonia an unlikely cause of fever or pulmonary infiltrates.
- ✓ Stable ventilator settings argue against clinically significant pneumonia.
- ✓ The absence of new infiltrates on X ray makes pneumonia unlikely, but their presence is not specific.
- ✓ CPIS a score of >6 suggests VAP but has significant limitations: (Sensitivity only 46% and specificity only 60%)
- ✓ Tracheal aspirate should be sent for Gram stain and culture before starting or changing antibiotic for VAP. However, bacteria grown in the tracheal aspirate may represent colonization or VAT (ventilator associated tracheitis) rather than VAP. Quantitative culture of tracheal aspirate showing **≥10^6 CFU per ml** is considered significant growth. Tracheal aspirate culture has **high sensitivity but only moderate specificity**; false positive results are common. So, the decision to treat should not be based on the results of tracheal aspirate culture alone.
- ✓ Even a single dose of effective antibiotic can reduce colony counts below the diagnostic threshold. So, it is very important to send tracheal aspirate culture before starting or changing antibiotics. Colony counts above the diagnostic threshold in a sample taken while the patient is on antibiotic therapy suggests that the current antibiotics are ineffective, particularly if the patient has already received 3 days of therapy already. If the patient is already receiving an antibiotic when he develops VAP, select an antibiotic from a different class.
- ✓ Tracheal colonization with Staph aureus or Gram negatives may persist even though lower airways may be eradicated with treatment. So, post treatment cultures are not recommended unless there is deterioration (fever, leucocytosis, worsening infiltrate, worsening in oxygenation)
- ✓ Enterococci and candida are often isolated from the sputum /tracheal aspirate in hospitalized patients. They are generally colonizers and *should not be* treated with antibiotics.
- ✓ **The absence of an MDR pathogen in the culture of tracheal aspirate or high-quality sputum eliminates the need for MDR coverage,** allowing empirical antibiotic therapy to be de-escalated.
- ✓ Treatment failure is not uncommon in VAP, especially that caused by MDR pathogens. VAP caused by MRSA is associated with a 40% clinical failure rate when treated with standard-dose vancomycin. Success rates may be higher if vancomycin dose is adjusted to achieve trough levels between 15 to 20 but there is higher risk of renal impairment. Linezolid has 15 % higher efficacy compared to even adjusted dose vancomycin. VAP due to Pseudomonas has a 40–50% failure rate, no matter what the regimen.

Duration of therapy: Seven days of therapy is as effective as longer therapy for most patients (IDSA 2016). Longer duration of therapy may be indicated in selected patients (lung abscess, empyema, MSSA or MRSA bacteraemia, immunocompromise, etc.)

Respiratory Infections - VAP and HAP – Empiric Therapy								
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**					**Remarks**
RES-04	***Ventilator associated pneumonia***	*Acinetobacter* *P. aeruginosa* *Klebsiella* *S. aureus (MSSA or MRSA)* *Other GNB like E. coli*	**I**	Cefoperazone-Sulbactam 2g IV, TID-QID or 3g Q12H *(preferred in early VAP, stable patient and limited disease).*	**AND**	Consider Amikacin 15 to 20mg/kg IV OD *(particularly if patient is hemodynamically unstable or BL/BLI is the primary antibiotic)* Add Vancomycin (15 to 20mg/kg Q8-12H) *(if MRSA is considered). See page for vancomycin dosing*		• Cefoperazone /sulbactam may be preferable to piperacillin tazobactam as sulbactam has intrinsic effect against Acinetobacter. • Acinetobacter is an important pathogen in VAP in India. • See dose of cefoperazone-sulbactam @page 81. • **See separate document for treatment of carbapenem resistant GNB**
				OR Piperacillin-tazobactam 4.5g IV Q6H *(Preferred in early VAP, stable patient and limited disease)*				
				OR Meropenem 1-2g IV Q8H *preferred in hemodynamically unstable patient*				
			II	*(If patient has current or recent carbapenem exposure)* Polymyxin B: 25,000 units/kg LD followed by 12,500 units/kg Q12h	**AND**	Consider adding a second antibiotic depending on local susceptibility see Add Vancomycin (15 to 20mg/kg Q8-12H) *(if MRSA is considered). See vancomycin dose @page 77.*		
				OR Colistin 9MU LD followed by 4.5MU IV Q12H				
RES-05	***Hospital-acquired pneumonia (but not VAP)***	*P. aeruginosa* *Klebsiella* *S. aureus (MSSA or MRSA)* *Other GNB like E. coli*	**I**	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H	**AND**	Consider amikacin 15 to 20mg/kg IV OD *(particularly if patient is hemodynamically unstable or BL/BLI is the primary antibiotic)* Add Vancomycin (15 to 20mg/kg Q8-12H) *(if MRSA is considered). See page 77 for vancomycin dosing*		
				OR Piperacillin-tazobactam 4.5g IV Q6H				
				OR Meropenem 1-2g IV Q8H *(preferred in hemodynamically unstable patients)*				
			II		*If patient has current or recent carbapenem exposure, follow* II *under VAP*			

Respiratory Infections – Empiric Therapy								
#	**Clin. Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**					**Remarks**
RES-06	***Aspiration Pneumonia*** *(Community acquired)*	***Oral Anaerobes**: (Bacteroides spp., Porphyromonas spp., Prevotella melaninogenica, Fusobacterium spp., and anaerobic gram-positive cocci)* ***Aerobes**: Streptococcus spp. and H. influenzae*	**I**	Amoxicillin-Clavulanate 1.2g IV Q8H followed by 625mg PO TDS for a minimum of 7 days Piperacillin/tazobactam 3.375gm IV q 6h for severe disease				• Management of witnessed / documented aspiration event: In patients with witnessed aspiration who develop fever, leucocytosis and pulmonary infiltrates within the first 48 hours have only chemical pneumonitis and do not benefit from antibiotic therapy. Majority show rapid clinical improvement with clearing of X-ray with supporting therapy alone. However, those with severe aspiration may have a fulminant course and die shortly after aspiration. • Secondary worsening after initial rapid improvement occurs only in one-fourth due to ARDS or bacterial infection complicating chemical injury. • Antibiotics are indicated in -severely ill patients -no significant improvement in 48 hours -secondary worsening after initial improvement.
RES-06	***Aspiration Pneumonia*** *(Hospital acquired)*	***Oral Anaerobes**: (Bacteroides spp., Porphyromonas spp., Prevotella melaninogenica, Fusobacterium spp., and anaerobic gram-positive cocci)* ***Aerobes:** Pseudonas, enteric gram-negative bacilli, less commonly S. aureus*	**I**	Piperacillin tazobactam 4.5g IV Q6H **OR** Meropenem 1g IV Q8H		**AND**	Add Vancomycin (15 to 20mg/kg Q8-12H) *(if MRSA is considered) See page for vancomycin dosing*	
RES-07	***Lung Abscess Primary*** *(Aspiration related)*	***Oral Anaerobes** (Bacteroides spp., Porphyromonas spp., Prevotella melaninogenica, Fusobacterium spp., and anaerobic gram-positive cocci)* *Sometimes along with aerobes Streptococcus spp. and H. influenzae*	**I**	Amoxicillin-Clavulanate 1.2g IV Q8H followed by 625mg PO TDS				• Most primary lung abscesses occur as a complication of aspiration pneumonia. Aspiration results from impaired consciousness / swallowing. • The presence of putrid discharge, indolent course, and necrotizing pneumonia should raise the suspicion for anaerobic bacterial involvement. • Duration of therapy: minimum of 3 to 4 weeks, and up to 14 weeks as per clinical improvement and radiological clearance. • Metronidazole is not recommended as monotherapy.
RES-07	***Lung Abscess Secondary***		**I**	Piperacillin tazobactam 4.5g IV Q6H **OR** Meropenem 1g IV Q8H		**AND**	Add Vancomycin (15 to 20mg/kg Q8-12H) *(if MRSA is considered). See vancomycin dose @page 77.*	• Empirical therapy depends on the setting.
	Septic pulmonary embolism	*S. aureus (IE) and F. necrophorum (Lemierre) in Septic pulmonary embolism*						
	Secondary to airway obstruction	*Usually, S. aureus and GNB in airway obstruction*	As per primary source / pathogen					
	Immunocompromised patients	*S. aureus, GNB, Nocardia, Aspergillus, Mucor in immunocompromised*						
	Endemic infections	*M. tuberculosis, NTM and Strongyloides are endemic*						

Respiratory Infections – Empiric Therapy

#	Clin. Diagnosis	Etiology	Empirical Antibiotic Therapy			Remarks
RES-08	**Acute Pharyngitis**	*Viral* *Group A Beta Haemolytic Strep (S. pyogenes)*	Very few require antibiotic therapy. See remarks			● 90% of pharyngitis in adults and 70% in children are viral ● Bacterial pharyngitis is self-limiting; only reason for treatment is prevention of acute rheumatic fever; primarily children between 5 and 15 at risk. ● Use Centor criteria to decide on need for antibiotics
			I	Penicillin V 500mg IV BD x 10d		
				OR	Amoxicillin 500mg PO BD x 10d	
				OR	Azithromycin 500mg PO OD x 3d or 500mg on D1 and 250mg on D2 to D5	
				OR	Cephalexin 500mg PO BD 10d	
RES-09	**Acute Sinusitis**	*Strep. pneumoniae* *H. influenzae* *M. catarrhalis*	Very few require antibiotic therapy. See remarks			● 90 to 98% of acute rhinosinusitis is viral; resolves by 5 to 10 days. Purulent secretions are also common in viral infection. ● Only 1% of viral acute rhinosinusitis in adults and only 5% in children are complicated by secondary bacterial infection. ● Bacterial infection may be considered in the following 3 situations; ○ Symptoms persisting beyond 7 to 14 days ○ Severe symptoms persisting for at least 3 to 4 days ○ Secondary worsening after initial improvement ● Macrolide and tetracycline resistance among pneumococci is more than 25 percent in several regions of India
			I	Amoxicillin-Clavulanate 875/125mg PO BD or 500/125 PO TID x 5 -7d		
				OR	Cefpodoxime 200mg BD x 5 -7 days	
				OR	Cefdinir 300mg BD x 5 -7 days	
				OR	Amoxicillin 500mg PO TID or 1g PO TID	
			II	Azithromycin 500mg PO OD x 5d		
				OR	Doxycycline 100mg PO BD x 5 -7d	
RES-10	**Acute Otitis Media**	*Strep. pneumoniae* *H. influenzae* *M. catarrhalis*	I	Amoxicillin-Clavulanate 875/125mg PO BD or 500/125mg TID x 5-10d		● Macrolide and tetracycline resistance among pneumococci is more than 25 percent in several regions of India ● Duration 5 to 7 days for mild to moderate cases and 10 days for severe cases
				OR	Cefpodoxime 200mg BD x 5 -10d	
				OR	Cefdinir 300mg BD x 5 -10d	
				OR	Amoxicillin 500mg TID or 1g TID x 5 -10d	
			II	Azithromycin 500mg OD x 5d		
				OR	Doxycycline 100mg PO x 5 -10d	
RES-11	**Acute Bronchitis**	*Viral*	--			● Symptomatic treatment only ● Antibiotics not required
RES-12	**AECOPD** *presence of at least 2 of the following 3 symptoms:* *a. Increased dyspnoea,* *b. Increased sputum volume, and* *c. Increased sputum purulence*	*Strep. pneumoniae* *H. influenzae* *M. catarrhalis* *Respiratory viruses*	I	Amoxicillin-Clavulanate 875/125mg PO BD or 500/125mg PO TID x 5d		● Start oseltamivir if clinical suspicion of influenza is high ● Antimicrobials for all patients who require mechanical ventilation ● Pseudomonas, Enterobacteriaceae are less common; they are seen in patients with severe COPD and extensive antibiotic exposure.
				OR	Cefpodoxime 200mg BD x 5d	
				OR	Cefdinir 300mg BD x 5d	
			II	Azithromycin 500mg PO OD x 3d		
				OR	Doxycycline 100mg PO BD x 5d	

X. Urinary Tract Infections – Principles of Antibiotic Therapy

Definitions (IDSA)

Cystitis: Growth of 10^3 colony-forming units (CFU)/mL or more of a uropathogen in a symptomatic individual *(IDSA; sensitivity 90% and specificity 90%)*

Pyelonephritis: Growth of 10^4 colony-forming units (CFU)/mL or more of a uropathogen in a symptomatic individual *(IDSA; sensitivity 90% and specificity 90%)*

Uncomplicated vs complicated UTI

Uncomplicated urinary tract infection (UTI) refers to infection in a structurally and functionally normal urinary tract in an immunocompetent host. The generally accepted definition of complicated UTI includes infection in the presence of factors that predispose to persistent or relapsing infection, such as foreign bodies (e.g., calculi, indwelling catheters, or other drainage devices); obstruction; immunosuppression; renal failure; renal transplantation; and urinary retention from neurologic disease. In addition, infection in men, pregnant women, children, and patients who are hospitalized or in health care–associated settings may be considered complicated.

Asymptomatic bacteriuria: Growth of ≥ 10^5/ml of a uropathogen is required for diagnosis of asymptomatic bacteriuria. Pyuria may or may not be present in asymptomatic bacteriuria. Single positive culture is enough in men but two positive cultures with the same organism 2 to 3 weeks apart are required for women. Screening or treating for asymptomatic bacteriuria is not recommended except in select situations (pregnancy, before urological procedure associated with mucosal bleeding and possibly early post- transplant period)

CAUTI (Catheter associated UTI):

Growth of ≥10^3 colony forming units (CFU)/mL of uropathogen **with symptoms or signs** compatible with UTI without other identifiable source in a patient with indwelling urethral, indwelling suprapubic, or intermittent catheterization.

Catheter associated asymptomatic bacteriuria

Growth of ≥10^5 colony forming units (CFU)/mL of uropathogen **without symptoms or signs** compatible with UTI without other identifiable source in a patient with indwelling urethral, indwelling suprapubic, or intermittent catheterization.

In the catheterized patient, pyuria does not differentiate ASB from CAUTI, but its absence suggests that CAUTI is not the cause of symptoms.

Symptoms:

The typical symptoms of cystitis are dysuria, urinary frequency, and urgency. Nocturia, hesitancy, suprapubic discomfort and gross haematuria are often noted as well.

Unilateral back or flank pain generally indicates pyelonephritis.

Fever indicates pyelonephritis or prostatitis rather than cystitis alone.

Lower tract symptoms (cystitis symptoms) may not be present in pyelonephritis.

Diagnosis:

Culture:

Urine sample for culture should be processed immediately (<u>within 1 to 2 hours</u>). Otherwise, it may be refrigerated at 4 degree <u>C up to 24 hours</u>; but once removed from refrigeration, it should be processed immediately.

False-positive cultures are caused by contamination at the time of collection or delay in processing

Urine examination

Pyuria is defined as > 10WBC/mm^3 of uncentrifuged urine or > 5 to 10 WBC/hpf in the sediment of centrifuged urine.

Urine dipsticks

Urine dipsticks for pyuria and bacteriuria are useful screening tests.

- Test for pyuria has a sensitivity of 75%–96% and a specificity of 94%–98% as compared to microscopy
- A positive test implies pyuria but not necessarily UTI
- A negative test for pyuria (leukocyte esterase test) makes UTI unlikely.
- Negative tests for bacteriuria (nitrite test) are common in UTI because of low titres of bacteria

However, when the clinical history is strongly suggestive of UTI, even negative results for both tests do not reliably rule out infection.

Urinary Tract Infections – Principles of Antibiotic Therapy *(continued)*

Aetiology

Uncomplicated cystitis/pyelonephritis

E. coli accounts for nearly 90% of isolates; *Klebsiella, Proteus, Enterococcus*, and *Citrobacter* species, along with other organisms, for 5–10%. *Staphylococcus saprophyticus accounts for* 5–15% of cystitis in the West, primarily in young sexually active women; does not cause pyelonephritis; uncommon in India

Complicated cystitis/pyelonephritis:

E. coli remains the predominant organism; *Pseudomonas, Klebsiella*, *Proteus*, *Citrobacter*, *Acinetobacter*, and *Morganella* are also are frequently isolated. Gram-positive bacteria (e.g., enterococci and *Staphylococcus aureus*) and yeasts also are important pathogens.

Duration of therapy

Cystitis

Duration of therapy is 3 to 7 days

Complicated cystitis

7 to 14 days

Uncomplicated pyelonephritis

Total duration of antimicrobial therapy generally ranges from 5 to 10 days, depending on the rapidity of clinical response and the antimicrobial chosen to complete the course. Fluoroquinolones are given for 5 to 7 days, cotrimoxazole for 7 to 10 days, and beta-lactams for 7 to 10 days.

Complicated pyelonephritis

Longer duration with antibiotic durations may be warranted in selected patients with complicated UTI (e.g. in patients who have a nidus of infection (such as non-obstructive stone that cannot be removed, men in whom prostatitis cannot be excluded)

Relapse

Relapse of cystitis or pyelonephritis requires 2 weeks of therapy

Duration of therapy - summary table*

Drug	Uncomplicated cystitis	Uncomplicated pyelonephritis
Nitrofurantoin	5 to 7	Not appropriate
Quinolone	3	5 to 7
Cotrimoxazole	3 to 7	7 to 10
Beta lactam	5 to 7	7 to 10

*Relapse of cystitis or pyelonephritis requires 2 weeks of therapy

Evidence suggests that cell wall–active agents such as penicillins and cephalosporins are less effective in eradicating infection from the kidneys or the urinary tract as compared to TMP-SMX, fluoroquinolones, or aminoglycosides

Nitrofurantoin is available in two formulations

1. Formulation containing a combination of nitrofurantoin monohydrate and nitrofurantoin macrocrystals: dose 100 mg bd for treatment

2. Formulation containing nitrofurantoin macrocrystals alone: dose 50 mg 6 h for treatment

Bacteraemic UTI

Bacteraemia develops in 20–30% of cases of pyelonephritis. Amikacin may not be effective for bacteraemic pyelonephritis. Bacteraemia per se does not warrant longer therapy

Candiduria

Candiduria is common in catheterized patients especially in those with risk factors (being in ICU, broad spectrum antibiotic therapy, diabetes). It may be asymptomatic, or cause cystitis or pyelonephritis or even sepsis. Asymptomatic candiduria does not warrant treatment except in those at risk for disseminated disease (neutropenia, clinical instability, scheduled for urologic procedure, low birth weight). Fluconazole is the choice. Flucytosine or parenteral amphotericin B is an alternative. Newer azoles and echinocandins have low level urinary excretion and are not recommended.

Urinary Tract Infections – Empiric Therapy

#	Clinical Diagnosis	Aetiology	Empirical Antibiotic Therapy				Remarks
UTI-01	***Cystitis uncomplicated*** *Signs and symptoms (e.g., dysuria, urgency, Frequency), pyuria (>10 WBC/hpf) AND positive urine culture ≥10^3 CFU/mL* *Uncomplicated: female, non-pregnant, no urologic abnormalities, no stones, no catheter, no urological instrumentation*	*E. coli accounts for 90% of isolates; Klebsiella, Proteus, Enterococcus, and Citrobacter species, along with other organisms, for 5–10%.*	I	Nitrofurantoin (depending upon the formulation), 100mg PO BD x 5d or 50 mg Q6H			• Duration of treatment is longer in pregnant women. (see complicated cystitis) • Fosfomycin can be considered for multi-drug-resistant E. coli (3 g as a single dose or 3 g once every 2 to 3 days for 3 doses) • Staphylococcus saprophyticus: accounts for 5–15% of cystitis in the West, primarily in young sexually active women; it does not cause pyelonephritis; It is uncommon in India
				OR	Amoxicillin-Clavulanate 625mg PO TDS x 5-7d		
			II	Ciprofloxacin 500mg PO BD x 3d			
				OR	Cotrimoxazole 960mg (one DS tablet) BD x 3-5d		
UTI-02	***Cystitis complicated*** *(Male gender, pregnancy, underlying GU disease, e.g., neurogenic bladder, renal stones, hydronephrosis etc.)*	*E. coli predominates but Pseudomonas Klebsiella, Proteus, Gram-positive bacteria, and yeasts*		As above but for 7-14d			• Definitive therapy is advisable. • E. coli predominates but Pseudomonas, Klebsiella, Proteus, Citrobacter, Acinetobacter, and Morganella are also frequently isolated. Gram-positive bacteria (e.g., enterococci, Staphylococcus aureus) and yeasts also are important pathogens.
UTI-03	***Pyelonephritis*** *Signs and symptoms (e.g., fever, flank pain) AND pyuria AND positive urine culture ≥10^4CFU/mL.* *Many patients will have other evidence of upper tract disease (i.e. leukocytosis, WBC casts, or abnormalities upon imaging)*	*Similar to uncomplicated cystitis*	I	Amikacin 15mg/kg OD IV (for mild to moderate illness only)			• In pyelonephritis with suspected bacteraemia amikacin monotherapy should not be used. • Quinolone for 5 to 7 days • Cotrimoxazole for 7 to 10 days • Beta lactam for 7 to 10 days • Bacteraemia per se does not warrant longer therapy
			II	Piperacillin-Tazobactam 4.5g IV Q6H	±	Amikacin 15mg/kg IV OD	
				OR Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H			
				OR Ertapenem 1g IV OD			
				OR Meropenem 1g IV Q8H			
UTI-04	***Pyelonephritis Complicated*** *(Male gender, pregnancy, underlying GU disease, e.g., neurogenic bladder,renal stones, hydronephrosis etc.)*	*Similar to complicated cystitis*	I	Amikacin 15 mg/kg IV OD (for mild to moderate illness only)			• In pyelonephritis with suspected bacteraemia, amikacin monotherapy should not be used. • Duration as above • Longer duration with antibiotic durations may be warranted in selected patients with complicated UTI (e.g. in patients who have a nidus of infection (such as non-obstructive stone that cannot be removed, men in whom prostatitis cannot be excluded)
				OR Piperacillin-Tazobactam 4.5g IV Q6H	±	Amikacin 15mg/kg IV OD	
				OR Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H			
			II	Meropenem 1g IV Q8H			

Urinary Tract Infections – Empiric Therapy							
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**				**Remarks**
UTI-05	***Urosepsis*** *UTI with features of sepsis*	*Similar to either complicated or uncomplicated UTI, depending on patient characteristics*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H	**±**	Amikacin 15mg/kg IV OD	• In pyelonephritis with suspected bacteraemia, amikacin monotherapy should not be used.
				OR Cefoperazone-Sulbactam 2g IV TID-QID or 3g 12 H			
			II	Meropenem 1-2g IV Q8H			
UTI-06	***Perinephric abscess*** *Usually secondary to pyelonephritis; occasionally secondary to bacteraemia*	*Secondary to pyelonephritis*		As for pyelonephritis (uncomplicated or complicated)			• Diagnostic aspiration is needed to identify the pathogen for all abscesses irrespective of the size, if this is not established through urine or blood culture. • Drainage is recommended for abscesses 5 cm or greater in size; drainage is optional for 3 to 5 cm, and is usually not required for abscesses less than 3 cm. • Surgical drainage is indicated of percutaneous drainage fails or contra indicated. • Duration of therapy is at least 2 to 3 weeks; duration depends on extent of the disease, clinical response to therapy and fall in inflammatory markers
		Secondary to bacteraemia		As per the primary focus			
UTI-07	***Renal abscess*** *Commonly secondary to pyelonephritis; less commonly secondary to bacteraemia*	*Secondary to pyelonephritis: as for pyelonephritis*		As for pyelonephritis (uncomplicated or complicated)			• Drainage is indicated for abscesses greater than 5 cm; abscesses less than 5 cm usually need drainage only if medical therapy fails • Duration of therapy is at least 2 to 3 weeks; duration depends on extent of the disease, clinical response and fall in inflammatory markers
		Secondary to bacteraemia		As per the primary focus			
UTI-08	***CAUTI*** *Signs and symptoms (Fever, suprapubic or flank pain) AND positive urine culture ≥10^3 CFU/mL from a catheterised sample*	*E. coli predominates but Pseudomonas. Klebsiella, Proteus, Citrobacter, Acinetobacter, and Morganella also frequently isolated. Gram-positive bacteria (e.g., enterococci and Staphylococcus aureus) and yeasts also are important pathogens.*		As for complicated cystitis or pyelonephritis depending on clinical features			.

XI. Gastrointestinal Infections – Empiric Therapy

#	Clinical Diagnosis	Aetiology	Empirical Antibiotic Therapy			Remarks
GIT-01	**Non bloody diarrhoea**	*Rotavirus, Norovirus, NTS(Non Typhoidal Salmonella), Shigella spp., ETEC, Campylobacter, V. cholera and C. difficile.*	Frequently self-limiting and may not be bacterial. Antibacterial not usually indicated. Antibiotic indicated only in severe disease. Empirical therapy may be justified in patients with high fever, significant abdominal pain, persistent or worsening symptoms. However high fever and significant abdominal pain occur in viral diarrhoea also (rotavirus or norovirus)			• Antibiotic is indicated in selected cases only. Majority of patients do not require antibiotic therapy. • Antibiotics indicated only in patients with severe disease or immunocompromised status. • Consider acute gastroenteritis due to preformed toxin as well (prominent vomiting is usual; Bacillus cereus, Staphylococcus aureus and Clostridium perfringens). • Profuse "rice-water" stools suggest cholera or a similar toxigenic process. • Ciprofloxacin resistance is very common in Salmonella, Shigella and Campylobacter. • For Campylobacter, azithromycin is the drug of choice. Beta-lactams are ineffective.
			I		Azithromycin 500mg OD x 3d [Non Typhoidal Salmonella (NTS), Shigella, Campylobacter]	
				OR	Cefixime 200mg BD or 400mg OD x 3-5d (Shigella)	
				OR	Ciprofloxacin 500mg BD x 5d (High rates of resistance in NTS, Shigella, Campylobacter in India)	
			II		Ceftriaxone 2g IV OD x 3-5d for severe disease (NTS, Shigella; but beta lactams not effective for Campylobacter)	
GIT-02	**Bloody diarrhoea (dysentery) with fever**	Common with *Shigella.* Less common with *Campylobacter, NTS, Yersinia, V. parahemolyticus* and *C. Difficile* (In *E. histolytica* dysentery<*40%* have fever)	I		Ceftriaxone 2g IV OD x 5d (for Shigella and NTS). Azithromycin 500mg OD x 3d (for Shigella, NTS, Campylobacter) For other pathogens see table.	• Bloody stools without faecal leukocytes should alert the laboratory to the possibility of infection with Shiga toxin–producing enterohemorrhagic Escherichia coli. • Tenesmus (painful rectal spasms with a strong urge to defecate but little passage of stool) may be a feature of cases with proctitis, as in shigellosis or amebiasis.
GIT-03	**Bloody diarrhoea without fever**	*E. histolytica*	I		Metronidazole for amoebic dysentery	• EHEC infection causes little or no fever. • Tenesmus (painful rectal spasms with a strong urge to defecate but little passage of stool) may be a feature of cases with proctitis, as in shigellosis or amebiasis.
		EHEC	I		Avoid antibiotics for EHEC	

Gastrointestinal Infections – Diarrhoea – General Principles

Definition: Diarrhoea is defined as the passage of loose or watery stools, typically at least three times in a 24-hour period.
Acute diarrhoea is defined as diarrhoea of ≤14 days in duration, in contrast to persistent (>14 days and ≤30 days) or chronic (>30 days) diarrhoea.
Dysentery is defined as diarrhoea with visible blood is stools. Dysentery is commonly associated with fever and abdominal pain.

- ***Mild*** diarrhoea (<3 unformed stools/day, minimal associated symptomatology).
- ***Moderate*** diarrhoea (3 to 6 unformed stools/day &/or systemic symptoms)
- ***Severe*** diarrhoea (>6 unformed stools/day, temp >101°F, tenesmus, blood in stool, or faecal leukocytes)

Microbiology
Common pathogens in acute community-acquired diarrhoea: Rotavirus, Norovirus, Salmonella, Shigella, enterotoxigenic E. coli, Campylobacter; consider V. cholera in outbreaks.
Consider acute gastroenteritis due to preformed toxin as well (prominent vomiting is usual; Bacillus cereus, Staphylococcus aureus and Clostridium perfringens).
Nosocomial diarrhoea: C. difficile
Persistent diarrhoea in immunocompromised (most likely causes vary depending on type of immunocompromise): Giardia, Cryptosporidium, Cyclospora, Isospora, Microsporidia, Cytomegalovirus.

Symptomatic therapy
Antimotility drug: Loperamide may be used for rapid control of symptoms. It can be used for mild to-moderate diarrhoea but should be avoided in inflammatory diarrhoea (do not give if there is fever, significant abdominal pain, or bloody stools)
Antibiotic therapy
Carefully assess the patient before prescribing antimicrobials. Most infectious diarrhoea is self-limited and only requires supportive management.
Treatment with antibiotics is not recommended for most mild & moderate disease. Antibiotic use may lead to adverse outcomes (e.g. 20 fold higher risk of HUS with Shiga toxin-producing E. Coli, increased risk of chronic carrier state in NTS infection). Viral pathogens, such as Rotavirus and Norovirus commonly cause diarrhoea and do not require antibiotics. A significant proportion of diarrhoea is due to viral pathogens (rotavirus in children and Norovirus in adults).

Non bloody diarrhoea
Frequently self-limiting and may not be bacterial. Antibacterial not usually indicated. Antibiotic is indicated only in severe disease. Empirical therapy may be justified in patients with high fever, significant abdominal pain, persistent or worsening symptoms. However high fever and significant abdominal pain occur in viral diarrhoea also (rotavirus or norovirus)
Consider acute gastroenteritis due to preformed toxin as well (prominent vomiting is usual; Bacillus cereus, Staphylococcus aureus and Clostridium perfringens).
Bloody stools with fever
Patients with bloody stools and fever require antibiotics (indicates infection with Shigella, entero-invasive E. coli, Campylobacter, Yersinia, or Vibrio parahaemolyticus; Clostridium difficile is also a consideration). Amoebic dysentery is also a consideration.
Bloody stools without fever
Consider enterohaemorrhagic E. coli; avoid antibiotics as they increase the risk of HUS by 20-fold. Consider amoebic dysentery (as only < 40% of patients have fever).

Gastrointestinal Infections – Diarrhoea – Pathogen specific therapy	
Organism and indications for treatment	***Treatment***
Cholera Watery diarrhoea Rice water stools	Doxycycline 300mg OD OR Azithromycin 1g OD OR Ciprofloxacin 500mg BD X 3d
Shigella spp. Treatment indicated in only moderate or severe cases.	Azithromycin 500mg OD X 3d OR Ceftriaxone 2g IV OD X 3d OR Cefixime 400mg OD x 3-5d. High rates of quinolone resistance in India. Increasing ceftriaxone resistance in India Ciprofloxacin resistance 51%[1] and 62%[2] Ceftriaxone resistance 21%[1] and 45%[2] (1. Jain 2020, Dharwad, Karnataka, 2. Taneja 2021, PGI, Chandigarh: high rates of ceftriaxone resistance could be due to referral bias)
Non-typhoid Salmonella spp. (NTS) Less than 5% of all patients with *Salmonella* gastroenteritis develop bacteraemia, Treat only *severe or invasive disease* or if patient *at risk* for developing invasive disease: • Severe illness requiring hospitalization, • Age < 6 months or > 50 years, • Bacteraemia, • Presence of prostheses, • Valvular heart disease, • Severe atherosclerosis, • Malignancy or other immunocompromise Antibiotic use may increase the risk of chronic intestinal carriage following salmonellosis.	Ceftriaxone 2g IV OD Azithromycin 500mg PO OD x 3d (limited studies) Ciprofloxacin 500mg PO BD (if susceptible) TMP/SMX 160/800mg PO BD (if susceptible) Amoxicillin 1g PO TID (if susceptible) Duration: 3-7d; 7-14d for bacteraemia High rates of quinolone resistance in India Ciprofloxacin R in 1/3, Azithromycin R 7%, Ceftriaxone R 6% (Jacob 2020, CMC Vellore)
Enteric fever Fever, anorexia, abdominal pain, nausea, vomiting. Diarrhoea is seen in about 25%; constipation is less common.	Azithromycin 1g PO OD x 5d or 1g on D1, f/b 500mg OD for the next 6 days may be used for mild to moderate disease; it also covers scrub typhus and leptospirosis, the two other common bacterial AUFI. Cefixime 400mg OD x 14d (defervescence may be slower and failure rates may be higher as compared to azithromycin) Ceftriaxone 2g IV Q12H x 10-14d is the choice for severe or complicated enteric fever. Ciprofloxacin resistance is very common in India. Amoxicillin 1g PO TID or 2g IV Q6H x 14d OR Cotrimoxazole DS 1 tab BD x 7-14d may be used after AST report as most strains have become sensitive now.

Gastrointestinal Infections – Diarrhoea – Pathogen specific therapy *(continued)*	
Organism and indications for treatment	***Treatment***
Campylobacter spp. Frequently self-limiting Treatment recommended only for: • Severe illness, • Age < 6 months or > 50 years • Gross blood in stool, • High fever • Worsening or relapsing symptoms • Pregnancy, • Immunocompromised host	Azithromycin 500mg PO OD X 1-3d High rates of quinolone resistance in India. (Beta lactams are ineffective against Campylobacter)
E. coli (enterotoxigenic, enteropathogenic, entero-invasive) or empiric therapy of traveller's diarrhoea	Ciprofloxacin 500mg PO BD X 1-3d
Enterohemorrhagic E. coli (Shiga toxin producing E. coli, including E. coli 0157:H7)	Treatment not recommended. Antibiotic use associated with 20-fold higher risk of HUS
Vibrio parahaemolyticus Associated with shellfish consumption. Treatment recommended for severe illness	Ciprofloxacin 500mg PO BD x 3d OR Doxycycline 100mg PO BD x 3d OR Ceftriaxone 2g IV OD x 3d
Yersinia spp. Treatment recommended for: • Immunocompromised host • Bacteraemia • Pseudo appendicitis syndrome	Ciprofloxacin 500mg PO BD x 3d OR Doxycycline 100mg PO BD x 3d OR (Not for bacteraemia) TMP/SMX 160/800mg PO BD x 3-5d (if susceptible)
Amoebic dysentery Studies in Bangladesh show that E. histolytica infection commonly leads to diarrhoea without dysentery; only 15 to 33% of symptomatic patients develop dysentery. Amoebic dysentery, or amoebic colitis, is defined as diarrhoea with **mucus *or*** visible or microscopic **blood** in a patient with E. histolytica infection. *Fewer than 40%* of patients with amoebic dysentery have *fever.* The abdominal pain in amoebic dysentery may be of such severity that an acute abdomen suspected	Metronidazole 500-750mg PO/IV TID x 5-10d AND Diloxanide furoate 500mg TDS x 10d.

Gastrointestinal Infections – Empiric Therapy						
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**			**Remarks**
GIT-04	***SBP (Primary)*** *Fever, non-localising symptoms (malaise, fatigue, encephalopathy), findings in peritoneal fluid> 250PNMs/µL, without an evident intra-abdominal surgically treatable source of infection.*	*E.coli, Klebsiella, Pneumococci, Streptococci*	**I**	Cefotaxime 2g IV Q8H		• 5 days of therapy is adequate, if there is prompt clinical response, even in those with bacteraemia. • Change of therapy or longer therapy may be needed in those with inadequate response. Repeat ascitic fluid PMN count should guide therapy in them. • Polymicrobial infection should prompt suspicion of GI perforation. • E. coli is the most common organism. About 80% of E. coli produce ESBL.
				OR	Ceftriaxone 2g IV Q24H	
				OR	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H	
				OR	Piperacillin-Tazobactum 4.5g Q6H	
			II	Ertapenem 1g IV OD		
				Meropenem 1g IV Q8H *(Use only if Ertapenem is not available as Pseudomanas coverage is unnecessary)*		
GIT-05	***Secondary Bacterial Peritonitis*** Contamination of peritoneal cavity by bowel flora due to bowel perforation, ruptured appendix, ruptured diverticula, ischemic bowel, leaking surgical anastomosis, intraabdominal abscess etc. *Abdominal pain, fever, features of peritonitis (guarding, rebound tenderness)*	*Enterobacteriaceae , Anaerobes (Bacteroids spp)* *Enterococci Candida* *Role of Enterococci and Candida in mixed infections is controversial*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H		• Emergency surgery to eliminate source of contamination, reduce bacterial load & prevent recurrence. • With adequate source control, 4 days of antibiotic therapy after achieving source control is adequate. • The role of enterococci and Candida species in mixed infections is controversial. • Routine coverage against enterococci is not necessary for patients with community-acquired intraabdominal infections. • When Candida is isolated in intraabdominal infections, antifungal therapy is indicated only in select situations (recent immunosuppressive therapy, postoperative infection or recurrent intraabdominal infection) • For Candida, fluconazole iv 800mg loading dose followed by 400mg maintenance dose needs to be added.
				OR	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H	
				OR	Ertapenem 1g IV OD	
				OR	Meropenem 1g IV Q8H (especially for health care related, or hospital acquired secondary peritonitis)	

Gastrointestinal Infections – Empiric Therapy							
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**				**Remarks**
GIT-06	***Liver Abscess– Amoebic***	*E. histolytica*	**I**	Metronidazole 500mg IV Q6H-Q8H or 800 mg PO TID x 7-10d.			• Primary management is medical. Ultrasound guided drainage is indicated only in selected patients: large abscesses, signs of imminent rupture and no response to medical treatment. • Drainage is indicated for: • (1) high risk of abscess rupture; (2) large left lobe liver abscess; (3) failure to respond to medical therapy within 5-7 days; and (4) cannot differentiate from a pyogenic liver abscess. • Pigtail drainage may achieve faster response as compared to needle aspiration alone, but long-term outcomes are similar.
GIT-07	***Liver Abscess - Pyogenic***	*Enterobactericeae, Anaerobes (Bacteroids spp) Enterococci* *Melioidosis has emerged as an important pathogen. Often there is evidence of disease elsewhere such as pneumonia, skin soft tissue infection, arthritis and splenic abscess.*	**I**	Inj. Piperacillin-Tazobactam IV 4.5g Q6H			• Unlike amoebic liver abscess, pyogenic liver abscesses require drainage in most patients, in addition to antibiotic therapy, • General rule for those with single abscess is:(UTD) • ≤5 cm: percutaneous aspiration • > 5 cm: pigtail drainage • > 10 cm: surgical drainage, usually • Individualised approach is needed for patients with multiple or multilocular abscesses. • Duration of antibiotic therapy is 4 to 6 weeks. If there is good response, consider oral step down after 2 to 3 weeks of initial parenteral therapy; in selected cases initial parenteral therapy may be shortened to 5 to 7 days. (M). Duration of therapy should be guided by adequacy of drainage, fever, pain, WBC count and CRP. (UTD) • Most pyogenic liver abscesses are polymicrobial, with enteric gram-negative bacilli and anaerobic species predominating. • In Asia, monomicrobial infection with hypervirulent Klebsiella pneumoniae is an important cause of primary liver abscess.
				OR	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H		
			II	Meropenem 1g IV Q8H			

Gastrointestinal Infections – Empiric Therapy						
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**			**Remarks**
GIT-08	***Acute Cholecystitis***	*Enterobactericeae, Anaerobes (Bacteroids spp) Enterococci*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H		• Duration of therapy • Medially managed patients • Duration if therapy: till signs and symptoms of acute cholecystitis have resolved. • Surgically managed patients • Stop the day after surgery
				OR	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H	
				OR	Ertapenem 1g IV OD	
				OR	Meropenem 1g IV Q8H *(Use only if Ertapenem is not available as Pseudomanas coverage is unnecessary)*	
GIT-09	***Cholangitis***	*Enterobactericeae, Anaerobes (Bacteroids spp) Enterococci*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H		
				OR	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H	
				OR	Ertapenem 1g IV OD	
				OR	Meropenem 1g IV Q8H *(Use only if Ertapenem is not available as Pseudomanas coverage is unnecessary)*	
GIT-10	***Intra-abdominal abscess***	*Enterobactericeae, Anaerobes (Bacteroids spp) Enterococci*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H		
				OR	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H	
				OR	Ertapenem 1g IV OD	
				OR	Meropenem 1g IV Q8H *(Use only if Ertapenem is not available as Pseudomanas coverage is unnecessary)*	

XII. Skin and Soft Tissue Infections – General Principles

- Classification of skin and soft tissue infections
    - Cellulitis – purulent
    - Cellulitis – non-purulent
    - Folliculitis, Carbuncle, Furuncle
    - Abscess
    - Impetigo
    - Diabetic Foot Infection
    - Snake bite,
    - Animal and Human bite
    - Necrotizing Fasciitis
- Cellulitis is an acute spreading infection of the skin that involves the subcutaneous tissues.
- Predisposing factors: previous trauma, often minor (laceration, abrasion, puncture wound) or underlying skin lesion (furuncle, ulcer, dermatophyte infection)
- Involvement is often extensive, with marked erythema, warmth, and swelling.
- Regional lymphadenopathy is common, and bacteraemia can occur.
- Local abscesses may develop, and small patches of overlying skin may subsequently undergo necrosis.
- Two types of cellulitis exist. Non-suppurative cellulitis and suppurative cellulitis
- A nidus of suppuration with surrounding cellulitis (suppurative cellulitis) is a hallmark of staphylococcal cellulitis, although this diagnostic finding is not always present.
- Always elevate affected extremity. Not elevating the limb often leads to treatment failure.
- Improvement of erythema can take days, especially in patients with lymphedema, because dead bacteria in the tissues continue to induce Inflammation.
- Very early post-operative wound infection within 6 to 48 hours after surgery is characteristic of streptococcal infection; in contrast, postoperative staphylococcal infection will not be evident for at least several days after surgery.
- Neutropenic, solid organ transplant, and cirrhotic patients may have cellulitis due to Gram-negative organisms. Consider expanding coverage in these cases.
- If eschar, consider angio-invasive organisms (GNB, aspergillosis, mould).
- Clindamycin resistance is common among streptococci and staphylococci in India and is best avoided. 33% of streptococci are resistant to clindamycin. Among Staphylococci, 21% of MSSA and 34% of MRSA resistant to clindamycin. (ICMR 2021)
- Doxycycline resistance is very common among streptococci, but MSSA and MRSA are largely susceptible to doxycycline [Resistance to tetracycline in India: Streptococci 58% R (JIPMER 2018) but MSSA 8% R and MRSA 15% R (ICMR 2021)]

Skin and Soft Tissue Infections – Cellulitis						
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**			**Remarks**
SSI-01	***Mild Non-purulent Cellulitis*** *Non suppurative cellulitis is defined as cellulitis with intact skin and no evidence of purulent drainage.*	*Beta-haemolytic streptococci (e.g., group A, B, C, G), Staph aureus*	**I**	Tab. Cephalexin 500mg PO Q6H		• Duration of therapy for cellulitis is 5-7 days. • Extension of antibiotic therapy (up to 14 days) may be warranted in the setting of severe infection, slow response to therapy, or immunosuppression. • Clindamycin resistance is common in India and is best avoided
				OR	Tab. Amoxicillin/clavulanate 625mg PO Q8H or 875mg PO Q12H	
			III	Tab. Clindamycin 300mg PO Q8H		
SSI-02	***Moderate to Severe Non-purulent Cellulitis******	*Beta-haemolytic streptococci (e.g., group A, B, C, G), Staph aureus*	**I**	Amoxicillin/clavulanate 1.2g IV Q8H		
				OR	Cefazolin 1 to 2g IV Q8H	
			II	Clindamycin 600mg IV Q8H x 5-7d		
SSI-03	***Mild Purulent cellulitis*** *Suppurative cellulitis is defined as cellulitis with purulent drainage or exudates in the absence of a drainable abscess.*	*Usually caused by S. aureus (MSSA and MRSA).*	**I**	Cap. Doxycycline 100mg PO BD		• Duration of therapy for cellulitis is 5-7 days. • Clindamycin resistance is common in India and is best avoided
			II	**OR**	Clindamycin 300mg PO Q8H	
				OR	Clindamycin 600mg IV Q8H (if parenteral therapy is needed)	
SSI-04	***Moderate to Severe Purulent cellulitis***	*Usually caused by S. aureus (MSSA and MRSA).*	**I**	Cloxacillin 2g IV Q6H (If MRSA is NOT a consideration)		• Duration of therapy total of 5 to 14 days; oral switch once there is clinical improvement and there are no signs of systemic toxicity
				OR	Cephazolin 1-2g IV Q8H (If MRSA is NOT a consideration)	
			II	Vancomycin 15mg/kg IV Q12H (If MRSA is a consideration)		
				OR	Linezolid 600mg PO/IV Q12H (If MRSA is a consideration)	

**These recommendations do not apply to diabetic foot infection, infection following traumatic injuries, human bite, animal bite, surgical site infection etc. (refer to the appropriate sections).*

Skin and Soft Tissue Infections – Cellulitis

#	Clinical Diagnosis	Aetiology	Empirical Antibiotic Therapy					Remarks
SSI-05	**Carbuncle, furuncle and abscess**	*As for purulent cellulitis*		*As for purulent cellulitis*				• Duration of therapy 5 to 7 days
SSI-06	**Impetigo (non-bullous)** *honey-crust lesions*	*Streptococcus pyogenes. Staph aureus may be secondary colonizer*		Penicillin V 250-500mg IV Q6H X 5d				• If lesions are few, topical therapy with mupirocin ointment may be used. • Duration of therapy 7 days
				OR	Amoxicillin 500mg PO TID x 5d			
SSI-07	**Impetigo (bullous)** *(After rupture, thin varnish like crust)*	*Staphylococcus*	**I**	Cloxacillin 500mg PO Q6H				• Duration of therapy 7 days • Clindamycin resistance is common among streptococci and staphylococci in India and is best avoided
				OR	Tab. Cephalexin 500mg PO Q6H			
				OR	Tab. Amoxicillin/clavulanate 625mg PO Q8H or 875mg PO Q12H			
				OR	Doxycycline 100mg PO BD x 7d			
			II	Tab. Clindamycin 300mg PO Q8H				
SSI-08	**Cellulitis** ***(suspected Vibrio vulnificus infection)*** *(Sea water laceration)*	*Vibrio vulnificus*		Ceftriaxone 1g IV Q24H *(For severe disease or mild disease with risk factors for severe disease)*		**AND**	Doxycycline 100mg PO BD OR ciprofloxacin 500 mg BD	• Rare, but rapidly fatal if untreated.
				Doxycycline 100mg PO BD OR ciprofloxacin 500 mg BD *(For mild disease, without risk factors for severe disease)*				
SSI-09	**Cellulitis** ***(suspected Aeromonas infection)*** *(Fresh water laceration)*	*Aeromonas hydrophila,*	**I**	Ceftriaxone 2g IV OD		**±**	Doxycycline 100 mg PO BD (add for severe disease)	
				OR	Ciprofloxacin 500mg PO BD			
SSI-10	**Infected wound- post trauma** Mild to moderate	*Polymicrobial: MSSA, MRSA, aerobic and anerobic streptococci, enteric gram negative bacilli, C.perfringens, C.tetani.* *If water exposure, Pseudomonas spp and Aeromonas spp. Also, Acinetobacter.*	**I**	Cephalexin 500mg PO Q6H				• Should focus on covering strep and staph. • Clindamycin resistance is common among streptococci and staphylococci in India and is best avoided. • About 33% of streptococci resistant to clindamycin. About 14% of MSSA and 39% of MRSA resistant to clindamycin. (ICMR 2018). About 58% and 79% of streptococci resistant to tetracyclines (JIPMER, AIIMS) • Wound debridement should be done as needed
				OR Amoxicillin-clavulanate 625mg PO TID or 875mg PO BD				
			II	**OR**	Doxycycline 100mg PO BD			
				OR	Clindamycin 300-450mg PO Q8H			
	Severe, requiring hospitalization			Piperacillin-tazobactam 4.5g Q6H OR Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H		**AND**	Vancomycin 15mg/kg Q12H	

Skin and Soft Tissue Infections – Diabetic Foot

Diabetic foot infection:
Look for evidence of infection (features of inflammation or purulence)
Non infected ulcers do not need antibiotics
Cultures should not be taken from clinically non-infected wounds
If there is evidence of infection, then antibiotics are indicated depending on the severity
(see international working group on diabetic foot classifications in the table)

Establishing the pathogen
Deep tissue specimens, obtained aseptically at surgery, usually contain only the true pathogens.
Material obtained through curettage (tissue scraping) with a curette or scalpel from *the base of a debrided ulcer* or needle aspirate of purulent secretions are reasonable alternatives.
Swabs of superficial lesions often yield contaminants.
If swabs are the only available method, they should be taken only after debriding and cleaning the wound.

Acute infections in previously untreated patients are usually caused by aerobic gram-positive cocci (often as a monomicrobial infection), but deep or chronic wounds may harbour polymicrobial flora, including gram-negative and anaerobic bacteria

S. aureus and β-haemolytic streptococci are common pathogens for cellulitis or mildly infected ulcers.

Additional pathogens need to be considered as follows:	
Chronic or previously treated ulcers	Enterobacteriaceae
Macerated lesions	Non-enteric gram-negative bacilli such as *P. aeruginosa*
Chronic refractory ulcers, particularly those with gangrene	Enterococci, diphtheroids, anaerobes, and even fungi

Severe or limb threatening infections should be treated with broad spectrum therapy covering MSSA, MRSA, streptococci, Enterobacteriaceae and anaerobes.

Skin and Soft Tissue Infections – Diabetic Foot								
#	**Clinical Diagnosis**	**Etiology**	**Empirical Antibiotic Therapy**					**Remarks**
SSI-11	***Diabetic foot ulcer: non infected*** *No evidence of inflammation except erythema ≤0.5 cm around the ulcer*			No role for antibiotics Address the biomechanical problems causing the ulcer; consider correcting deformities leading to ulceration such as hammer toes or hallux valgus; use appropriate footwear;				
SSI-11	***Diabetic foot infection: mild*** *(Inflammation of skin and subcutaneous tissue, > 0.5 cm but ≤ 2 cm around ulcer, no systemic signs).* *Features of inflammation include erythema, warmth, tenderness, pain or purulence*	*Staphylococcus aureus* *Streptococcus pyogenes*	**I**	Cloxacillin 500-1000mg PO Q6H				• Duration of therapy one to two weeks • Clindamycin resistance is common among streptococci and staphylococci in India and is best avoided.
				Amoxicillin-Clavulanate 625mg PO TDS				
				Cephalexin 500mg PO QID				
				Clindamycin 300mg PO TDS				
			II	Cefazolin 1g IV Q8H				
				Clindamycin 600mg IV Q8H				
SSI-11	***Diabetic foot infection: moderate*** *One of the following:* *1. Inflammation spreading > 2 cm beyond ulcer margin,* *2. Involvement tissues deep to subcutaneous tissue such as involvement of muscle, bone, joint or tendon* *3. Presence of systemic signs – fever, malaise, etc. but without sepsis*	*Polymicrobial (Staph aureus, Group A Strep, aerobic GNB, anaerobes)*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H/Q8H		**AND**	Add Vancomycin 15mg/kg Q12H *(if MRSA is considered)*	• Surgical consultation for drainage or debridement. • Source control is mainstay • Duration; 1 to 2 weeks • Some patients may require extended treatment (up to four weeks). • If the patient underwent resection of the entire area of infection, a brief course of oral antibiotic therapy (less than five days) following surgery is usually sufficient • Consider and evaluate for underlying osteomyelitis
				OR Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H				
SSI-11	***Diabetic foot infection: severe*** *(with features of sepsis)*	*Polymicrobial (Staph aureus, GroupA Strep, aerobic Gram-negative bacilli, anaerobes)*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H/Q8H		**AND**	Consider Amikacin15-20mg/kg IV OD *(particularly if BL/BLI is the primary antibiotic or the patient is in shock)* Add Vancomycin 15mg/kg Q12H *(if MRSA is considered)*	
				OR	Cefoperazone-Sulbactam 2g IV TID-QID or 3g Q12H			
				OR	Meropenem 1-2g IV Q8H			

Skin and Soft Tissue Infections - Bites						
#	**Clinical Diagnosis**	**Etiology**	**Empirical Antibiotic Therapy**			**Remarks**
SSI-12	***Snake Bite Cellulitis***		**I**	Amoxycillin/clavulanate 625mg PO TID or 875mg PO BD or 1.2g IV Q8H		
				OR	Piperacillin-Tazobactam 4.5g IV Q6H/Q8H	
SSI-12	***Animal and human bites***	*Pasteurella multocida* should be covered in cat and dog bites.	**I**	Amoxicillin/clavulanate 625mg PO TDS or 875 mg PO BID or 1.2g IV Q8H		
				OR	Moxifloxacin 400mg PO/IV Q24H (in case of penicillin allergy)	

Skin and Soft Tissue Infections - Necrotizing Fasciitis

Requires emergency surgical evaluation.
Prompt debridement is required.
Antibiotics are only an adjunct.
Any of the following should raise suspicion of necrotizing fasciitis rather than simple cellulitis.

- Pain out of proportion to physical findings
- Severe systemic features such as hypotension or organ dysfunction.
- Bluish, maroon, or black discolouration of the skin
- Skin necrosis
- Haemorrhagic bullae
- Areas of impaired sensation
- Progress of erythema beyond the initially marked margins
- Positive probe test: on inserting a probe through a limited incision, the probe easily reaches down to the deep fascia.
- Features suggesting necrotizing fasciitis (NEJM 2017, 377: 2253):
  [tachycardia, >120 beats/min; hypotension; elevated CK level; CRP, >15 mg/dl; LRINEC score >6]

Type I necrotizing fasciitis (polymicrobial)
Mixed infection of anaerobes with aerobes
Anaerobes, most commonly Bacteroides or Peptostreptococcus with one or more facultative anaerobes such as non-group A streptococci, Enterobacteriaceae.

Usually seen in the in the elderly or in those with underlying illnesses.
Predisposing factors include diabetic or decubitus ulcers, hemorrhoids,
rectal fissures, episiotomies, and colonic or urologic surgery or gynecologic procedures.
Often associated with gas in the tissue and thus is difficult to distinguish from gas gangrene.

Type II necrotizing fasciitis (monomicrobial, Gram positive)
Most common pathogen is Group A beta haemolytic streptococcus (Streptococcus pyogenes). Occasionally, MRSA may cause NF. MSSA has not been associated with NF. Unlike type I infections, Type II infections may occur *in any age group and in persons without any underlying illness.* Gas in the tissue is usually absent in NF due to S. pyogenes or Staph aureus. NF due to *S. pyogenes* is associated muscle necrosis in 20 to 40 % of patients and is preferably called as NSTI (necrotizing soft tissue infection).

Type III necrotizing fasciitis (monomicrobial, Gram negative; Vibrio or Aeromonas)
Vibrio vulnificus
Aeromonas

Skin and Soft Tissue Infections - Necrotizing Fasciitis							
#	**Clinical Diagnosis**	**Etiology**	**Empirical Antibiotic Therapy**				**Remarks**
SSI-13	***Necrotizing fasciitis-Type I*** *(Polymicrobial, mixed anaerobic and aerobic)*	*Anaerobes (most commonly Bacteroides or Peptostreptococcus) with one or more facultative anaerobes (non-group A streptococci, Enterobacteriaceae)*	**I**	Piperacillin-Tazobactam 4.5g IV Q6H **OR** Meropenem 1-2g IV Q8H	**AND**	Clindamycin 600-900mg IV Q8H/Q6H	• Duration of therapy is 2 weeks or longer; needs to b individualized. • Clindamycin should probably be included in all cases of NF (inhibition of exotoxin and M protein synthesis; enhances opsonization and phagocytosis)
SSI-14	***Necrotizing fasciitis-Type II*** *(Monomicrobial, Group A Streptococcus)*	*Gram positive organisms- most common are Streptococcus pyogenes (Group A Streptococcus) and occasionally by MRSA. MSSA has not been associated with NF.*	**I**	Penicillin G 4MU IV Q4H	**AND**	Clindamycin 600-900mg IV Q8H/Q6H	
SSI-15	***Necrotizing fasciitis-Type III*** *(Monomicrobial, Gram negative)*	*Aeromonas hydrophila,* *Fresh water laceration*	**I**	Ceftriaxone 2g IV OD **OR** Ciprofloxacin 500mg PO BD	**AND**	Doxycycline 100mg IV/PO BD	• Aeromonas : Duration: 7 to 10 days for wound infections and 14 days for bacteraemia • Vibrio vulnificus: Duration of therapy 5 to 7 days with oral antibiotic in those with mild illness without underlying disease; longer duration therapy is warranted with Initial IV therapy in those with severe disease or underling disease. Minocycline may be used instead of doxycycline
		Vibrio vulnificus *Sea water laceration*	**I**	Ceftriaxone 2g IV OD	**AND**	Doxycycline 100mg IV/PO BD **OR** Ciprofloxacin 500mg PO BD	

XIII. Bone and Joint Infections

- Hematogenous spread is the most common route in all age groups.
- Arthritis is monoarticular in 80% to 90% of cases. The most common joint affected is the knee in adults. Hip, shoulder, wrist, and ankle are frequently involved. In children, hip infections predominate.
- In adults, fever and malaise are common but high fever with rigors and shaking chills are typically not apparent.
- Polyarticular arthritis is observed in 10% to 20% of patients, especially in those with rheumatoid arthritis, immunosuppression, or prolonged or intense bacteremia, and is usually caused by S. aureus.
- De-escalate once the causative organism is identified.
- Synovial fluid analysis:
    - Synovial fluid WBC count of >50,000 WBC/cu.mm has a sensitivity of 62% and specificity of 92%. A synovial fluid WBC count of >25,000 WBC/cmm has a higher sensitivity of 77% but lower specificity of 73%.
    - Synovial fluid culture in patients who have not previously received antibiotics will yield bacterial growth up to 80% to 90% of the time; however, Gram staining is diagnostic in only 50% of these cases. False-positive Gram stains of synovial fluid can occur because of artefacts from stain, mucin, and cellular debris.

Septic arthritis

George J, 2019, CMC Vellore, 2006 to 2017; 70 adults; 49 had positive synovial culture; among the 49, 38% grew staph and 27% grew streptococci; GNB grew in 31%; 38% of the staph were MRSA and nearly half the GNB were Pseudomonas. History of fever in 75%, leucocytosis only in 66%, gram stain showed organisms in only 42%

Duration: For staph aureus without bacteraemia: at least 14 days of parenteral therapy followed by oral therapy for additional 7 to 14 days.

For staph aureus with bacteraemia: 4 weeks of parenteral therapy

Melioidosis is an important pathogen that causes acute arthritis (monoarthritis or polyarthritis). Often there is evidence of disease elsewhere such as pneumonia, skin soft tissue infection and visceral abscess such as liver abscess and splenic abscess.

Extremity osteomyelitis: Srinivas T, 2011; JIPMER, 2005 to 2010, children 1 to 12 with multifocal osteomyelitis; neonates excluded. Organisms cultured from surgical sample: 22 out of 26 were gram positive cocci (staph aureus 19, streptococcus pyogenes 2; enterococcus 1); 8 out of 26 were GNB (6 enteric GNB, one Acinetobacter, one Pseudomonas); 68% of staph was MRSA.

Duration: If no residual infected bone: at least 2 to 5 days after last debridement if there is no concomitant soft tissue infection at the operative site; 10 to 14 days if there is concomitant soft tissue infection. If there is residual infected bone: continue antibiotic until debrided bone has been covered by vascularized soft tissue (usually takes at least six weeks from the last debridement). Highly bioavailable oral antibiotic may be used after initial parenteral therapy to complete the course.

Vertebral osteomyelitis: S. aureus most common in acute vertebral osteomyelitis but a variety of other organisms may cause.

Consider TB and Brucella in subacute or chronic vertebral osteomyelitis.

Bone and Joint Infections

#	Clin. Diagnosis	Aetiology	Empirical Antibiotic Therapy					Remarks
BJI-01	***Septic Arthritis-Acute Monoarthritis*** *Gram stain should be used to guide empiric therapy*	*<u>Infants (<3 months):</u> Staph aureus, gram negative bacilli, group B streptococcus*	**I**	Vancomycin	**AND**	Meropenem 1g IV Q8H		• See General Principles *Gram stain of the synovial fluid should be used to guide empiric therapy* Gram positive cocci: Vancomycin Gram negative bacilli: Piperacillin tazobactam No organisms: As given under empiric therapy Gram negative diplococci: Ceftriaxone (to cover Neisseria)
						OR	Piperacillin tazobactam 4.5g IV Q6H	
		<u>Children and adults:</u> S. aureus, streptococci (group A, B, C, G, F). aerobic Gm-neg. bacilli.	**I**	Vancomycin	**AND**	Piperacillin tazobactam 4.5g IV Q6H		
						OR	Meropenem 1g IV Q8H	
		Melioidosis has emerged as an important pathogen	**I**	Ceftazidime 2g IV Q8H or Meropenem 1-2g IV Q8H				
BJI-02	***Arthritis-Acute polyarticular arthritis***	*Gonococci, acute rheumatic fever; viruses (chikungunya, hepatitisB, rubella, parvo B19); Staphylococcus and streptococcus.*	Depends on suspected aetiology. For Gonococcus, Ceftriaxone 2g IV OD For melioidosis, see above					
BJI-03	***Chronic Monoarthritis***	*Tuberculosis, Brucella*						
BJI-04	***Prosthetic Joint Infection***	*MSSA, MRSA, CONS, Enterococci, GNB such as E coli, Klebsiella, Pseudomonas, Acinetobacter etc*	Empirical therapy is very difficult; every effort must be made to establish the causative pathogen. It is preferable to wait till culture reports are available. Avoid empiric therapy unless the patient has features of sepsis.					• Therapy should be as per the possible organism.
BJI-05	***Extremity Osteomyelitis***	*Staph aureus, Streptococci, Gram-neg bacilli*	Vancomycin 15 to 20mg/kg IV Q8H (Maximum of 2g per dose)		**AND**	Piperacillin tazobactam 4.5g IV Q6H		• S. aureus is most common • <u>Infants < 4 months</u>: S. aureus, Gm-neg. bacilli, Group B strep. • <u>Infants > 4 months, children, adults</u>: Staph aureus, Group A strep. Gm-neg bacilli
BJI-06	***Vertebral Osteomyelitis +/- epidural abscess***	*S.aureus, others*	Vancomycin 15 to 20mg/kg IV Q8H (Maximum of 2g per dose)		**AND**	Piperacillin tazobactam 4.5g IV Q6H		• See General Principles

XIV. Nervous System Infections

#	Clinical Diagnosis	Aetiology	Empirical Antibiotic Therapy				Remarks
CNS-01	***Acute Bacterial Meningitis*** *(fever, altered sensorium, neck stiffness, CSF showing neutrophilic pleocytosis, low sugar and high protein)*	*Usually caused by pneumococcus meningococcus and Hemophilus influenzae* *Listeria in selected situations*	**I**	Inj. Ceftriaxone 2g IV Q12H **OR** Inj. Cefotaxime 2g IV Q4H Duration of therapy: 7 days for uncomplicated meningococcal meningitis; at least 7 days for Haemophilus meningitis; 10 to 14 days for pneumococcal or staphylococcal meningitis and at least 3 weeks for meningitis with gram negative bacilli.	**AND**	Vancomycin 15 to 20mg/kg IV Q8H (Maximum of 2g per dose) **AND** Ampicillin 2g Q4H to cover Listeria in selected situations	• See treatment of pneumococcal infections @page 8. • Draw blood culture and administer the first dose of antibiotic as early as possible. • If there are no indications for CT, go ahead with LP. • Do CT before LP only if there are clear indications. • Addition of high dose ampicillin in those at risk for Listeria (<1 month, >55 years, pregnancy, alcoholism, and immunosuppression). • Add Inj. Metronidazole 500mg IV Q6H, if coexistent otitis/sinusitis. • Dexamethasone 0.15 mg/kg IV Q6H for 2 to 4 days may be given. First dose of steroid should be given just before or at least along with the first dose of antibiotic. It has been found to be useful primarily in pneumococcal meningitis. *Benefit is less clear in studies from developing countries.*
	Meningitis following neurosurgery	*MSSA, MRSA, CoNS, enteric GNB, Pseudomonas*	**I**	Vancomycin 15 to 20mg/kg IV Q8H (Maximum of 2g per dose)	**AND**	Meropenem 2g IV Q8H	
	Meningitis following penetrating head trauma without basal skull fracture	*Enteric GNB, Pseudomonas, MSSA, MRSA, CoNS,*	**I**	Vancomycin 15 to 20mg/kg IV Q8H (Maximum of 2g per dose)	**AND**	Meropenem 2g IV Q8H	
	Meningitis following basal skull fracture	*Pneumococcus Hemophilus influenza, Streptococcus pyogenes*	**I**	Inj. Ceftriaxone 2g IV Q12H **OR** Inj. Cefotaxime 2g IV Q4H	**AND**	Vancomycin 15 to 20mg/kg IV Q8H (Max of 2 g per dose)	
	Infected EVD or VP shunt	*MSSA, MRSA, CoNS, Cutibacterium acnes, enteric GNB*	**I**	Vancomycin 15 to 20mg/kg IV Q8H (Maximum of 2g per dose)	**AND**	Meropenem 2g IV Q8H	Duration 10 to 14 days; some recommend 21 days for GNB. If repeat cultures are positive, treat for 10 to 14 days following the last positive culture

Nervous System Infections

#	Clinical Diagnosis	Aetiology	Empirical Antibiotic Therapy				Remarks
CNS-02	**Brain Abscess** *(Fever, altered sensorium, focal deficit, raised ICT features)*						• Prompt needle spiration or surgical excision recommended as soon as possible to obtains samples before initiation of antibiotic therapy and to reduce the size of the collection. Send for culture for aerobes, anaerobes, fungi, and *Mycobacterium tuberculosis.* • Prompt drainage is essential except in unencapsulated abscess (cerebritis), <2cm abscess, inaccessible site of lesion or moribund patient • Empiric treatment is guided by suspected source and underlying condition. While therapy should be adjusted based on culture results, anaerobic coverage should ALWAYS be continued even if none are grown. • Use meropenem if ESBL producing Gram negative pathogens such as E. coli or Klebsiella are suspected. • Rule out TB, Nocardia, Aspergillus, Mucor • Duration is for 6 to 8 weeks or longer depending on initial size, causative pathogen and response to therapy. • A minimum of 6–8 weeks of parenteral antibiotic therapy is necessary. Role of additional oral therapy is not well defined. Shorter duration (3 to 4 weeks) is sufficient in those who undergo complete excision of the abscess. • Dexamethasone should not be given routinely. It is usually reserved for patients with substantial peri-abscess oedema and associated mass effect and increased ICP. • Dose of dexamethasone is 10 mg stat followed by 4 mg q 6H. It should be tapered as rapidly as possible; steroid delays encapsulation of the abscess and may increase the risk of rupture of the abscess. • Prophylactic anticonvulsant therapy is indicated and should be continued for at least 3 months after resolution of the abscess; then taper off or, if EEG is abnormal, continue long term
	Associated with cyanotic heart disease:	*Streptococci (esp. viridans)*	**I**	Penicillin G 40MU IV Q4H **OR** Ceftriaxone 2g IV Q12H			
	Associated with dental infection:	*Streptococci, anaerobes*	**I**	Penicillin G 40MU IV Q4H **OR** Ceftriaxone 2g IV Q12H	**AND**	Inj. Metronidazole 7.5mg/Kg (Usually 500mg) IV Q8H/Q6H	
	Associated with lung abscess, empyema, bronchiectasis:	*Streptococci, anaerobes*	**I**	Penicillin G 40MU IV Q4H **OR** Ceftriaxone 2g IV Q12H	**AND**	Inj. Metronidazole 7.5mg/Kg (Usually 500 mg) IV Q8H/Q6H	
	Associated with sinusitis:	*Streptococci, anaerobes, Enterobacteriaceae,* S. aureus	**I**	Ceftriaxone 2g IV Q12H	**AND**	Inj. Metronidazole 7.5mg/Kg (Usually 500 mg) IV Q8H/Q6H	
	Associated with chronic otitis:	*Streptococci, anaerobes, Enterobacteriaceae*	**I**	Ceftriaxone 2g IV Q12H	**AND**	Inj. Metronidazole 7.5mg/Kg (Usually 500 mg) IV Q8H/Q6H	
	Unknown source or bacteraemia or associated with infective endocarditis	*S. aureus, streptococci,* Enterobacteriaceae, anaerobes	**I**	Ceftriaxone 2g IV Q12H	**AND**	Inj. Metronidazole 7.5mg/Kg (Usually 500mg) IV Q8H/Q6H **AND** Vancomycin if MRSA is likely 15 to 20 mg/kg Q8H (single dose not exceeding 2 g)	
	Following penetrating trauma or neurosurgery:	*S. aureus*, CONS, streptococci, Enterobacteriaceae *Pseudomonas spp*	**I**	Vancomycin 15 to 20 mg/kg Q8H (single dose not exceeding 2 g)	**AND**	Meropenem 2g IV Q8H	

Nervous System Infections							
#	**Clinical Diagnosis**	**Aetiology**		**Empirical Antibiotic Therapy**			**Remarks**
CNS-03	***Subdural Empyema*** *Fever, raised ICT features, neck stiffness, seizure, focal deficit*						• Urgent drainage is essential through craniotomy, craniectomy or burr hole. • Duration of antibiotic therapy: minimum of 3 to 4 weeks after drainage of subdural empyema • Patients with associated cranial osteomyelitis may require • longer therapy
	Following sinusitis	*Aerobic and anaerobic streptococci, staphylococci, enteric gram-negative bacilli, anaerobes*	I	Inj. Ceftriaxone 2g IV Q12H	**AND**	Metronidazole 500mg IV Q8H/6H **±** Vancomycin 15 to 20mg/kg IV Q8H (single dose not exceeding 2 g)	
	Following neurosurgery or head injury	*Staphylococci, Gram negative bacilli including Pseudomonas.*	I	Inj. Meropenem 2g IV Q8H	**AND**	Vancomycin 15 to 20mg/kg IV Q8H (single dose not exceeding 2 g)	
CNS-04	***Epidural Abscess*** *Fever, raised ICT features, neck stiffness, seizure, focal deficit*						• Prompt neurosurgical drainage is indicated • Gram's stain and culture of the pus obtained at the time of drainage should be used to adjust initial empirical antibiotic therapy • Duration of therapy is 3 to 6 weeks after the drainage. • Associated osteomyelitis may require longer therapy
	Following sinusitis or otitis	*Aerobic and anaerobic streptococci, other anaerobes such as* Cutibacterium *and* Peptostrepto coccus *species.*	I	Inj. Ceftriaxone 2g IV Q12H	**AND**	Inj. Metronidazole 500mg IV Q8H/Q6H **±** Vancomycin 15 to 20mg/kg IV Q8H (single dose not exceeding 2g)	
	Following neurosurgery	*MSSA, MRSA, Gram negative bacilli including Pseudomonas*	I	Inj. Meropenem 2g IV Q8H	**AND**	Vancomycin 15 to 20mg/kg IV Q8H (single dose not exceeding 2 g)	

XV. Cardiac Infections – Principles of Empirical Therapy for Infective Endocarditis

Empirical antibiotic therapy should be given *only to* patients who are *acutely ill.* Otherwise, ***it is preferable to wait*** for initial culture reports for 48 to 72 hours and draw additional cultures for establishing the causative agent. This waiting period may allow the effect of any prior antibiotics to wear off **allowing better yield in subsequent cultures**

Empirical therapy for acutely ill patients should be given ***only after at least two, preferably three blood cultures*** are obtained from separate venipunctures and ideally spaced over 30 to 60 minutes. Each venipuncture should draw 10 ml and should be injected into BacT/Alert aerobic bottle.

Likely organisms and antibiotic choice depend on the setting (acute vs subacute, NVE vs PVE, community acquired vs health care associated vs nosocomial, individual factors such as IVDU or immunosuppressed and geographic region).

For patients with acute clinical presentation (ie, symptomatic for days), therapy should cover *S. aureus*, beta-hemolytic streptococci, and aerobic gram-negative bacilli.

For patients with a subacute clinical presentation (ie, symptomatic for weeks), therapy should cover *S. aureus*, viridans group streptococci, HACEK organisms, and enterococci.

The three most common pathogens in Indian studies are streptococci, staphylococci and enterococci. Some studies show small but significant contribution from non-HACEK Gram negative pathogens and fungi. However, data are not shown separately for NVE vs PVE or for community acquired IE vs heath care associated IE.

Empirical therapy with penicillin G/ampicillin covers enterococci but does not cover staphylococci. Empirical therapy with ceftriaxone will not cover enterococci but provides modest MSSA coverage. Vancomycin covers MSSA, MRSA as well as enterococci. Coverage for non-HACEK gram negative bacilli is important in health care associated IE and early PVE

For staphylococci, beta lactams are highly preferable to vancomycin if the organism is susceptible and if the patient is not severely allergic. Strongly consider penicillin desensitization for allergic patients.

Use of vancomycin: Vancomycin dose is based on actual body weight. Recommended dose for IE varies; 15 mg Q12H for streptococcal or enterococcal endocarditis; 15–20mg/kg Q8-12H for staphylococcal endocarditis. Single doses should not exceed 2 g. Adjust for trough level of 10–15 µg/mL for streptococcal and enterococcal infections and 15–20 µg/mL for staphylococcal infections.

A loading dose of 20 to 35 mg/kg is recommended in critically ill patients with suspected or documented serious MRSA infections (use actual body weight; should not exceed 3,000 mg).

Use of aminoglycosides: For empirical therapy of infective endocarditis, traditional thrice daily dosing should be used.

For definitive therapy, single daily aminoglycoside dosing may be used for streptococcal infection but not for enterococcal infection.

Gentamicin and streptomycin demonstrate synergism with beta lactams for enterococci. But ***tobramycin and amikacin do not demonstrate such synergism***.

Ideal body weight is used to calculate doses of gentamicin and streptomycin per kilogram (men = 50 kg + 2.3 kg per inch over 5 feet; women = 45.5 kg + 2.3 kg per inch over 5 feet)**.**

Culture negative endocarditis: Prior antibiotic therapy and inadequate microbiologic techniques are major reasons in developing countries.

In the absence of prior antibiotic therapy, it is unlikely that infection due to *S. aureus*, CoNS, enterococci, or Enterobacteriaceae will present with negative blood cultures; thus, in this situation, recommended empirical therapy *need not target* these organisms. Therapy should target fastidious organisms; nutritionally variant bacteria now designated *Granulicatella* and *Abiotrophia* species need to be considered.

Some fastidious organisms occur in characteristic geographic settings (e.g., *C. burnetii* and *Bartonella* species in Europe, *Brucella* species in the Middle East). Non-candida fungi also need to be considered (e.g., Aspergillus in PVE).

Recent evidence suggests that HACEK organisms are not fastidious and can be easily isolated when incubated for at least five days with current automated continuously monitored liquid broth blood culture systems (such as BacT/alert).

IE mimics such as NBTE/marantic endocarditis, atrial myxoma also need to be considered.

Cardiac Infections – Initial Empiric Therapy							
#	**Clinical Diagnosis**	**Aetiology**	**Empirical Antibiotic Therapy**				**Remarks**
CVS-01	***Native Valve Endocarditis (Community acquired –ACUTE)***	*Staph aureus*	I	Vancomycin 15-20mg/kg IV Q8-12H	**+/-**	Gentamicin 1mg/Kg IV Q8H	• See overview of cardiac infections/infective endocarditis
			II	Ceftriaxone 2g IV Q24H	**+/-**	Gentamicin 1mg/Kg IV Q8H	
CVS-02	***Native Valve Endocarditis (Community acquired – SUBACUTE)***	*Streptococcus viridans, other streptococci, Staph aureus, Enterococcus, HACEK,*	I	Ceftriaxone 2g IV Q24H	**+/-**	Gentamicin 1mg/Kg IV Q8H	
			II	Vancomycin 15-20mg/kg Q8-12H	**+/-**	Gentamicin 1mg/Kg IV Q8H	
CVS-03	***Native Valve Endocarditis (Health care associated)***	*Staph aureus including MRSA, GNB*	I	Vancomycin 15-20mg/kg Q8-12H	**AND**	Gentamicin 1mg/Kg IV Q8H	
			II	Vancomycin 15-20mg/kg Q8-12H	**AND**	Meropenem 1-2g IV Q8H	
CVS-04	***Prosthetic Valve Endocarditis- Early (<1 year)***	*CONS, Staph aureus, GNB, Enterococci, Candida*	I	Vancomycin 15-20mg/kg Q8-12H	**AND**	Gentamicin 1mg/Kg IV Q8H as well as Meropenem 1-2g IV Q8H	
CVS-05	***Prosthetic Valve Endocarditis- Late (>1 year)***	*As for Community acquired NVE, CONS*	I	Vancomycin 15-20mg/kg Q8-12H	**AND**	Gentamicin 1mg/Kg IV Q8H	
CVS-06	***Culture Negative Infective Endocarditis***						• See overview • Consult ID

Cardiac Infections – Definitive Therapy

#	Aetiology	Antibiotic Therapy				Remarks
CVS-07	Streptococci-Penicillin Susceptible, *S. gallolyticus* (MIC ≤0.12 µg/mL)	Penicillin G (2–3 MU IV Q4H for 4 weeks)				• Can use ampicillin or amoxicillin (2 g IV q4h) if penicillin is unavailable. • Can use ceftriaxone in patients with non-immediate penicillin allergy. • Use vancomycin in patients with severe or immediate β-lactam allergy. • Avoid 2-week regimen when risk of aminoglycoside toxicity is increased and in prosthetic–valve or complicated endocarditis. • Acute infective endocarditis is often hospital acquired and requires Gram Negative coverage
		OR	Ceftriaxone 2g IV OD for 4 weeks)			
		OR	Vancomycin (15mg/kg IV Q12H for 4 weeks)			
		OR	Penicillin G (2-3 MU IV Q4H) or ceftriaxone (2g IV OD) for 2 weeks	**AND**	Gentamicin (3mg/kg QID IV or IM, as a single dose or divided into equal doses Q8H for 2 weeks)	
CVS-08	Streptococci-Relatively Penicillin Resistant (MIC >0.12 µg/mL and <0.5 µg/mL)	Penicillin G (4 MU IV Q4H) or ceftriaxone (2g IV OD) for 4 weeks		**AND**	Gentamicin (3 mg/kg QID IV or IM, as a single dose or divided into equal doses Q8H for 2 weeks)	• Can use ampicillin or amoxicillin (2 g IV q4h) if penicillin is unavailable. Penicillin alone at this dose for 6 weeks or with gentamicin during the initial 2 weeks is preferred for prosthetic–valve endocarditis caused by streptococci with penicillin MICs of ≤0.1 µg/mL. • Use vancomycin if unable to tolerate penicillins. Ceftriaxone alone or with gentamicin can be used in patients with non-immediate β-lactam allergy.
		OR	Vancomycin (15mg/kg IV Q12H for 4 weeks)			
CVS-09	Streptococci-Moderately Penicillin Resistant; (MIC, ≥0.5 µg/mL and <8 µg/mL); *Granulicatella*, *Abiotrophia*, or *Gemella* spp.	Penicillin G (4-5 MU IV Q4H) or ceftriaxone (2g IV OD) for 6 weeks		**AND**	Gentamicin (3mg/kg QID IV or IM as a single dose or divided into equal doses Q8H for 6 weeks)	• Preferred for PVE caused by streptococci with penicillin MICs of >0.1 µg/mL.
		Vancomycin (15mg/kg IV Q12H for 4 weeks)				

Cardiac Infections – Definitive Therapy *(continued)*

#	Aetiology	Antibiotic Therapy		Remarks
CVS-10	Enterococci	Penicillin G 4-5 MU IV Q4H plus gentamicin 1mg/kg IV Q8H, both x 4–6 weeks		**For Enterococci** • Can treat NVE for 4 weeks if symptoms last <3 months. Treat NVE with >3 months of symptoms and all PVE for 6 weeks. Can abbreviate gentamicin course in some patients. Can use streptomycin (7.5 mg/kg q12h) in lieu of gentamicin if there is not high-level resistance to streptomycin • Can use amoxicillin in lieu of ampicillin (same dose) • Use vancomycin plus gentamicin only for penicillin-allergic patients (preferable to desensitize to penicillin) and for isolates resistant to penicillin/ampicillin. • Use for ampicillin plus ceftriaxone for E. faecalis isolates with or without high-level resistance to gentamicin and streptomycin or for patients at high risk for aminoglycoside nephrotoxicity (creatinine clearance rate <50 mL/min). **For Staphylococci** • Can use penicillin (4 mU q4h) if isolate is penicillin susceptible (i.e., does not produce β-lactamase); 6-week course preferred • Can use cefazolin regimen for patients with non-immediate penicillin allergy; 6-week course preferred • Only use vancomycin for patients with immediate (urticarial) or severe penicillin allergy; see text regarding addition of gentamicin, fusidic acid, or rifampin. A 6-week course is preferred. • No role for routine use of rifampin. Consider high dose daptomycin treatment (see text) for MRSA with vancomycin MIC >1.0 or persistent bacteremia during vancomycin therapy. • Use gentamicin during the initial 2 weeks; determine susceptibility to gentamicin before initiating rifampicin. • if patient is highly allergic to penicillin, use regimen for MRSA; if β-lactam allergy is of the minor non-immediate type, cefazolin can be substituted for cloxacillin, flucloxacillin.
		OR	Ampicillin 2g IV Q4H plus gentamicin 1mg/kg IV Q8H, both x 4–6 weeks	
		OR	Vancomycin 15mg/kg Q12H plus gentamicin 1mg/kg IV Q8H, both x 6 weeks	
		OR	Ampicillin 2g IV Q4H plus ceftriaxone 2g IV Q12H, both x 6 weeks	
CVS-11	MSSA infecting native valves (no foreign devices)	Cloxacillin or flucloxacillin 2 g IV q4h x 4–6 weeks		
		OR	Cefazolin 2 g IV q8h x 4–6 weeks	
		OR	Vancomycin 15 mg/kg Q8-12H x 4–6 weeks	
	MRSA infecting native valves (no foreign devices)	Vancomycin 15mg/kg IV Q8H-Q12H		
		OR	Daptomycin 8-10mg/kg daily x 6 weeks	
CVS-12	MSSA infecting prosthetic valves	Cloxacillin or flucloxacillin 2 g IV q4h x 6–8 weeks		
		AND	Gentamicin 1 mg/kg IM or IV q8h x 2 weeks	
		AND	Rifampicin 300 mg PO q8h x 6–8 weeks	
	MRSA infecting prosthetic valves	Vancomycin 15mg/kg IV Q12H x 6-8 weeks		
		AND	Gentamicin 1g/kg IM/IV Q8H x 2 weeks	
		AND	Rifampicin 300mg PO Q8H x 6-8 weeks	
CVS-13	HACEK Organisms	Ceftriaxone 2 g/d IV as a single dose x 4 weeks		• Can use another third-generation cephalosporin at comparable dose. • If the isolate is susceptible, ciprofloxacin (400 mg IV q12h) can be used.
		OR	Ampicillin/sulbactam 3 g IV q6h x 4 weeks	

XVI. Duration of Antimicrobial Therapy

Duration of Antibiotic therapy	
Infectious syndrome	**Duration of therapy (conditions apply)**
Gram negative bacteraemia, *without complicating factors, does not apply for CRBSI*	7 days (see conditions)[1]
Staphylococcal aureus bacteraemia with low risk of complications	2 weeks
Staphylococcal aureus bacteraemia with high risk of complications	4 to 6 weeks
Community acquired pneumonia	5 to 7 days
Hospital acquired pneumonia	7 days
Ventilator associated pneumonia	7 days
Cystitis Depends on gender and class of antibiotic, longer for males Females: FQ 3 days, cotrimoxazole 3 days, beta lactams 5 to 7 days Males: FQ 5 days, cotrimoxazole 7 days, beta lactams 7 days FQ = fluoroquinolone	3 to 7 days
Pyelonephritis (fluoroquinolones: 5 to 7 days, cotrimoxazole, beta lactams:7 to 10 days)	5 to 10 days
Catheter associated UTI	7 to 10 days
Skin and Soft tissue infections	5 days
Secondary peritonitis, with source control	4 to 7 days
Surgical antibiotic prophylaxis	Single dose
Meningitis, uncomplicated	10 to 14 days
Meningitis, uncomplicated (pathogen identified):	
Meningococcus and H. influenzae	7 days
Pneumococcus	10 to 14 days
Group B Streptococcus	3 weeks
Listeria	3 weeks
Gram Negative bacteria	3 weeks

For patients with uncomplicated Enterobacterales bacteraemia who show clinical improvement in 48 to 72 hours and who do not have any of the complicating factors (endovascular, bone, joint, or CNS infection, uncontrolled source of infection, major immunocompromising condition), a 7-day course is sufficient.

Oral switch may be considered in select patients who have defervesced and have remained afebrile for 48 hours. The oral agent should have excellent bioavailability, and the isolate should be fully susceptible to the agent. Fluroquinolones are an excellent choice for susceptible pathogens.

XVII. Antibiotic Dosing for selected antimicrobials

1. Polymyxins dosing

There are two polymyxin antibiotics: Colistin also called as Polymyxin E, and Polymyxin B.

Generally, both the polymyxins show poor penetration into the lungs, pleura, bones, and central nervous system. They are not effective against many Gram-negative pathogens.

Susceptible organisms:

Escherichia coli, Klebsiella pneumoniae, some Enterobacter spp, Pseudomonas aeruginosa, and Acinetobacter baumannii. Other susceptible organisms include Haemophilus influenzae, Bordetella pertussis, Legionella pneumophila, Salmonella spp, Shigella spp, and the majority of Stenotrophomonas maltophilia strains. However, acquried resistance in these organisms may lead to resistance to polymyxins

Resistant organisms:

The following organisms are **intrinsically resistant** to polymyxins. Proteus spp, Providencia spp, Serratia marcescens, Morganella morganii, Burkholderia cepacia, Moraxella catarrhalis are all intrinsically resistant to polymyxins. Organisms with variable resistance include Aeromonas, Vibrio, Prevotella, and Fusobacterium spp

Polymyxin B is generally preferred over colistin for all indications with the following exceptions.

- Urinary tract infections
- Adjunctive inhalation therapy for pneumonia caused by MDR gram-negative bacilli.

For all indications other than UTI, Polymyxin B is preferred over colistin because of equivalent efficacy, faster attainment of target serum concentrations, less inter-patient variability in pharmacokinetics, no need for dose adjustment in renal impairment and lower risk of renal toxicity. Polymyxin B achieves adequate plasma levels more rapidly and reliably than colistin. Polymyxin B is administered as the active agent, whereas colistin is a prodrug that has to be converted to the active agent.

For UTI, colistin is preferred over polymyxin B; colistin achieves high urinary concentrations but polymyxin urinary concentrations are low.

For both the polymyxins, treatment of CNS infection requires adjunctive intrathecal therapy; administration by nebulization should be considered for treatment of pneumonia

	Polymyxin B	Colistin
Loading dose	20,000 to 25,000 units/kg	1,20,000 IU per kg; maximal is 9 million units
Maintenance dose	12,500 to 15,000 units/kg Q12H	4.5 million units Q12H (but see weight-based dosing table; maximal dose per day is 11 million units)
Dosing in obese individuals (Body weight > 30% above ideal body weight)	Use adjusted body weight ($ABW_{0.4}$) *	Use ideal body weight (IBW)*
Dosing in renal impairment/RRT	No adjustment required	Adjustment required
Conversion	1 mg = 10,000 units	1 mg CBA = 2.4 mg CMS = 30,000 IU CMS

CBA: colistin base activity; CMS: colistin methane sulfonate; IBW: Ideal Body Weight ; ABW: Adjusted Body Weight. $ABW_{0.4}$= ideal body weight + 40% of (total body weight – ideal body weight).
*Reference: Pharmacotherapy 2017;37(11):1415–1431.

Dosing in renal impairment		
	Daily dose of CMS for plasma colistin $C_{ss,avg}$ of 2 mg/ L[c]	
Creatinine clearance mL/minute[b]	mg CBA/day	Million IU/day
0	130	3.95
5 to < 10	145	4.40
10 to < 20	160	4.85
20 to < 30	175	5.30
30 to < 40	195	5.90
40 to < 50	220	6.65
50 to < 60	245	7.40
60 to < 70	275	8.35
70 to < 80	300	9.00
80 to < 90	340	10.30
≥ 90	360	10.90

CBA= colistin base activity; CMS= colistin methanesulfonate; $C_{ss,avg}$ = average steady-state plasma concentration;
[a]To achieve a desired target plasma colistin $C_{ss,avg}$ of 2 mg/L for patients with narrow windows of creatinine clearance. Reproduced from reference 6 with minor modifications.
[b]Adjusted body weight should be used to estimate creatinine clearance.
[c]Daily dose administered in two divided dosed 12 hours apart.

Creatinine clearance calculated using Cockcroft and Gault formula. Use adjusted body weight for calculating creatinine clearance. $ABW_{0.4}$= ideal body weight + 40% of (total body weight – ideal body weight)

Patients on intermittent HD

On a non-dialysis day, administer a CMS dose of 130 mg CBA/day (~3.95 million IU/day). On a dialysis day, administer a supplemental dose of 1.2 million IU or 1.6 million IU for a 3- or 4-hour IHD session, respectively. If possible, the supplement to the baseline (non-dialysis) daily dose should be administered with the next regular dose, after the dialysis session has ended. Conduct IHD sessions as late as possible within a CMS dosage interval to minimize the amount of colistin and CMS lost to the extracorporeal system.

CRRT

Add 10% (of 130 mg) per hour of CRRT to the baseline daily dose of 130 mg. Thus, after 24 hrs the suggested daily dose is 442 mg (divided q12h).

SLED

Add 10% (of 130 mg) per hour of SLED to the the baseline daily dose of 130 mg. Example: the daily dose for a patient receiving 10 hours of SLED would be 260 mg (divided q12h).

Inhalation therapy:

As per International Consensus Guidelines for the Optimal Use of the Polymyxins (Tsuji et all 2019). It is recommended that patients requiring IV polymyxin therapy for suspected or documented XDR gram-negative HAP or VAP should receive adjunctive polymyxin aerosol therapy (weak recommendation, low-quality evidence). It is recommended that for polymyxin aerosol therapy, either colistin or polymyxin B is appropriate (weak recommendation, very low-quality evidence)

Colistin:

1.5 to 2.25 million IU or 50-75 mg CBA in 3-4 mL saline via vibrating mesh nebulizer 2-3 times/day (As per Sanford). 2.25 to 4.5 million IU or 75 to 150 mg CBA Q12H, every 12 hours; doses up to 167 mg every 8 hours have been described (As per UpToDate)

Polymyxin B

Dose not well defined. 5,00,000 units (0.5 million IU) or 50mg, every 12 hours (UpToDate). Studies have used doses up to 25,000 IU/kg or 2.5 mg/kg per day per day for inhalation. Ex vivo PK/PD studies indicate that 41200 IU per kg (4.12 mg/kg) Q8H is the optimal dose to achieve the recommended epithelial lining fluid concentrations at 20 µ/ml 8 h after nebulisation. The optimal dose for 70 kg person should be 3 million IU or 300 mg per dose (Rouby et al (2022))

Optimizing ventilator settings for nebulisation (Rouby et al (2022))

Ventilator settings should be optimized during the nebulization phase to reduce extrapulmonary deposition: Controlled ventilation with 40 l/min constant inspiratory flow (avoid pressure support and spontaneous breathing), respiratory frequency 12–15/min, I/E=1:2, inspiratory pause 20%, positive-end expiratory pressure 5–10 cmH_2O; asynchronies between the patient and the ventilator should be avoided by administering a short propofol sedation. Turning off the conventional humidifier is not mandatory. However, using a dry inspiratory circuit (HME filter instead of conventional humidifier) for each nebulisation may increase the lung delivery.

Intrathecal and intraventricular therapy for meningitis and ventriculitis (susceptible gram-negative organisms; adjunct to systemic therapy)

Use preservative free solution of the drug

When an intraventricular dose is administered via a ventricular drain, clamp drain for 15 to 60 minutes after administration. Polymyxin B: 50,000 IU once daily, in combination with systemic therapy. Colistin: 4.2 mg CBA/day = 10 mg CMS/day = 1,25,000 IU/day, once daily, in combination with systemic therapy

2. Aminoglycoside dosing

Which body weight to is to be used for aminoglycoside dose calculation: actual body weight, ideal body weight or adjusted body weight?
It depends on the body weight category.
If the patient is underweight, normal weight or only moderately overweight, use actual body weight.
But, if the actual body weight (TBW) exceeds ideal body weight by more than 25 % (that is TBW > 1.25 times the IBW), use adjusted body weight.
Note: actual body weight is also known as total body weight or TBW

Adjusted Body Weight (ABW)
ABW = IBW + 0.4 (TBW minus IBW)
Adjusted body weight is obtained by adding 40% of the excess body weight to ideal body weight.
Excess body weight is TBW minus IBW.
ABW is equal to IBW plus 40% of the excess weight.

Ideal Body Weight (IBW) formula:
Males: IBW = 50 kg + 2.3 kg for each inch over 5 feet.
Females: IBW = 45.5 kg + 2.3 kg for each inch over 5 feet.
(This formula can be used only for those who are 5 feet or more in height)

Administration
Intravenous administration of aminoglycosides should occur over at least 30 minutes for traditional intermittent and at least 60 minutes for extended-interval dosing. Because beta-lactams inactivate aminoglycosides in vitro, they should not be mixed in the same solution.

What are the two approaches to aminoglycoside doing? (using amikacin for example)
Traditional approach (also called multiple daily dose approach).
The total daily dose of 15 to 20 mg/kg is given as two or three divided doses (5mg/kg Q8H or 7.5mg/kg Q12H)
Once daily approach (also called extended interval dosing approach)
The total daily dose of 15 to 20 mg/kg is given as a single dose.

Which approach is preferred?
Aminoglycosides are concentration dependent antibiotics. Their efficacy is related to the Cmax/MIC ratio. Once daily approach achieves higher peak concentration (Cmax) and hence higher Cmax/MIC ratio. In addition, aminoglycosides have a prolonged post-antibiotic effect; the bactericidal effect persists for a long period even after the plasma level has fallen below the MIC, thus allowing longer intervals.
Once daily approach has been shown to be possibly more effective and possibly less toxic. So once daily approach is the preferred approach in most situations. However, this approach is not well recommended in certain indications and certain populations.

Once daily approach should be avoided in whom?
Children: Use of once daily approach is generally not recommended in children younger than three months of age because of decreases in drug clearance at this age.
Adults: Once daily approach is not recommended in patients with ascites, burns covering >20% of their body surface area, cystic fibrosis, endocarditis.
Pregnancy: Once daily approach is generally not recommended in pregnancy
Renal impairment: Avoid this approach in those with advanced renal impairment as well (creatinine clearance less than 50)

Dose adjustment in renal failure

For once daily approach

CrCl (mL/min)	Initial dosing interval
≥50	15mg/kg Q24H
< 50	Follow traditional dosing approach as per the table below

For traditional approach

CrCl (mL/min)	Initial dosing interval
CrCl ≥ 50	7.5 mg/kg q12h
CrCl 10 - 49	7.5 mg/kg q24h
CrCl < 10	7.5 mg/kg q48h
Haemodialysis	7.5 mg/kg q48h. In addition, give an extra dose of 3.75 mg/kg after dialysis (AD) If the q48h dose is due on a dialysis day, give the 7.5 mg/kg dose before dialysis and the extra 3.75 mg/kg after dialysis. Periodically check peak and trough serum levels.

3. Piperacillin-Tazobactam dosing

Adult doses are expressed as the combined amount of piperacillin and tazobactam.

Dose of Piperacillin- Tazobactam

The usual adult daily dose of cefoperazone is 13.5 to 18 grams per day administered in equally divided doses every 6 to 8 hours. Traditionally each dose is given over 30 mins. In severe infections or infections caused by less sensitive organisms, the total daily dose and/or frequency may be increased.

Daily dosing

- Mild to moderate infection: 13.5 g per day, in 4 divided doses
- Severe infection (Maximal daily dose) : 18 g per day, usually in 3 divided doses, in case of Pseudomonas aeruginosa infection, given as 4 divided doses

Piperacillin Tazobactam product availability:

Piperacillin Tazobactam combination is available in two dosage formulations.

- 4.5-gram Piperacillin Tazobactam, containing 4-gram Piperacillin and 500 mg tazobactam
- 2.25-gram Piperacillin Tazobactam, containing 2-gram Piperacillin and 250 mg tazobactam

Dosing in renal impairment

Piperacillin Tazobactam requires dose adjustment in renal failure:

CrCl (mL/minute)	Total dose per day	If the recommended dose is 3.375 g every 6 hours	If the recommended dose is 4.5 g every 6 hours
100 to <130	13.5 to 18 g	3.375 infused over 4 hours every 6 hours	4.5 g infused over 4 hours every 6 hours
40 to <100	13.5 to 18 g	3.375 g every 6 hours	4.5 g every 6 hours
20 to <40	9 to 13.5g	2.25 g every 6 hours	4.5 g every 8 hours **or** 3.375 g every 6 hours
<20	6.75 to 9g	2.25 g every 8 hours	4.5 g every 12 hours **or** 2.25 g every 6 hours

Dosing in liver disease or biliary obstruction

No dosage adjustment necessary.

Pharmacokinetics

Piperacillin (in combination with tazobactam) is excreted mainly in the urine. Peak concentrations are achieved in the serum immediately following completion of 30-minute infusion.

Both piperacillin and tazobactam are approximately **30% bound** to plasma proteins. The protein binding of either piperacillin or tazobactam is unaffected by the presence of the other compound. Protein binding of the tazobactam metabolite is negligible. Piperacillin and tazobactam are widely distributed into tissues and body fluids including intestinal mucosa, gallbladder, lung, female

reproductive tissues (uterus, ovary, and fallopian tube), interstitial fluid, and bile. Distribution of piperacillin and tazobactam into cerebrospinal fluid is low in subjects with non-inflamed meninges, as with other penicillin.

Piperacillin is metabolized to a minor microbiologically active desethyl metabolite. Tazobactam is metabolized to a single metabolite that lacks pharmacological and antibacterial activities.

Following single or multiple piperacillin and tazobactam for injection doses to healthy subjects, the plasma half-life of piperacillin and of tazobactam ranges from 0.7 to 1.2 hours and is unaffected by dose or duration of infusion. Both piperacillin and tazobactam are eliminated via the kidney by glomerular filtration and tubular secretion. Piperacillin is excreted rapidly as unchanged drug with **68%** of the administered dose excreted in the urine. Tazobactam and its metabolite are eliminated primarily by renal excretion with **80%** of the administered dose excreted as unchanged drug and the remainder as the single metabolite. Piperacillin, tazobactam and desethyl piperacillin are also secreted into the bile.

After the administration of single doses of piperacillin and tazobactam to subjects with renal impairment, the half-life of piperacillin and of tazobactam increases with decreasing creatinine clearance. Dosage adjustments for piperacillin and tazobactam for injection are recommended when creatinine clearance is below 40 mL/min in patients receiving the usual recommended daily dose of piperacillin and tazobactam for injection. Hemodialysis removes 30% to 40% of a piperacillin and tazobactam dose with an additional 5% of the tazobactam dose removed as the tazobactam metabolite. Peritoneal dialysis removes approximately 6% and 21% of the piperacillin and tazobactam doses, respectively, with up to 16% of the tazobactam dose removed as the tazobactam metabolite.

The half-life of piperacillin and of tazobactam increases by approximately 25% and 18%, respectively, in patients with hepatic cirrhosis compared to healthy subjects. However, this difference does not warrant dosage adjustment of piperacillin and tazobactam for injection due to hepatic cirrhosis.

Caution

- Probenecid prolongs the half-lives of piperacillin and tazobactam and should not be co-administered with piperacillin and tazobactam for injection unless the benefit outweighs the risk.
- Co-administration of piperacillin and tazobactam with vancomycin may increase the incidence of acute kidney injury. Monitor kidney function in patients receiving piperacillin and tazobactam and vancomycin.

4. Cefoperazone sulbactam (CPZ-SBT) dosing

Dose of cefoperazone (as per package insert of Pfizer)

The usual adult daily dose of cefoperazone is 2 to 4 grams per day administered in equally divided doses every 12 hours. In severe infections or infections caused by less sensitive organisms, the total daily dose and/or frequency may be increased. Patients have been successfully treated with a total daily dosage of 6–12 grams divided into 2, 3 or 4 administrations ranging from 1.5 to 4 grams per dose.

Usual daily dose is 2 to 4 g per day. Maximal daily dose is 6 to 12 g per day. Higher doses should be used in serious infections and infections with relatively resistant organisms (higher dose should be avoided in liver disease or biliary obstruction).

Dose of sulbactam

Usual dose of sulbactam is 2 to 4 gram per day. Doses up to 12 g per day have been used in Acinetobacter infections.

Cefoperazone sulbactam (CPZ-SBT) product availability

Cefoperazone sulbactam combination is available in two ratios:

1. 1:1 ratio (Example: Magnex 2g with 1 g each of cefoperazone and sulbactam and Magnex 1g with 500 mg each of cefoperazone and sulbactam)
2. 2:1 ratio (Example: Magnex 3g with cefoperazone 2g and sulbactam 1g and Magnex 1.5 g with cefoperazone 1g and sulbactam 500 mg)

Dosing in renal impairment

Cefoperazone does not require dose adjustment in renal failure. However, sulbactam requires dose adjustment in renal failure.

Sulbactam dose in renal failure

CrCl	Dose
> 30	2 to 4 g per day (up to 12 g per day)
15 to 30	2g per day (1g Q12H)
< 15	1g per day (500 mg Q12H)

The dose of sulbactam should not exceed 2g for creatinine clearance of 15 to 30 and 1g for for creatinine clearance less than 15.

Adjusting the dose of the combination for varying intended doses of cefoperazone

Desired CPZ dose per day	Desired SBT dose per day	Using 1:1 ratio product	Using 2:1 ratio product
2g	2 to 4g	CPZ-SBT 2g Q12H	
3g	2 to 4g	CPZ-SBT 2g Q8H	
4g	2 to 4g	CPZ-SBT 2g Q6H	CPZ-SBT 3g Q12H
6g	2 to 4g	CPZ-SBT 2g Q6H with plain CPZ 1g Q12H	CPZ-SBT 3g Q12H with plain CPZ 1g Q12H
8g	2 to 4g	CPZ-SBT 2g Q6H with plain CPZ 1g Q6H	CPZ-SBT 3g Q6H

If you want to give higher doses of sulbactam (up to 12 g per day) the dose may be adjusted using the above approach.

Dosing in liver disease or biliary obstruction

Cefoperazone is extensively excreted in bile. The serum half-life of cefoperazone is increased 2–4 fold in patients with hepatic disease and/or biliary obstruction. In general, total daily dosage above 4 g should not be necessary in such patients. If higher dosages are used, serum concentrations should be monitored.

Dosing in renal impairment

How will you adjust the dose of CFP-SBT in renal impairment for varying intended doses of cefoperazone?

Cr. clearance	Desired CPZ dose per day	Maximal SBT per day	Using 1:1 ratio product	Using 2:1 ratio product
CrCl: 15 to 30	2g	2 g	CPZ-SBT 2g Q12H	
	4g	2 g	CPZ-SBT 2g Q12H with plain CPZ 1g Q12H	CPZ-SBT 3g Q12H
	6g	2 g	CPZ-SBT 2g Q12H with plain CPZ 2g Q12H	CPZ-SBT 3g Q12H with plain CPZ 1g Q12H
	8g	2 g	CPZ-SBT 2g Q12H with plain CPZ 3g Q12H	CPZ-SBT 3g Q12H with plain CPZ 2g Q12H
CrCl: < 15	2g	1 g	CPZ-SBT 1g Q12H with plain CPZ 0.5g Q12H	CPZ-SBT 1.5 g Q12H
	4g	1 g	CPZ-SBT 1g Q12H with plain CPZ 1.5g Q12H	CPZ-SBT 1.5g Q12H with plain CPZ 1g Q12H
	6g	1 g	CPZ-SBT 1g Q12H with plain CPZ 2.5g Q12H	CPZ-SBT 1.5g Q12H with plain CPZ 2g Q12H

Dosing in combined hepatic dysfunction and significant renal disease,

In patients with combined hepatic dysfunction and significant renal disease, the dosage of cefaperazone should not exceed 1–2g daily without close monitoring of serum concentrations.

Pharmacokinetics

Cefoperazone is excreted mainly in the bile. Maximum bile concentrations are generally obtained between one and three hours following drug administration and exceed concurrent serum concentrations by up to 100 times.

Following a single intramuscular or intravenous dose, the urinary recovery of cefoperazone over a 12-hour period averages **20–30%.** No significant quantity of metabolites has been found in the urine. Repeated administration of cefoperazone at 12-hour intervals does not result in accumulation of the drug in normal subjects. Peak serum concentrations, areas under the curve (AUC's), and serum half-lives in patients with severe renal insufficiency are not significantly different from those in normal volunteers.

In patients with hepatic dysfunction, the serum half-life is prolonged and urinary excretion is increased. In patients with combined renal and hepatic insufficiencies, cefoperazone may accumulate in the serum.

Cefoperazone is extensively excreted in bile. The serum half-life of cefoperazone is increased 2–4 fold in patients with hepatic disease and/or biliary obstruction. In general, total daily dosage above 4g should not be necessary in such patients. If higher dosages are used, serum concentrations should be monitored.

Because renal excretion is not the main route of elimination of cefoperazone, patients with renal failure require no adjustment in dosage when usual doses are administered. When high doses of cefoperazone are used, concentrations of drug in the serum should be monitored periodically. If evidence of accumulation exists, dosage should be decreased accordingly.

In patients with both hepatic dysfunction and significant renal disease, cefoperazone dosage should not exceed 1–2 g daily without close monitoring of serum concentrations.

Caution

CPZ-SBT therapy may prolong INR. Monitoring is indicated when high doses are used or when there is liver disease or biliary obstruction or renal impairment or both.

5. Vancomycin dosing

Loading dose
(Loading dose is recommended only for patients with known or suspected severe Staphylococcus aureus infection)
Severe S. aureus infections include (but are not limited to) bacteraemia, endocarditis, osteomyelitis, prosthetic joint infection, pneumonia warranting hospitalization, or infection causing critical illness.

Loading dose	Load 20 to 35 mg/kg (rounded to the nearest 250 mg increment; not to exceed 3000 mg). Within this range, use higher per kg dose in critically ill and a lower per kg dose in the following: age ≥60, patients with risk factors for acute kidney injury. Use actual body weight for dose calculation; in the obese patients alone, use adjusted body weight)

Maintenance dose
If facility for vancomycin drug levels is not available:
Use a maintenance dose of 15 mg/kg Q12H for non-serious infections and 15mg/kg 8H for serious infections.

If facility for vancomycin drug levels is available:

Initial maintenance dose and interval	Typically, 15 to 20 mg/kg every 8 to 12 hours for most patients (based on actual body weight, rounded to the nearest 250 mg increment).
Subsequent maintenance dose and interval adjustments	Based on AUC-guided (preferred for severe infection) or trough-guided serum concentration monitoring.

Use of vancomycin plasma levels to guide therapy
Traditionally vancomycin dose has been guided by trough levels. A target serum trough concentration of 15 to 20 mcg/mL is warranted for patients with severe or deep-seated infection (such as bacteraemia, endocarditis, osteomyelitis, prosthetic joint infection, pneumonia warranting hospitalization, infection involving the central nervous system, or infection causing critical illness). A target trough of 10 to 15 mcg/mL is warranted for patients with non-severe infection (such as soft tissue infection). However, AUC/MIC ratio guided dosing is now preferred. Recommended AUC/MIC ratio target is 400–600 mg h/L.

Which body weight to is to be used for vancomycin dose calculation: actual body weight, ideal body weight or adjusted body weight?
It depends on the body weight category. If the patient is underweight, normal weight or only moderately overweight, use actual body weight. But, if the actual body weight (TBW) exceeds ideal body weight by more than 25 % (that is TBW > 1.25 times the IBW), use adjusted body weight.
Note: actual body weight is also known as total body weight or TBW

Adjusted Body Weight (ABW)
ABW = IBW + 0.4 (TBW minus IBW)
Adjusted body weight is obtained by adding 40% of the excess body weight to ideal body weight.
Excess body weight is TBW minus IBW.
ABW is equal to IBW plus 40% of the excess weight.
Ideal Body Weight (IBW) formula
Males: IBW = 50 kg + 2.3 kg for each inch over 5 feet.

Females: IBW = 45.5 kg + 2.3 kg for each inch over 5 feet.
(This formula can be used only for those who are 5 feet or more in height)

Importance of accurate dosing
Vancomycin has a narrow therapeutic window. Underdosage increases the probability of clinical failure, and overdosage increases the risk of nephrotoxicity.

Concurrent treatment with aminoglycosides or piperacillin-tazobactam further potentiates the risk of acute kidney injury.

Infusion concentration and infusion rate
Vancomycin should be diluted to a concentration of 5 mg/mL or less; concentrations up to 10mg/mL may be used if there is a need to restrict fluids. The rate of infusion should not exceed 10 to 15 mg/minute (e.g., 500 mg over at least 1/2 h; 1000 mg over at least 1 hour, etc.).
For patients who develop infusion reaction (histamine-mediated flushing during or immediately following infusion of vancomycin), a slower infusion rate or reduction in dose is warranted.

Vancomycin dose adjustment in renal failure
(Based on UpToDate, John Hopkins Guide and Sanford Guide). Guidelines recommend that dosing should be guided by vancomycin levels.

CrCl (mL/min)	Dosing interval
CrCl > 100	15-20 mg/kg q8-12h
CrCl 50 to 100	15-20 mg/kg q12h
CrCl 15 to 49*	10-20 mg/kg q24h
CrCl < 15*	10-20 mg/kg q48h

* Lower the clearance within this range, use lower per kg dose for that range

Dosing in Haemodialysis

15 - 20 mg/kg, then re-dose based on serum concentrations. Most individuals will require 5 - 10 mg/kg IV post-dialysis to maintain serum concentrations within goal range. Higher doses may be required in patients with residual renal function.

XVIII. Surgical Antibiotic Prophylaxis

This annexure is based on the following practice guidelines but adapted as per antimicrobial susceptibility patterns seen in India.

Clinical practice guidelines for antimicrobial prophylaxis in surgery. Surg Infect (Larchmt). 2013 Feb;14(1):73-156. doi: 10.1089/sur.2013.9999.

Endorsement:

This practice guideline is ***endorsed*** by the following organizations

Infectious Diseases Society of America (IDSA)

Surgical Infection Society (SIS)

American Society of Health-System Pharmacists (ASHP)

Society for Healthcare Epidemiology of America (SHEA).

Purpose of surgical antibiotic prophylaxis:

The purpose is to prevent the development of infection at the surgical site (surgical site infection).

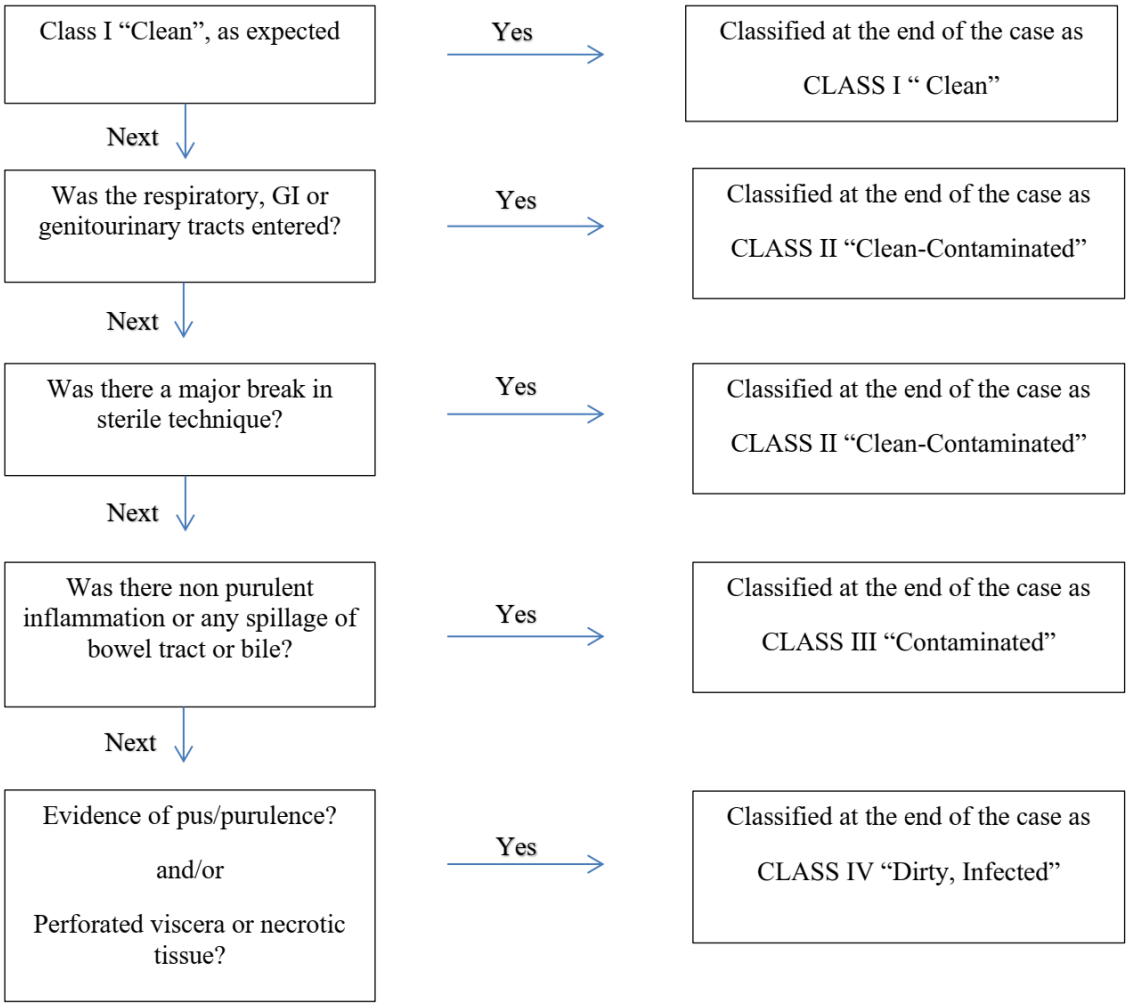
Wound classification:

Clean wounds are uninfected operative wounds in which no inflammation is encountered and the wound is closed primarily. By definition, a viscus (respiratory, alimentary, genital, or urinary tract) is not entered during a clean procedure.
Clean-contaminated wounds are operative wounds in which a viscus (respiratory, alimentary, genital, or urinary tract) is entered under controlled conditions and without unusual contamination.
Contaminated wounds are open, fresh accidental wounds, operations with major breaks in sterile technique, or gross spillage from a viscus. Wounds in which acute, nonpurulent inflammation was encountered also were included in this category.
Dirty wounds are old traumatic wounds with retained devitalized tissue, foreign bodies, or fecal contamination or wounds that involve existing clinical infection or perforated viscus.

Several studies have found a moderate correlation between the wound classification and the SSI rate. SSI rates according to wound class were:

Clean	1.3 to 2.9 %
Clean-contaminated	2.4 to 7.7 %
Contaminated	6.4 to 15.2 %
Dirty	7.1 to 40.0 %

End of procedure assessment of wound class



Pathogens causing surgical site infection:

The predominant organisms causing SSIs after clean procedures are skin flora, including *S. aureus* and coagulase-negative staphylococci (e.g., *Staphylococcus epidermidis*). In clean-contaminated procedures, including abdominal procedures and heart, kidney, and liver transplantations, the predominant organisms include gram-negative rods and enterococci in addition to skin flora.

Choice and the timing of the antibiotic:

Successful prophylaxis requires that

1. The antibiotic chosen should cover the pathogens likely to cause surgical site infection
2. The timing of administration of the antibiotic should ensure that ***serum and tissue concentrations*** of the antibiotic exceeds the MIC of organisms ***at the time of incision*** and the levels should be maintained above MIC ***throughout the duration of the procedure.*** There is no need to administer the antibiotic once the surgery has been completed.

Timing of the antibiotic:

For antibiotics that are given as IV push over 3 to 5 minutes or as short infusion, the infusion should be started ***not earlier than 60 minutes*** of expected time of surgical incision. The antibiotic administration infusion should be completed before the surgical incision to ensure that therapeutic levels of the antibiotic is present in the tissues at the time of skin incision. For antibiotics requiring prolonged infusion over 1 to 2 hours, such as vancomycin or fluoroquinolones, the infusion should be started ***not earlier than 120 minutes*** prior to the expected time of surgical incision. These drugs have long half-lives, and they are likely to sustain the levels above MIC for the period of surgery despite starting the infusions earlier.

Redosing of the antibiotic:

Redosing required depending on the duration of the surgery and the half-life to the drug chosen. Redosing is also required in the event of major intraoperative blood loss. Redosing may also be warranted if there are factors that shorten the half-life of the antimicrobial agent (e.g., extensive burns). The redosing interval should be calculated from the time of administration of the first dose and not from the time of surgical incision and not from the beginning of the procedure.

For antimicrobials with a short half-life (redosing in the operating room is recommended at an interval of approximately two times the half-life of the agent in patients with normal renal function. For example, cefazolin has a half-life of 2 hours, and it should be given every 4 hours, piperacillin tazobactam has a half-life of 1 hour and should be given every 2 hours.

No further doses are required once the surgery is completed.

Antibiotics with long half-life such as vancomycin do not need redosing.

Despite short half-life, aminoglycosides do not need redosing, because they have prolonged post-antibiotic effect.

In the event of major intraoperative blood loss additional dosage of prophylactic antibiotic should be considered after fluid replacement. (>1,500 ml in adults, children >25ml/kg in children is considered major blood loss). The dose should be repeated after for every 1500 ml blood loss (after fluid replacement)

Duration of antibiotic prophylaxis:

Surgical antibiotic prophylaxis should be generally limited to the duration of the surgery. At any cost, it should not be continued beyond 24 hours. Pre-operative antibiotic should be continued for several days only in case of *dirty or infected wounds*. However, this is treatment of infection and not prophylaxis. The practice of continuing antimicrobial prophylaxis until all invasive lines, drains, and indwelling catheters are removed *cannot be supported* due to concerns regarding the development of drug-resistant organisms, superinfections, and drug toxicity

Infusion times for antibiotics:

β-lactams (cephalosporins, ampicillin/sulbactam, piperacillin tazobactam): IV push over 3-5 minutes; all reach adequate skin levels within minutes--hence can be given just before surgery.

Vancomycin: infuse over 60 minutes; start at least 30 minutes before incision

Ciprofloxacin: infuse over 60 minutes

Clindamycin: infuse over 10-20 minutes.

Metronidazole: infuse over 10 minutes.

Antibiotics commonly used in surgical antibiotic prophylaxis

Recommended Doses and Redosing Intervals for Commonly Used Antimicrobials for Surgical Prophylaxis Adapted from ASHP/IDSA/SIS/SHEA 2013 Guideline				
Antimicrobial *(alphabetical order)*	**Adults**	**Pediatrics**[a]	**Half-life in Adults with Normal Renal Function** *(in hours)*	**Recommended Redosing Interval** *(From Initiation of Preoperative Dose)***, assuming a normal renal function**
Ampicillin/ sulbactam	3 g (ampicillin 2 g/sulbactam 1 g)	50 mg/kg of the ampicillin component	0.8–1.3	2
Cefazolin	2 g, 3 g for pts weighing ≥100 kg	30 mg/kg	1.2–2.2	4 (Cardiac surgery q 2h)
Cefoperazone-sulbactam	2 g		2 for cefo. 1 for sulb.	Every 2 to 4 hours (inadequate data)
Ceftriaxone	2 g[b]	50–75 mg/kg	5.4–10.9	Redosing not required
Ciprofloxacin [c]	400 mg IV	10 mg/kg	3–7	Redosing not required
Clindamycin	< 100 kg: 600 mg ≥ 100 kg: 900 mg	10 mg/kg	2–4	6
Ertapenem	1 g	15 mg/kg	3–5	Redosing not required
Fluconazole	400 mg	6 mg/kg	30	Redosing not required
Gentamicin[d]	5 mg/kg based on dosing weight (single dose)	2.5 mg/kg based on dosing weight	2–3	Redosing not required
Metronidazole	500 mg IV	15 mg/kg Neonates weighing < 1200 g should receive a single 7.5-mg/kg dose	6–8	12
Piperacillin/ tazobactam	3.375 g	Infants 2–9 mo: 80 mg/kg of the piperacillin component Children >9 mo and ≤40 kg: 100 mg/kg of the piperacillin component	0.7–1.2	2
Vancomycin	15 mg/kg < 70 kg = 1g 71-99 kg = 1.25g > 100 kg = 1.5 g	15 mg/kg	4–8	12

[a] The maximum paediatric dose should not exceed the usual adult dose.

[b] When used as a single dose in combination with metronidazole for colorectal procedures.

[c] While fluoroquinolones have been associated with an increased risk of tendinitis/tendon rupture in all ages, use of these agents for single-dose prophylaxis is generally safe.

[d] In general, gentamicin for surgical antibiotic prophylaxis should be limited to a single dose given preoperatively.

- Dosing is based on the patient's actual body weight.
- If the patient's actual weight is more than 20% above ideal body weight (IBW), the dosing weight (DW) can be determined as follows: DW = IBW + 0.4(actual weight – IBW).

The above table is based on the following, but modified as per antimicrobial resistance pattern in India

- *Bratzler DW, Dellinger EP, Olsen KM, et al. Clinical practice guidelines for antimicrobial prophylaxis in surgery. Am J Health Syst Pharm. 2013;70(3):195-283*
- *John Hopkins ABX guide*

Antimicrobial Recommendations for Surgical Prophylaxis

Type of Procedure	Recommended Agents [a]	Alternative Agents in Pts With β-Lactam Allergy
Cardiac		
Coronary artery bypass, median sternotomy	Cefazolin (add vancomycin if MRSA colonized/infected)	Clindamycin,[b] vancomycin[b]
Cardiac device insertion procedures (e.g., pacemaker implantation)	Cefazolin (add vancomycin if MRSA colonized/infected)	Clindamycin, vancomycin
Thoracic		
Noncardiac procedures, including lobectomy, pneumonectomy, lung resection, and thoracotomy	Cefazolin OR Ampicillin–sulbactam	Clindamycin,[b] vancomycin[b]
Esophageal surgery	Cefazolin PLUS metronidazole	Clindamycin
Video-assisted thoracoscopic surgery	Cefazolin OR Ampicillin–sulbactam	Clindamycin,[b] vancomycin[b]
Gastroduodenal[c]		
Procedures involving entry into lumen of gastrointestinal tract (bariatric, pancreaticoduodenectomy[d])	Ceftriaxone +/- metronidazole	Clindamycin or vancomycin PLUS aminoglycoside[e]
Procedures without entry into gastrointestinal tract (antireflux, highly selective vagotomy) Decide as per risk of infections		
Patients at low risk for infections	None	None
Patients at high risk for infections **High-risk:** patients with increased gastric pH (e.g., those receiving histamine H2-receptor antagonists or proton-pump inhibitors), gastroduodenal perforation, decreased gastric motility, gastric outlet obstruction, gastric bleeding, morbid obesity, or cancer.	Ceftriaxone	Clindamycin or vancomycin + PLUS aminoglycoside [e]
Biliary tract		
Open procedure	Ceftriaxone +/- metronidazole Ampicillin/sulbactam[f]	Clindamycin or vancomycin PLUS aminoglycoside[e] OR Metronidazole PLUS aminoglycoside[e]
Laparoscopic procedure		
Elective,	None	None

Type of Procedure	Recommended Agents [a]	Alternative Agents in Pts With β-Lactam Allergy
Patients at low risk for infections[g] (see foot note l)		
Elective, *Patients at high risk for infections*[g] **High-risk**: age >70 years, acute cholecystitis, non-functioning gallbladder, obstructive jaundice, common duct stones, diabetes, pregnancy, immunosuppression and other factors. (see foot note l) With cholangitis, treat as infection, not prophylaxis.	Ceftriaxone +/- metronidazole Ampicillin/sulbactam	Clindamycin or vancomycin PLUS aminoglycoside[e] or Metronidazole PLUS aminoglycoside[e]
ERCP	Use prophylaxis only if biliary drainage is not expected to be complete at the end of the procedure. In such case use: Piperacillin tazobactam If there is existing cholangitis, that needs to be treated	Most studies show that achieving adequate drainage will prevent postprocedural cholangitis or sepsis and no further benefit from prophylactic antibiotics; greatest benefit likely when complete drainage cannot be achieved. American Society of Gastrointestinal Endoscopy recommendations for ERCP: treat existing cholangitis; use prophylaxis if biliary drainage is (or anticipated to be) not complete.
Appendectomy for uncomplicated appendicitis	Ceftriaxone PLUS metronidazole	Clindamycin PLUS aminoglycoside[e] or Metronidazole PLUS aminoglycoside[e] or
Small intestine		
Nonobstructed	Ceftriaxone +/- metronidazole	Clindamycin PLUS aminoglycoside[e]
Obstructed	Ceftriaxone PLUS metronidazole	Metronidazole PLUS aminoglycoside[e]
Hernia repair (hernioplasty and herniorrhaphy)	Cefazolin	Clindamycin or vancomycin
Colorectal[h]	Ceftriaxone + metronidazole OR Piperacillin tazobactam	Clindamycin PLUS aminoglycoside[e]

Type of Procedure	Recommended Agents [a]	Alternative Agents in Pts With β-Lactam Allergy
		Metronidazole + aminoglycoside[e]
Head and neck Antimicrobial prophylaxis in head and neck surgery appears efficacious only for procedures involving oral/pharyngeal mucosa (e.g., laryngeal or pharyngeal tumor) but even with prophylaxis, wound infection rate can be high. No prophylaxis for uncontaminated head and neck surgery unless prosthetic device is to be implanted. Tympanostomy tubes do not require prophylaxis.		
Clean or adenoidectomy, parotidectomy, thyroidectomy, tonsillectomy	None	None
Clean with placement of a prosthesis, tumor debulking, mandibular facture repair	Ampicillin-sulbactam	Clindamycin[b]
Clean-contaminated cancer surgery	Cefazolin + metronidazole, Ampicillin/sulbactam	Clindamycin[b]
Neurosurgery		
Elective craniotomy, intrathecal pumps, cranioplasty and cerebrospinal fluid-shunting procedures	Cefazolin (and vancomycin if MRSA colonized/infected)	Vancomycin[b]
Laminectomy	Cefazolin	Clindamycin,[b] vancomycin[b]
Spinal fusion	Cefazolin (plus vancomycin if MRSA colonized/infected)	Vancomycin
Transsphenoidal surgery	Ceftriaxone	Moxifloxacin
Gynecologic Surgery	Ceftriaxone + metronidazole	Clindamycin + aminoglycoside[e]
Cesarean delivery	Ceftriaxone + metronidazole	Clindamycin + aminoglycoside[e]
Hysterectomy (vaginal or abdominal)	Ceftriaxone + metronidazole,	Clindamycin + aminoglycoside
Orthopedic		
Clean operations involving hand, knee, or foot and not involving implantation of foreign materials	None	None
Closed fractures Open reduction of closed fractures with implantation of internal fixation devices (e.g., nails, screws, plates, wires) or bone graft material	Ceftriaxone (plus vancomycin if MRSA colonized/infected)	Clindamycin,[b] vancomycin[b]

Type of Procedure	Recommended Agents [a]	Alternative Agents in Pts With β-Lactam Allergy
Open fractures Open reduction of open fractures	Treat as infected for 10 days	
Hip fracture repair	Cefazolin (plus vancomycin if MRSA colonized/infected May continue up to 24 hours	Clindamycin,[b] vancomycin[b]
Total joint replacement	Cefazolin (plus vancomycin if MRSA colonized/infected May continue up to 24 hours	Clindamycin,[b] vancomycin[b]
Spinal procedures with and without instrumentation	Cefazolin (plus vancomycin if MRSA colonized/infected) May continue up to 24 hours	Clindamycin,[b] vancomycin[b]
Arthroscopy	Cefazolin	Clindamycin
Urologic		
Cystoscopy	Not required Treat patients with UTI prior to procedure using an antimicrobial active against pathogen isolated if possible.	
Cystoscopy with manipulation (ureteroscopy, biopsy, fulguration, TURP)	Ceftriaxone OR Single dose of aminoglycoside (amikacin) Treat patients with UTI prior to procedure using an antimicrobial active against pathogen isolated if possible.	
Transrectal prostate biopsy	Ceftriaxone OR Single dose of aminoglycoside (amikacin)	
Clean without entry into the urinary tract	Cefazolin (the addition of a single dose of an aminoglycoside may be recommended for placement of prosthetic material [e.g., penile prosthesis])	Clindamycin,[b] vancomycin[b]

Type of Procedure	Recommended Agents [a]	Alternative Agents in Pts With β-Lactam Allergy
Clean, involving implanted prosthesis	Cefazolin or vancomycin + aminoglycoside	Clindamycin + aminoglycoside
Clean with entry into the urinary tract	Cefazolin or vancomycin PLUS the addition of a single dose of an aminoglycoside	Aminoglycoside[e] with or without clindamycin
Clean-contaminated (radical cystectomy, ileal conduit, cystoprostatectomy, anterior exenteration)	Ceftriaxone + metronidazole	Aminoglycoside[e] + clindamycin or vancomycin
Vascular[i]		
Carotid or brachiocephalic procedures	Not recommended	
Upper extremity procedures with grafts or lower extremity procedures	Cefazolin	Clindamycin or vancomycin
Abdominal aorta procedure or groin incision	Cefazolin + metronidazole	Vancomycin plus gentamicin
Liver transplantation	Piperacillin/tazobactam	Clindamycin or vancomycin + Gram negative coverage
Renal transplantation/live adult donor	Cefazolin	Clindamycin
Plastic surgery		
Clean with risk factors or clean-contaminated	Cefazolin, ampicillin/sulbactam (add vancomycin if MRSA colonized/infected)	Clindamycin,[b] vancomycin[b]

[a] For patients known to be colonized with methicillin-resistant Staphylococcus aureus, it is reasonable to add a single preoperative dose of vancomycin to the recommended agent(s).

[b] For procedures in which pathogens other than staphylococci and streptococci are likely, an additional agent with activity against those pathogens could be considered. For example, suppose surveillance data show that gram-negative organisms cause surgical-site infections (SSIs) for the procedure. In that case, practitioners may consider combining clindamycin or vancomycin with another agent such as aminoglycoside.

[c] Prophylaxis should be considered for patients at the highest risk for postoperative gastroduodenal infections, such as those with increased gastric pH (e.g., those receiving histamine H2-receptor antagonists or proton-pump inhibitors), gastroduodenal perforation, decreased gastric motility, gastric outlet obstruction, gastric bleeding, morbid obesity, or cancer. Antimicrobial prophylaxis may not be needed when the lumen of the intestinal tract is not entered.

[d] Consider additional antimicrobial coverage with infected biliary tract.

[e] Gentamicin or tobramycin.

[f] Due to increasing resistance of *Escherichia coli* to ampicillin/sulbactam, local population susceptibility profiles should be reviewed before use.

[g] Factors that indicate a high risk of infectious complications in laparoscopic cholecystectomy include emergency procedures, diabetes, long procedure duration, intraoperative gallbladder rupture, age of >70 years, conversion from laparoscopic to open cholecystectomy, American Society of Anesthesiologists classification of 3 or greater, episode of colic within 30 days before the procedure, reintervention in less than one month for a non-infectious complication, acute cholecystitis, bile spillage, jaundice, pregnancy, nonfunctioning gallbladder, immunosuppression, and insertion of prosthetic device. Because a number of these risk factors cannot be determined before surgical intervention, it may be reasonable to give a single dose of antimicrobial prophylaxis to all patients undergoing laparoscopic cholecystectomy.

[h] For most patients, a MECHANICAL bowel preparation combined with oral neomycin sulphate plus oral erythromycin base or with oral neomycin sulphate plus oral metronidazole should be given in addition to i.v. prophylaxis.

[i] Prophylaxis is not routinely indicated for brachiocephalic procedures. Although there are no data in support, patients undergoing brachiocephalic procedures involving vascular prostheses or patch implantation (e.g., carotid endarterectomy) may benefit from prophylaxis.

XIX. Infective Endocarditis Prophylaxis

<u>CONDITIONS</u> for which IE prophylaxis is recommended:

1. Previous endocarditis
2. Prosthetic heart valve.
3. Heart valve with prosthetic material (native valve repair using prosthetic material such as clip, annuloplasty ring)
4. Congenital heart disease
    - Unrepaired cyanotic CHD with or without palliative shunts and conduits.
    - *Completely repaired* congenital heart defect with prosthetic material or device, whether placed by surgery or by catheter intervention, *during the first 6 months* after the procedure (endothelization takes 6 months)
    - Repaired CHD *with residual defects at* the site or adjacent to the site of a prosthetic patch or prosthetic device (this inhibits endothelization)
5. A heart transplant with valvulopathy
6. Durable mechanical circulatory support device (ventricular assist device or artificial heart).

<u>CONDITIONS</u> for which IE prophylaxis is NOT recommended:

1. Valvular heart disease unless there is previous endocarditis.
2. Acyanotic congenital heart disease such as ASD, VSD, PDA (unless repaired *with residual defects at* the site or adjacent to the site of a prosthetic patch or prosthetic device)
3. ICD
4. Pacemaker

<u>PROCEDURES</u> for which IE prophylaxis is recommended:

Dental procedures

1. Procedures involving manipulation of the gingiva, periapical area of teeth, or perforation of the oral mucosa. Example: tooth extraction, drainage of a dental abscess, periodontal surgery, *subgingival* scaling and root planning, replantation of avulsed teeth, implant placement, apicoectomy
2. For routine dental cleaning such as scaling, prophylaxis is indicated, but the benefit is uncertain.

Respiratory procedures

Procedures that involve incision of respiratory mucosa. Example: tonsillectomy, and adenoidectomy. If there is an active infection, tailor antibiotic selection based on suspected/confirmed infection.

Procedures on infected skin or subcutaneous tissue. Example: abscess

<u>PROCEDURES</u> for which IE prophylaxis is NOT recommended:

Dental procedures

Anesthetic injections through uninfected tissue
Dental imaging
Placement of removable prosthodontic or orthodontic applications
Adjustment of orthodontic appliances or orthodontic brackets
Shedding deciduous teeth
Bleeding from trauma to lips or oral mucosa

(for routine dental cleaning such as scaling, prophylaxis is indicated, but the benefit is uncertain)

GI, reproductive organ, and GU procedures

Drug selection			
Situation	***Drug***	***Adult***	***Children***
Oral	Amoxicillin	2 g	50 mg/kg
Parenteral (if unable to take oral medication)	Ampicillin	2 g IM or IV	50 mg/kg IM or IV
	or		
	Cefazolin or ceftriaxone	1 g IM or IV	50 mg/kg IM or IV
Oral (If allergic to penicillin or ampicillin)	Cephalexin	2 g	50 mg/kg
	or		
	Azithromycin or clarithromycin	500 mg	15 mg/kg
	or		
	Doxycycline	100 mg	<45 kg, 2.2 mg/kg >45 kg, 100 mg
Parenteral (If allergic to penicillin or ampicillin and not able to take oral medication)	Cefazolin or Ceftriaxone	1 g IM or IV	50 mg/kg IM or IV
	Vancomycin	15 to 20 mg/kg IV, (up to 2 g)	15 mg/kg IV (up to 1 g)

The selected drug should be given 30 to 60 min prior to the procedure. Cephalosporins should not be used for patients with history of anaphylaxis, angioedema, or urticaria with penicillin or ampicillin.

XX. Bibliography

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